Molecular Biology

Topic 1 Introduction to Molecular Biology

Molecular Biology

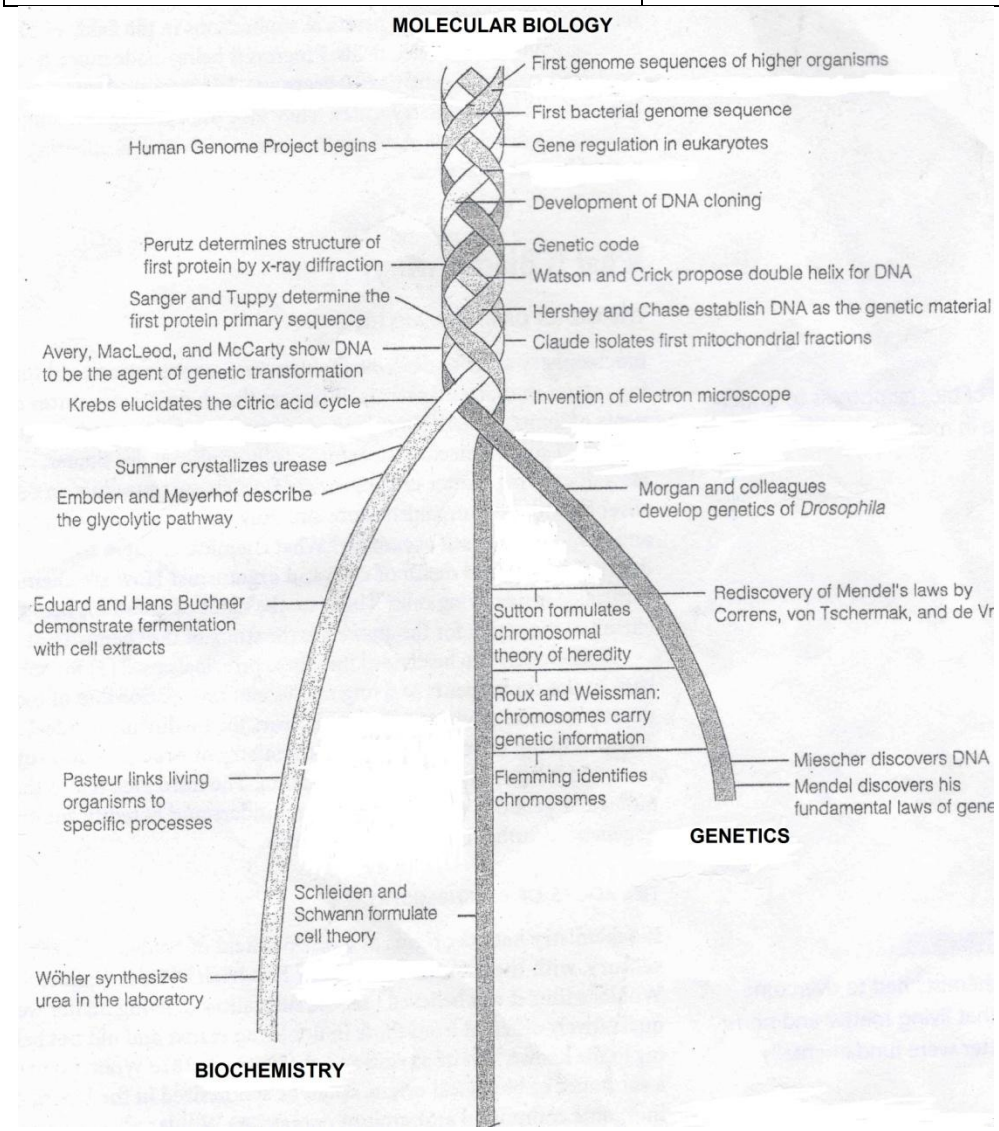
- In the broader terms, definition of molecular biology includes all aspects of the study of life from a molecular perspective.
- More precisely, the term "Molecular Biology" refers to the biology of the molecules related to genes, gene products and heredity.
- In other words, the term molecular biology is often substituted for a more appropriate term, M**olecular Genetics.**
- So molecular biology grew out of the disciplines of genetics and biochemistry.
- In the present age, world is in the midst of two scientific revolutions. One is information technology and the other is Molecular Biology.
- Both deal with the handling of large amounts of information.
- Molecular Biology has revolutionized the biological sciences as well especially in the fields of Health Sciences and Agricultural Sciences.

Topic 2 History of Molecular Biology

- Molecular Biology takes its roots mainly from the the disciplines of Genetics and Biochemistry.
- Some important discoveries which led to the emergence of Molecular Biology as a discipline include:-

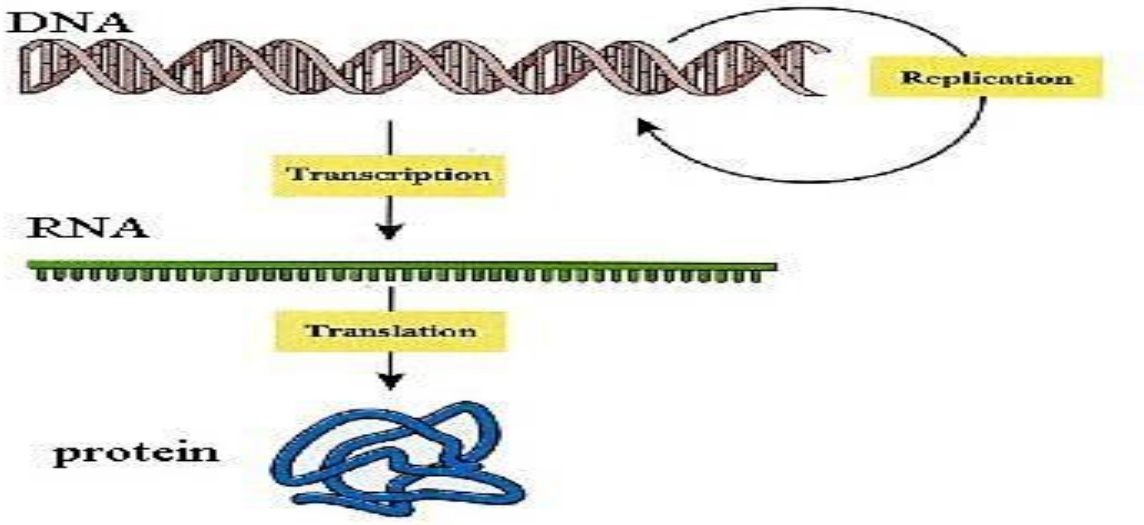
Biochemistry	Genetics
• Synthesis of Urea by Friedrich Wohler (1828). • The first enzyme diastase was discovered by Anselme Payen in 1833.	• Mendel's Experiments on garden pea by Gregor Mendel in 1865. • Chromosome theory of inheritance by Thomas Hunt Morgan in 1910.
• The Discovery of DNA by Friedrich Miescher in 1869. • Eduard Buchner discovered cell-free fermentation in1897. • James B. Sumner crystallized enzyme Urease in 1926.	• Barbara McClintock and Harriet Creighton provided physical evidence of recombination in 1931. • One-gene/one-enzyme hypothesis proposed by George Beadle and Edward Tatum in 1941.

• James Watson and Francis Crick discovered the double helical structure of DNA in 1953.	• Oswald Avery and his colleagues demonstrated in 1944 that DNA is the hereditary material. • It was further confirmed by Alfred Hershey and Martha Chase in 1952. • At this point, the discoveries in the two fields led to the foundation of a new discipline called Molecular Biology.

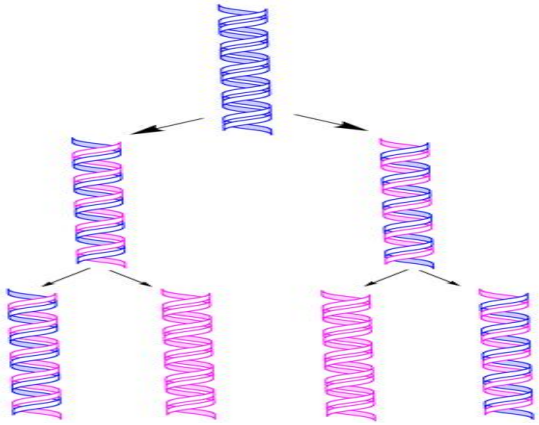


Topic 3 Achievements of Molecular Biology

In 1957, Francis Crick laid out the "Central dogma of molecular biology" which foretold the relationship between DNA, RNA, and proteins.



- In 1958, Mathew Meselson & Franklin Stahl proved that DNA replication was semi-conservative.



- Marshall Nirenberg and Gobind Khorana working independently cracked the code in the early 1960s.
- They found that 3 bases constitute a code word, called a **codon**, that stands for one amino acid.

Gene Cloning

- Since 1970s, scientists have learned to isolate genes, place them in new organisms and reproduce them by a set of techniques collectively known as **gene cloning**.

Genetically Modified Organisms (GMOs)

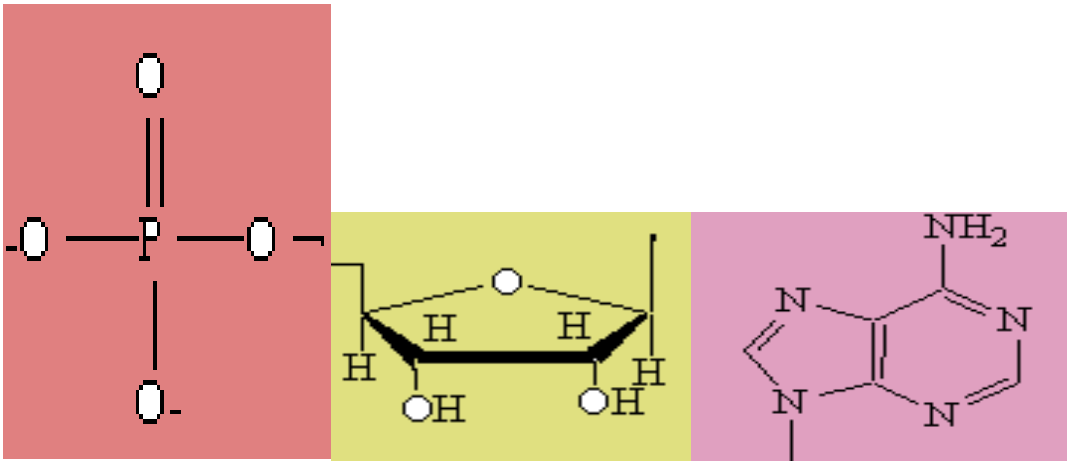
- The technique of Gene cloning led to the creation of a large number of genetically modified organisms with desirable characters.

Human Genome Project

- The Human Genome Project (HGP) was launched in 1990 and completed in 2003.

Topic 4 Nucleoside & Nucleotide

- A molecule containing all these three components is called a **nucleotide**.
- While a molecule without the phosphate group is called a **nucleoside**.



Nucleotide = Nucleoside + Phophoric acid

&

Nucleoside = Nucleotide – Phoshoric acid

Topic 4 Nucleic Acids

- Nucleic acids are important group of biomolecules which are responsible for storage & transmission of hereditary information.
- Like proteins and polysaccharides, nucleic acids are also polymeric compounds.
- The repeating units in the nucleic acids are Nucleotides.

Nucleic acid is divided into two categories:-

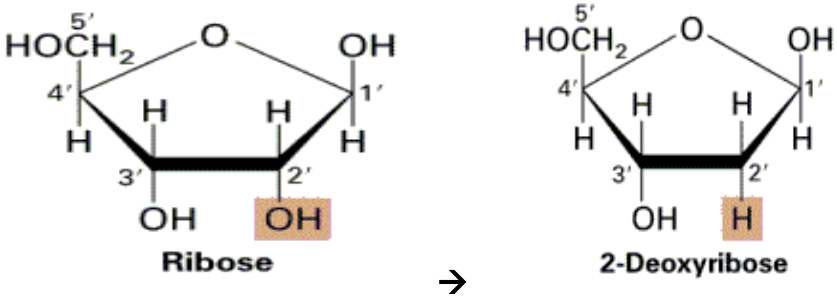
1. Ribonucleic acids (RNA)

2. Deoxyribonucleic acids (DNA)

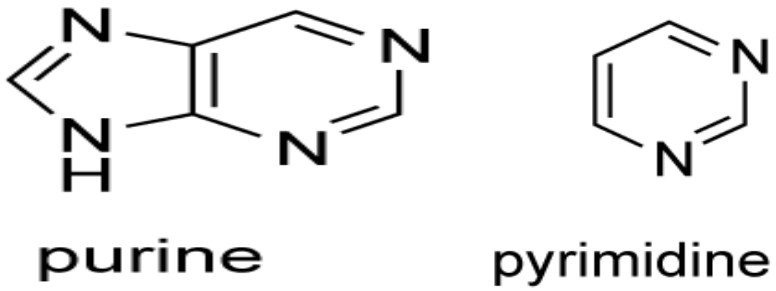
Topic 5 Chemical composition of DNA

- DNA is a polymer of Deoxyribonucleotides.
- Deoxyribonucleotides is composed of three components:
    - Deoxyribose
    - Nitrogenous Base
    - Phosphoric acid

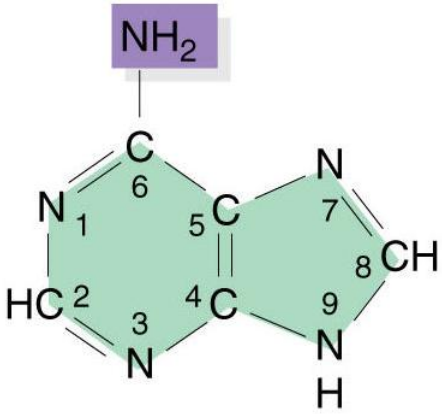
Deoxyribose (a pentose sugar derivative)



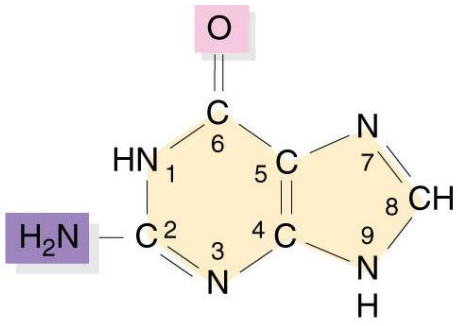
Nitrogenous Bases



Purines

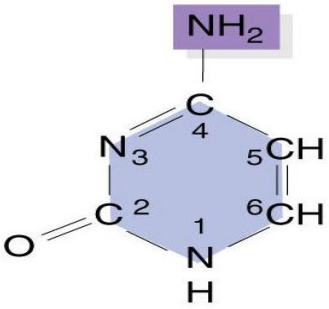


Adenine (A)

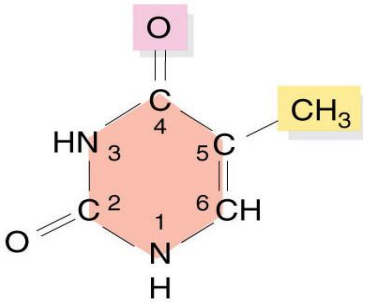


Guanine (G)

Pyrimidines

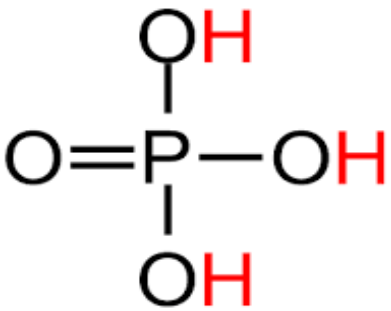


Cytosine (C)



Thymine (T)

Phosphoric acid

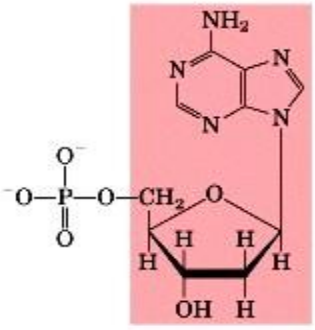


H_3PO_4

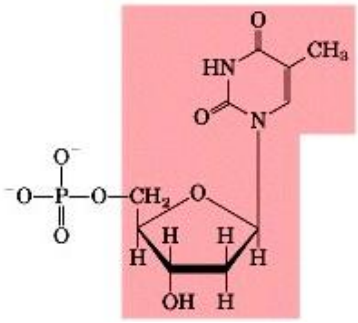
<u>Topic 6 Nucleoside & Nucleotide (above)</u>

<u>Topic 7 Types of Deoxyribonucleotides</u>

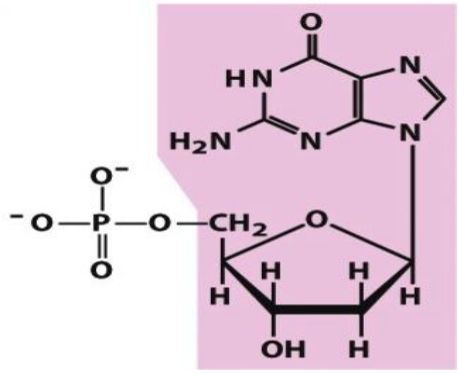
There are four types of Deoxy-ribonucleotides.



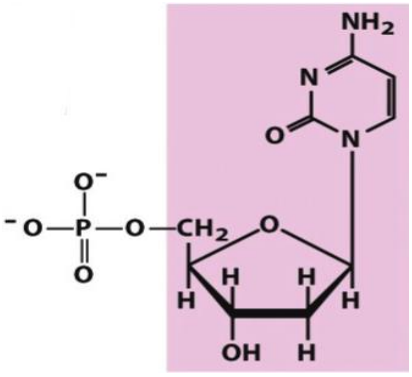
Deoxyadenosine → Deoxyadenylate



Deoxythymidine → Deoxythymidylate



Deoxyguanosine → Deoxyguanylate

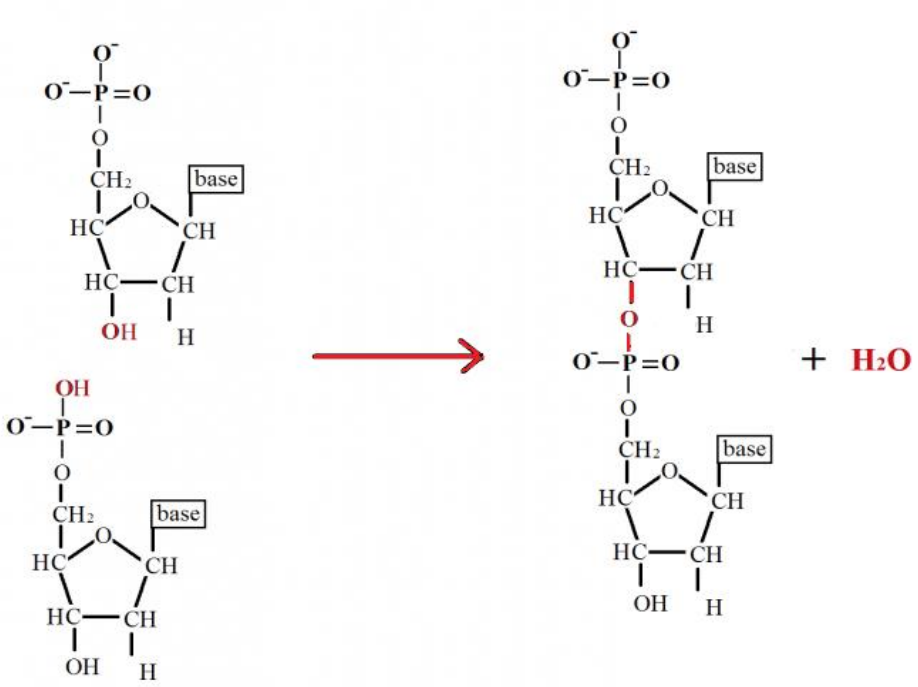


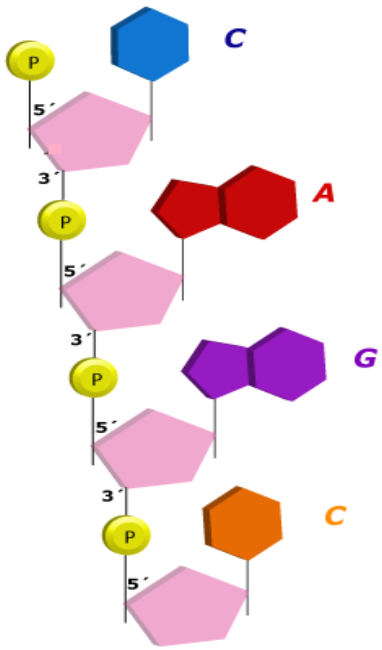
Deoxycytidine → Deoxycytidylate

- These four deoxy-ribonucleotides make the structural units of DNA.

Topic 8 : How do Deoxyribonucleotides Join?

The successive nucleotides of DNA are joined together through phospho-diester linkages.





Topic 9 : Structure of DNA

Work of Chargaff (Late 1940s)

- The discovery of the structure of DNA is one of the greatest events in the history of science. Erwin Chargaff and his colleagues provided a most important clue to the structure of DNA.

- The work of Chargaff led him to following conclusions, also called "Chargaff Rules":-

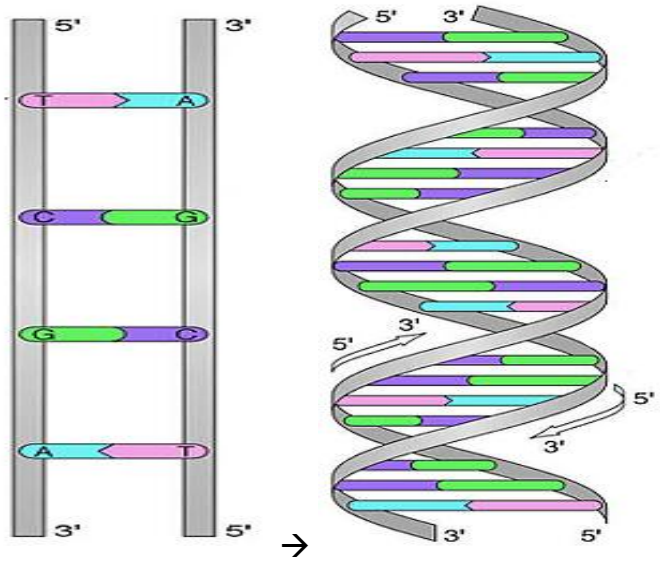
1. Base composition of DNA varies from one species to another.
2. The DNA isolated from different tissues of the same species has the same base composition.
3. The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment.
4. In DNA, the number of adenosine residues is equal to the number of thymidine (A=T) and the number of guanosine residues is equal to the number of cytidine (G=C).
5. It means that the sum of the purine residues equals the sum of the pyrimidine residues (AG=TC).

Topic 10 is not available at this moment.

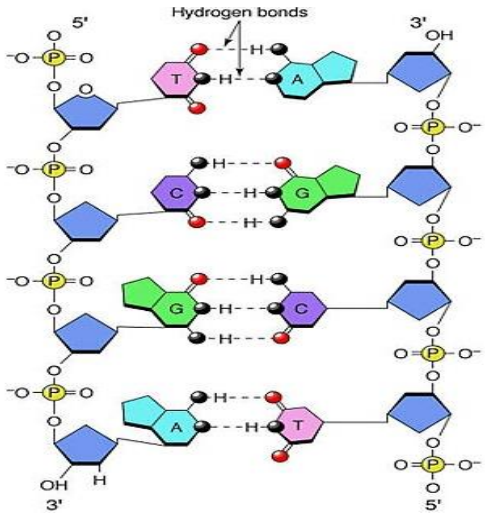
Topic 11 Structure of DNA (Work of Watson & Crick)

James Watson and Francis Crick postulated a three dimensional model of DNA structure in 1953. The major features of this model are as follow:-

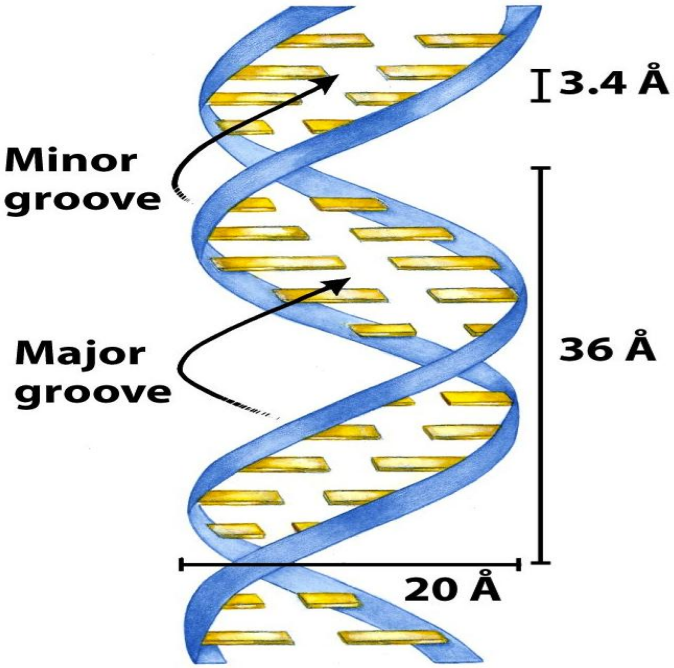
1. DNA consists of two helical polynucleotide strands which are wound around the same axis to form a right handed double helix.



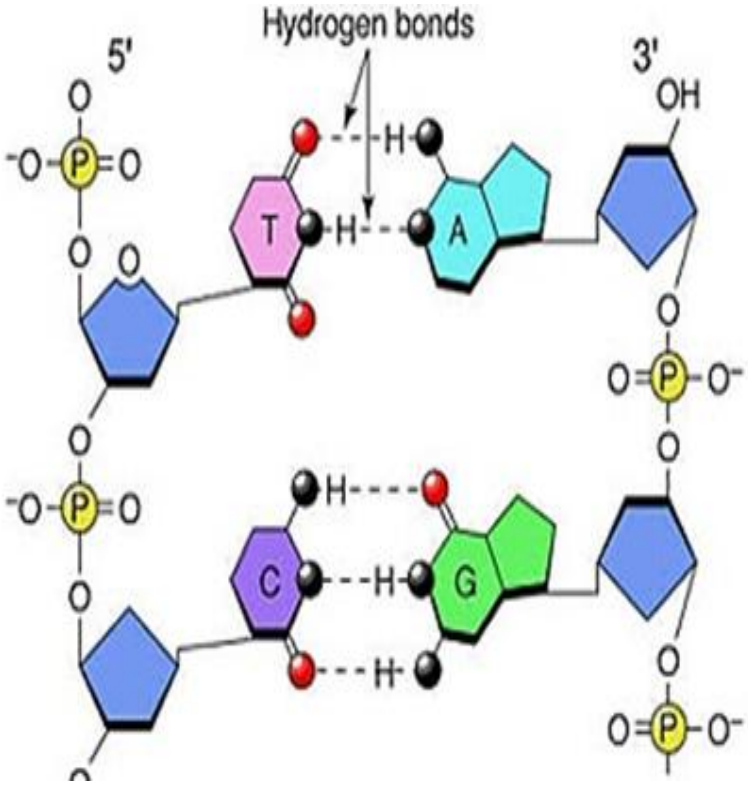
2. The hydrophilic backbones of alternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water
3. The nitrogenous bases of both strands are stacked inside the double helix lying perpendicular to the long axis of the helix.



4. The pairing and coiling of the two strands create a major groove and minor groove on the surface of the helix.



5. Each nucleotide base of one strand is paired in the same plane with a base of the other strand.

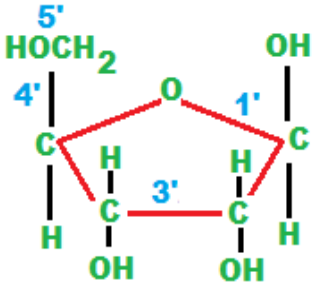


6. They also found that G pairs with C and A pairs with T due to the reason that they fit best within the structure.
7. The two strands in DNA are present in antiparallel orientation i.e; their 5, 3-phosphodiester bonds run in the opposite directions.
8. The two antiparallel strands of double-helical DNA are complementary to each other.

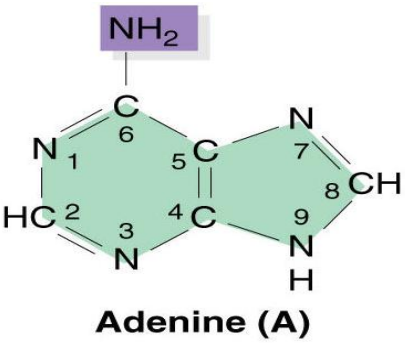
Topic 12 Chemical composition of RNA

- RNA (Ribonucleic acid) is a polymer of ribonucleotides.
- Each ribonucleotide is composed of three components:
    - A ribose sugar
    - A Nitrogenous Base
    - A Phosphoric acid

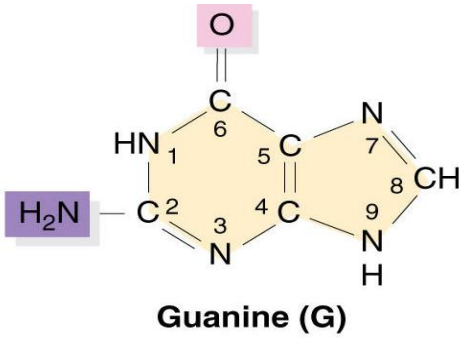
Ribose (a pentose sugar)



Nitrogenous Bases



Adenine (A)

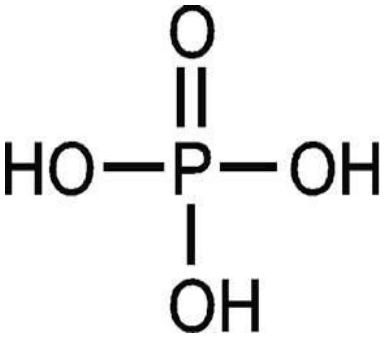


Guanine (G)

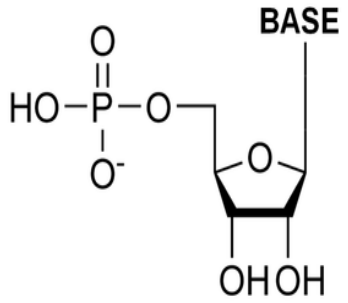
Cytosine (C)

Uracil (U)

Phosphoric acid

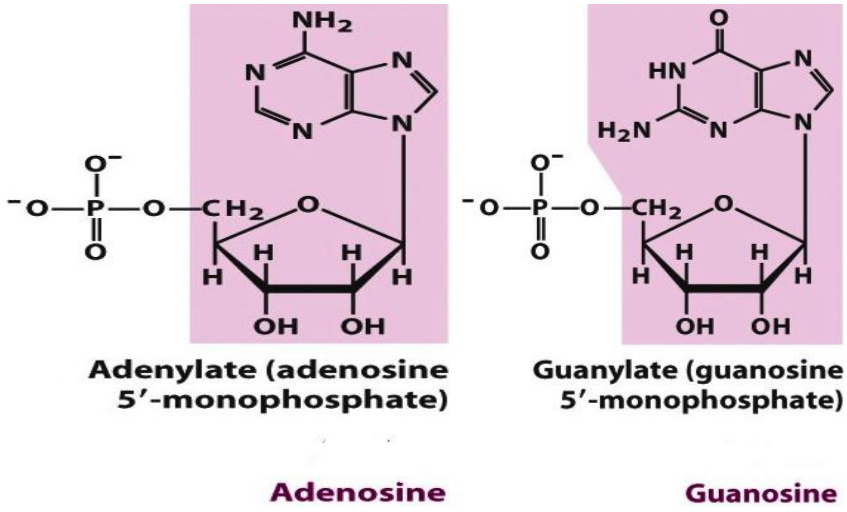


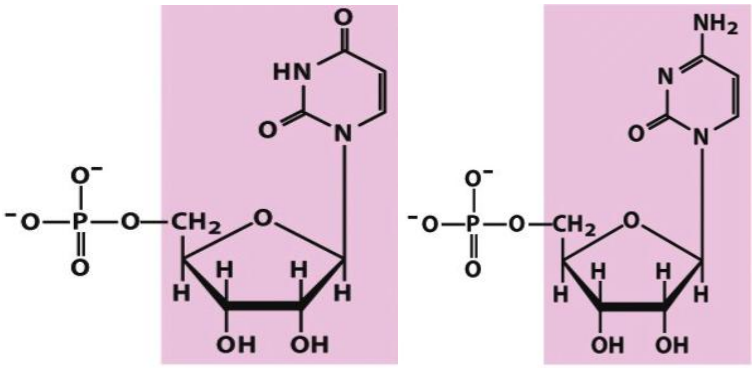
A Ribonucleotide



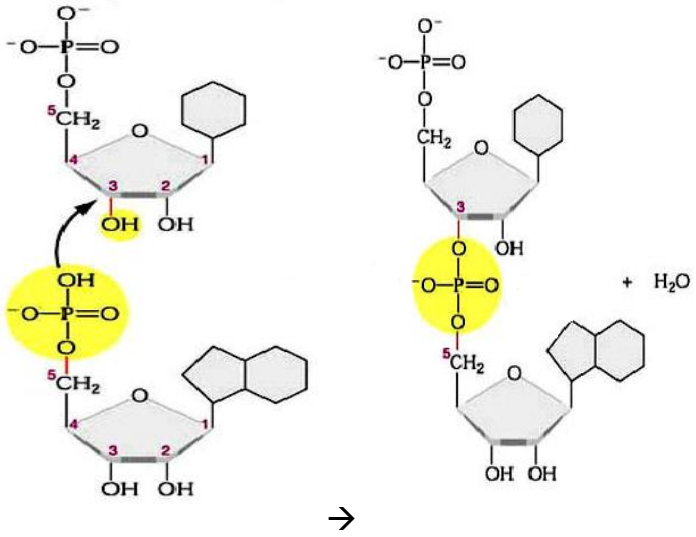
Topic 13 Types of Ribonucleotides

There are mainly four types of ribonucleotides depending upon the types of nitrogenous bases present in RNA.





How do Ribonucleotides Join?



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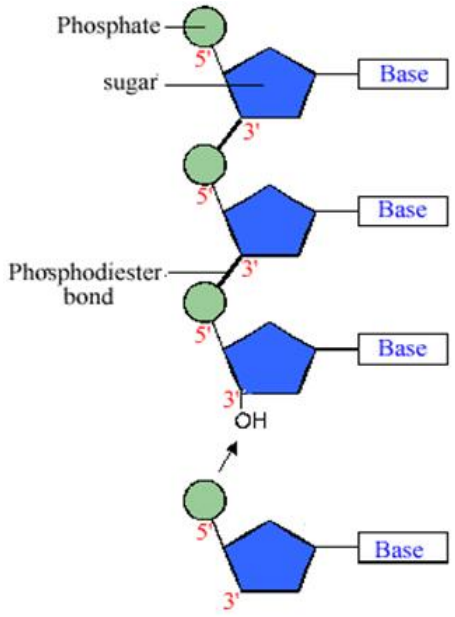
A Poly-Ribonucleotide



RNA molecules are much shorter

Topic 14 Types of RNAs

There are mainly three types of Ribonucleic acids (RNAs) present in the cells of living organisms.

- Messenger RNA (mRNA)
- Transfer RNA (tRNA)
- Ribosomal RNA (rRNA)

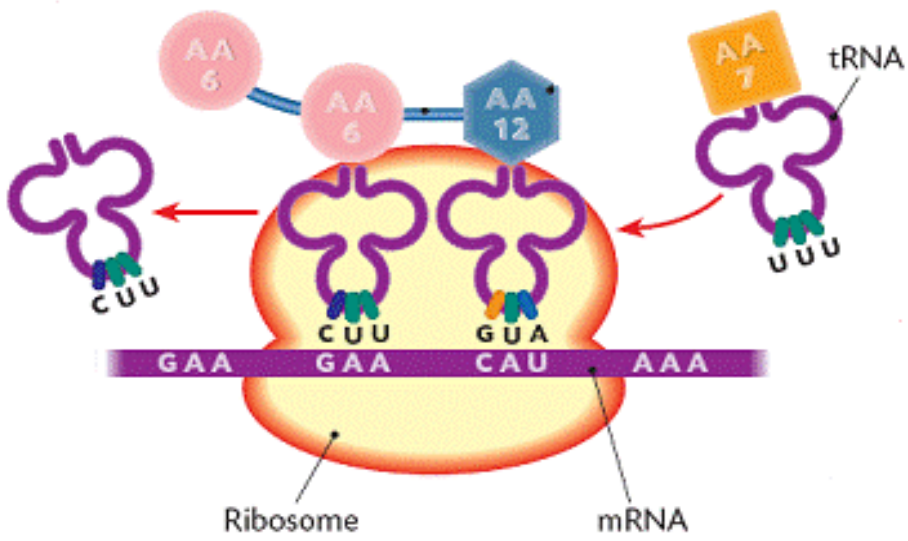
Messenger RNA (mRNA)

- It is the type of RNA that carries genetic information from DNA to the protein biosynthetic machinery of the ribosome.
- It provides the templates that specify amino acid sequences in polypeptide chains.
- The process of forming mRNA on a DNA template is known as **transcription**.
- It may be monocistronic or polycistronic.
- The length of mRNA molecules is variable and it depends on the length of gene.

Transfer RNA (tRNA)

- Transfer RNAs serve as adapter molecules in the process of protein synthesis.
- They are covalently linked to an amino acid at one end.

- They pair with the mRNA in such a way that amino acids are joined to a growing polypeptide in the correct sequence.



Ribosomal RNA (rRNA)

- Ribosomal RNAs are components of ribosomes.
- rRNA is a predominant material in the ribosomes constituting about 60% of its weight.
- It has a number of functions to perform in the ribosomes.

Topic 15 is not available at this moment.

Topic 16 Nature of Genetic Material

- After establishment of the fact that genes are the physical units located on the chromosomes. A major problem for the biologists was to find out the molecules responsible for carrying the hereditary information.

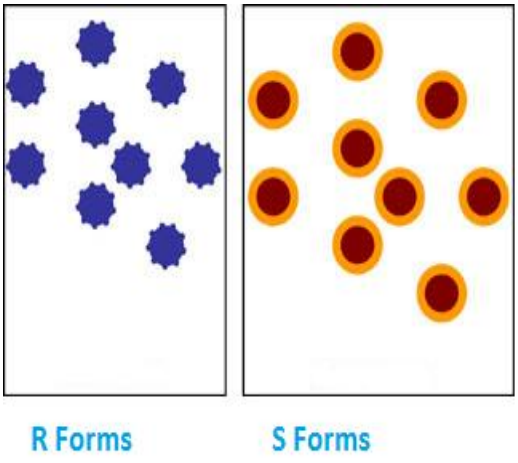
Characteristics of Genetic Material

- Genetic material must contain complex information.
- Genetic material must replicate faithfully.
- Genetic material must encode phenotype.
- Three sets of experiments provided pivotal evidence that DNA rather than protein is the hereditary material.

- Griffith's Experiments (1928)
- Avery's Experiments (1944)
- Hershey-Chase experiments (1952)

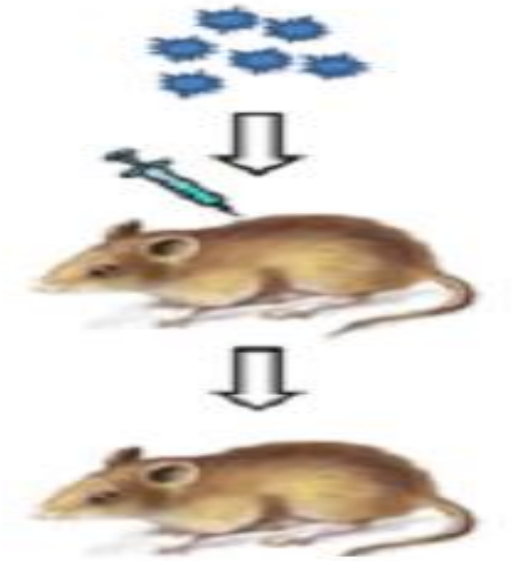
Topic 17 Griffith's Experiments

- The first clue that DNA was the carrier of hereditary information came from the work of Fred Griffith in 1928.
- He observed the phenomenon of Transformation in the bacterium *Streptococcus pneumonia.*
- Griffith had isolated different strains of *S. pneumonia.*
- In the virulent strain, each bacterium is surrounded by a polysaccharide coat.
- This polysaccharide coat makes the bacterial colony appear smooth when grown on an agar plate.
- These forms of *S. pneumonia* are referred to as S forms.
- Griffith found that these virulent forms occasionally get mutated to non-virulent forms, which lack a polysaccharide coat.
- They produced a rough appearing colony on an agar plate; these forms are referred to as R forms.



Topic 18 Transformation Experiments

Griffith carried out series of experiments using S-Form & R-Form strains of *S. pneumonia* and mouse as an experimental animal.



Possible Interpretations

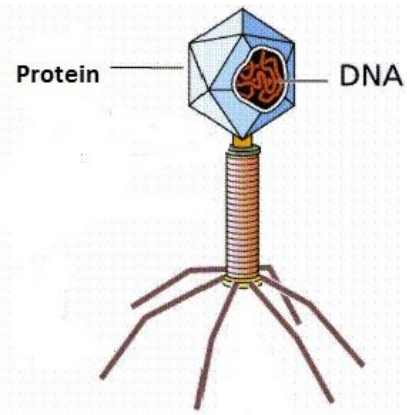
- It could have been the case that S- Type bacteria were not completely killed and a few live bacteria remained in the culture.
- A second interpretation was that the live R-Type bacteria had mutated to the virulent S form.
- Griffith finally concluded that R-Type bacteria had been *transformed.*
- Griffith theorized that some substance of the dead bacteria might be responsible for that transformation. He called that substance as the **transforming principle.**

Topic 19 is not available yet.

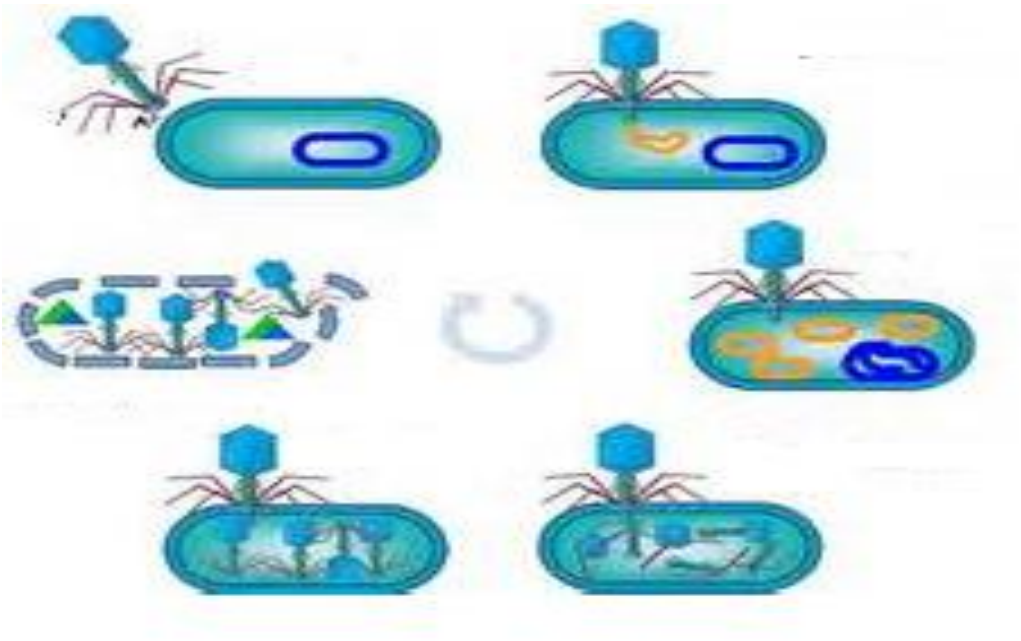
Topic 20 Hershey & Chase experiment

- Alfred Hershey and Martha Chase in 1952 provided second evidence that DNA is the genetic material.
- They were working on T2 which is a *bacteriophage* that infects the bacterium *Escherichia coli.*

Bacteriophage

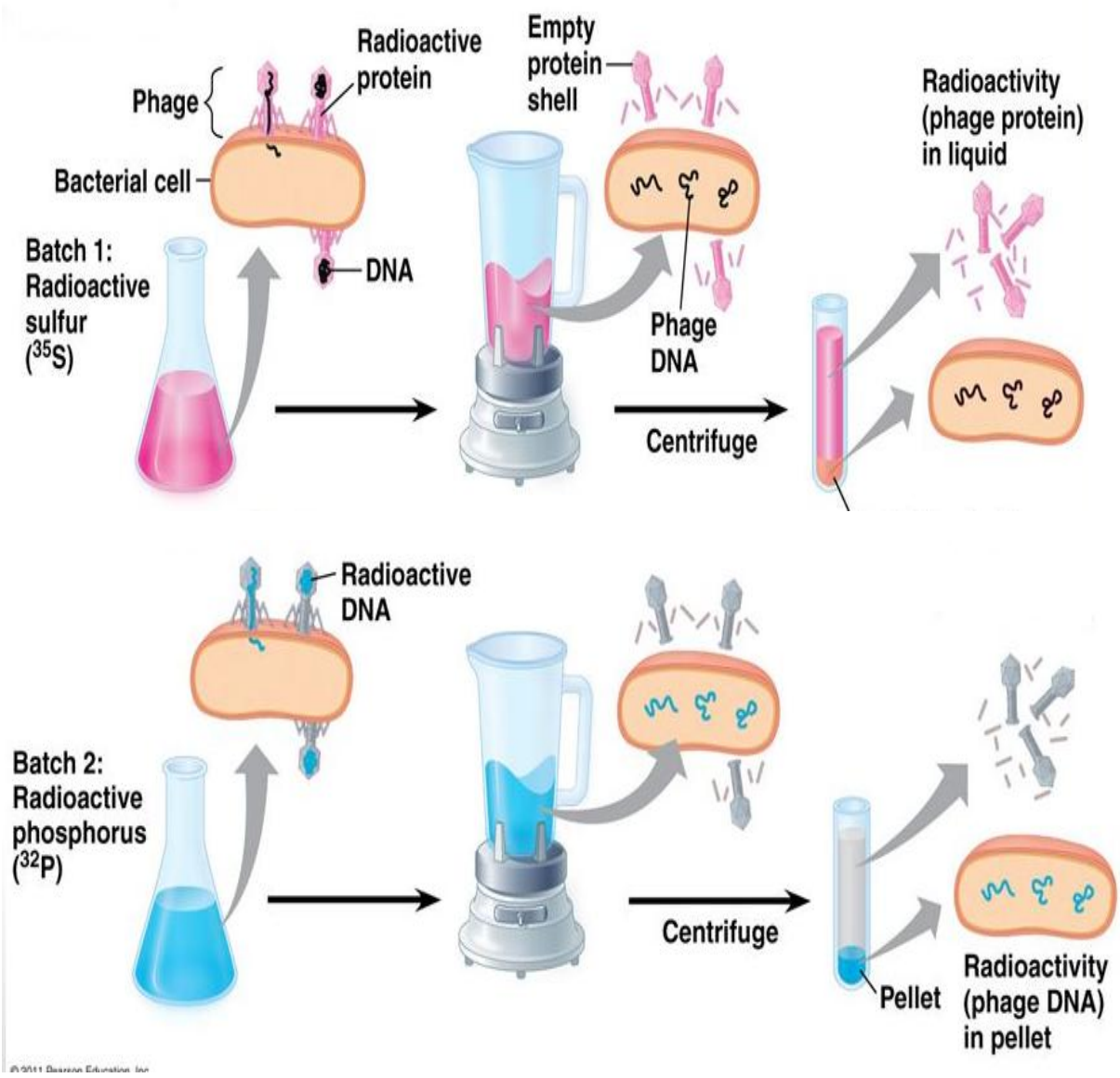


Life Cycle of a Bacteriophage



- They used radioactive isotopes of phosphorus and sulfur.
- DNA contains P but not S; so Hershey and Chase used ^{32}P to follow phage DNA during reproduction. Protein contains sulfur but not phosphorus; so they used ^{35}S to follow the protein.
- Hershey and Chase first grew *E. coli* in a medium containing ^{32}P and infected the bacteria with T2 so that all the new phages would have DNA labeled with ^{32}P.
- They grew a second batch of *E. coli* in a medium containing ^{35}S and infected these bacteria with T2 so that all these new phages would have protein labelled with ^{35}S.
- Hershey and Chase then infected separate batches of unlabeled *E. coli* with the ^{35}S- and ^{32}P-labeled phages.
- Then they placed the *E. coli* cells in a blender and sheared off the empty protein coats from the cell walls.
- They separated out the protein coats and cultured the infected bacterial cells.
- Eventually, the cells burst and new phage particles emerged.
- When phages labeled with ^{35}S infected the bacteria, most of the radioactivity separated with the protein coats.
- When new phages emerged from the cell, they contained almost no radioactivity.

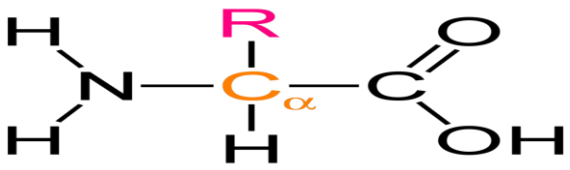
- So?
- When phages labelled with ^{32}P infected the bacteria, and removed the protein coat, radioactivity was present in the cells.
- When new phages emerged from the cell, they were also radioactive.
- So?



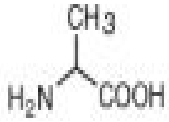
Topic 21 Chemical composition of proteins

- Proteins are polymers of amino acids.

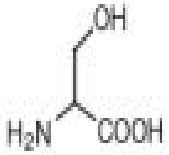
- They range in size from small to very large.
- All the proteins are made up of twenty different types of amino acids. So these amino acids are called standard amino acids.



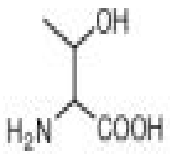
Glycine (Gly, G)



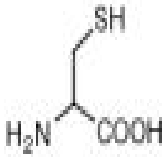
Alanine (Ala, A)



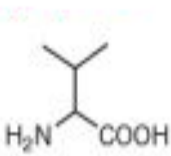
Serine (Ser, S)



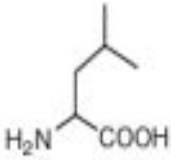
Threonine (Thr, T)



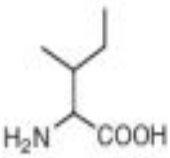
Cysteine (Cys, C)



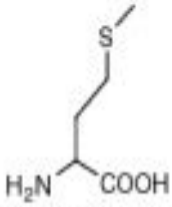
Valine (Val, V)



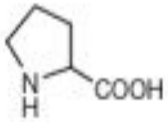
Leucine (Leu, L)



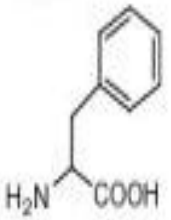
Isoleucine (Ile, I)



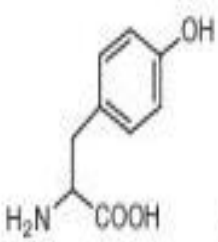
Methionine (Met, M)



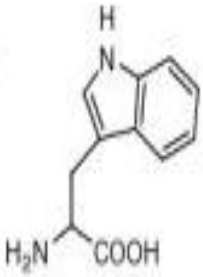
Proline (Pro, P)



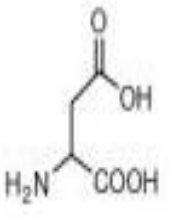
Phenylalanine (Phe, F)



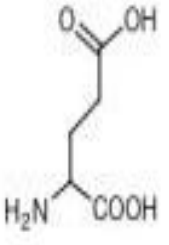
Tyrosine (Tyr, Y)



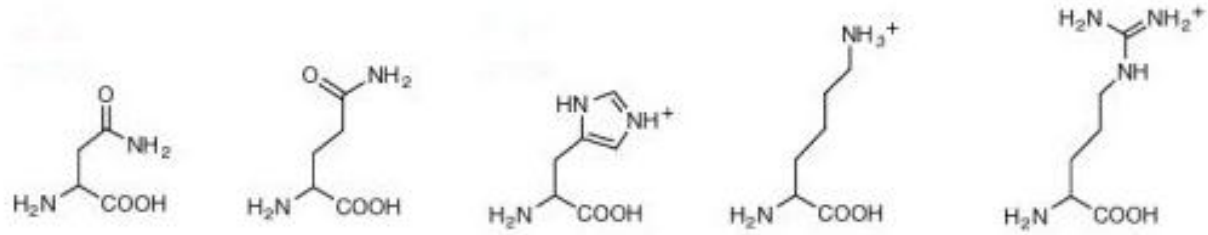
Tryptophan (Trp, W)



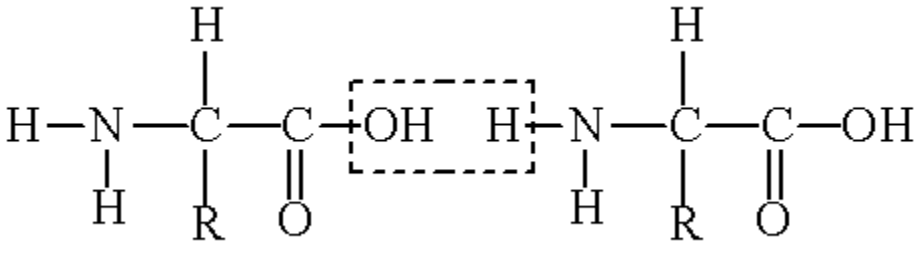
Aspartic Acid (Asp, D)



Glutamic Acid (Glu, E)



- In a protein molecule, each amino acid residue is joined to its neighbour by a specific type of covalent bond which is called Peptide Bond.



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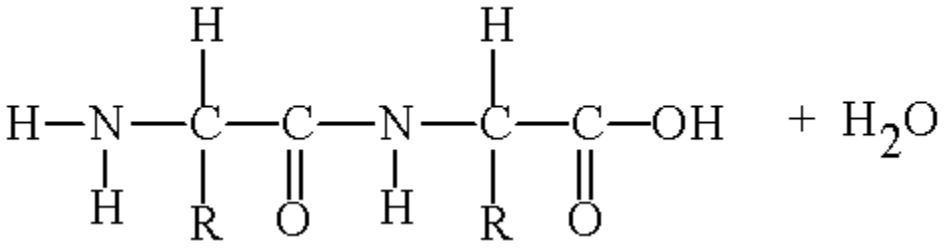


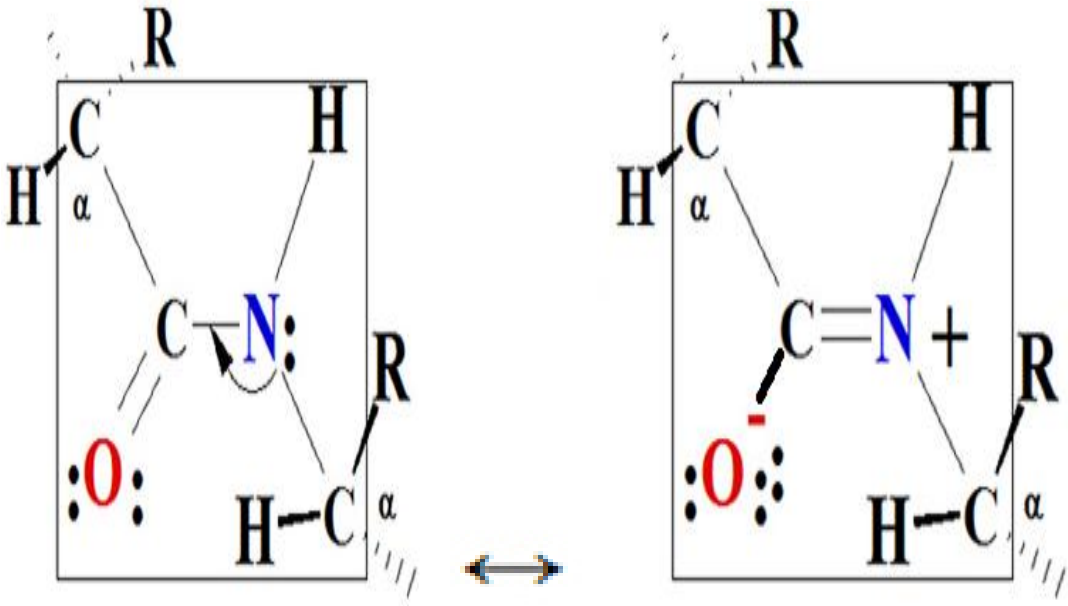
- Amino acids can successively join to form dipeptides, tripeptides, tetra peptides, oligo peptides and polypeptides.

Topic 22 Primary structure of proteins

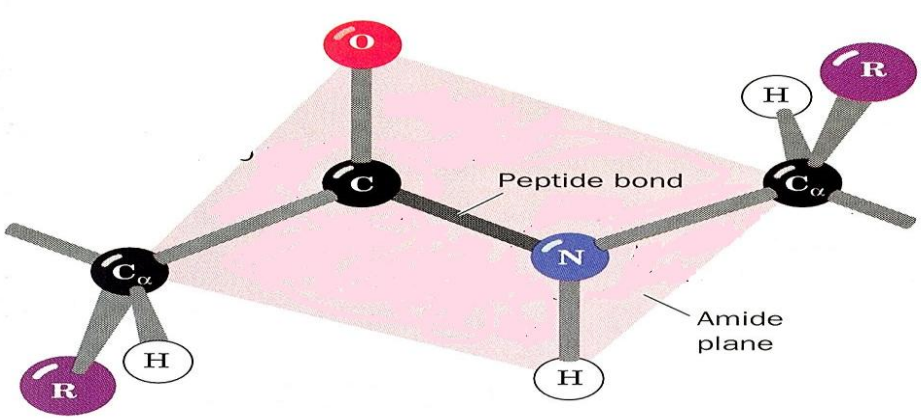
- Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain.
- Each type of protein has a unique amino acid sequence.

Peptide Bond Is Rigid and Planar

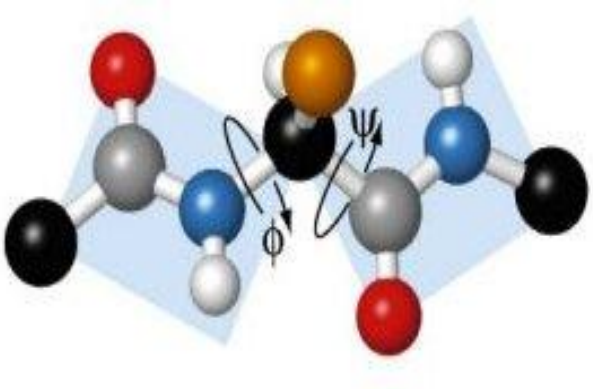
- Linus Pauling and Robert Corey carefully analysed the peptide bond.
- Their findings laid the foundation for our present understanding of protein structure.
- They demonstrated that the peptide C - N bond is somewhat shorter than the C - N bond in a simple amine.



- The six atoms of the peptide group are co-planar i.e., lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen Trans to each other.



- Pauling and Corey concluded that the peptide C - N bonds are unable to rotate freely because of their partial double-bond character.
- Rotation is permitted about the N - αC and the αC - C bonds.

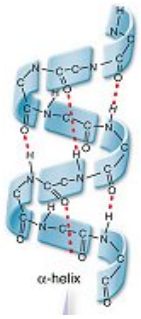


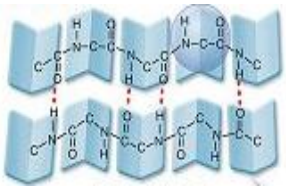
- The bond angles resulting from rotations at C are labelled ϕ (phi) for the N - αC bond and ψ (psi) for the αC - C bond.
- In principle, ϕ and ψ can have any value between +180 & -180.

<u>Topic 23 Secondary structure of proteins</u>

- Secondary structure of proteins refers to the local conformation of some part of a polypeptide.
- A few types of secondary structures are particularly stable and occur widely in proteins.

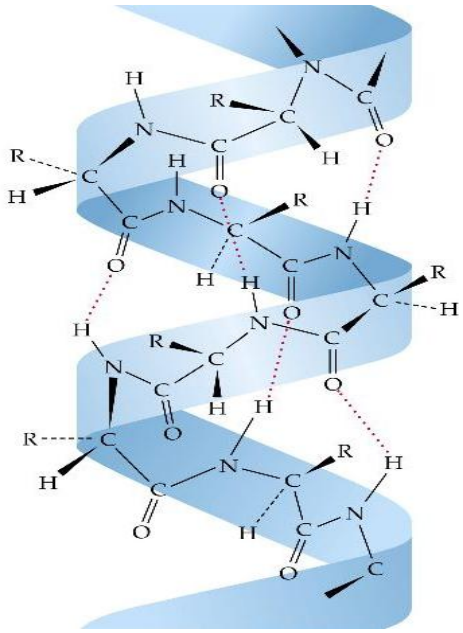
The most prominent are:-

1. α-helix 

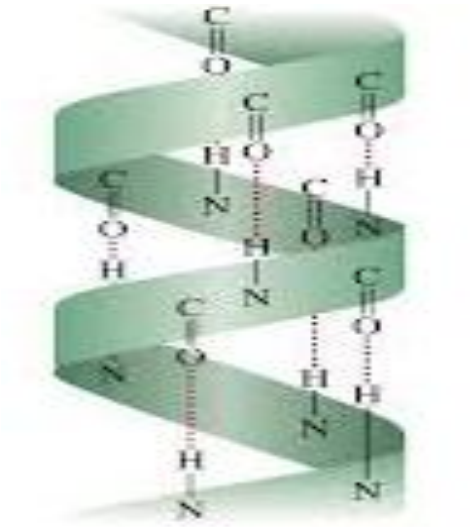
2. β- conformations. 

Topic 24 (α- Helix)

- The simplest arrangement which a polypeptide chain could assume with its rigid peptide bonds is a helical structure, which Pauling and Corey called the α-helix.



- The helical twist of the *α*-helix found in all proteins is right-handed.
- The repeating unit is a single turn of the helix, which extends about 5.4 Å (includes 3.6 amino acid residues) along the long axis.
- The amino acid residues in a helix have conformations with psi = –45 to –50 and phi = –60.
- A helix makes optimal use of internal hydrogen bonds.

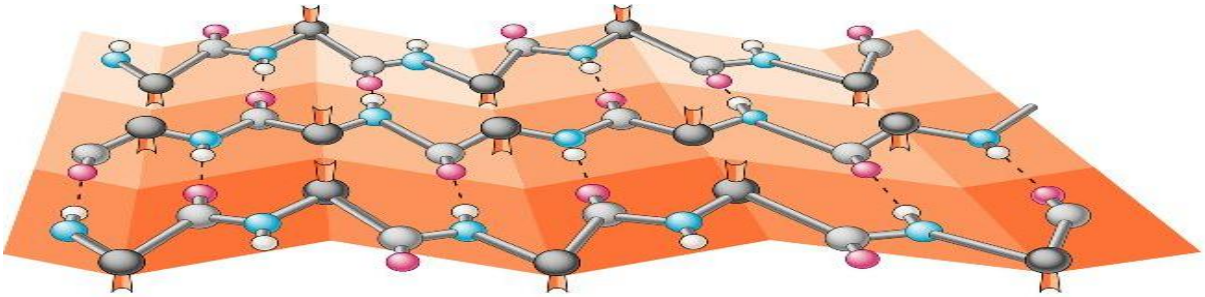


The structure is stabilized by a hydrogen bond between the hydrogen atom attached to the electronegative nitrogen atom of a peptide linkage and the electronegative carbonyl oxygen atom of the fourth amino acid on the amino-terminal side of that peptide bond.

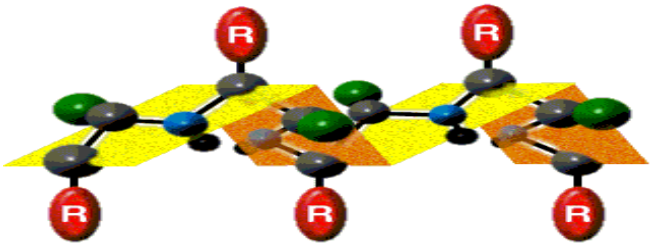
- About one-fourth of all amino acid residues in polypeptides are found in α-helices while in some proteins it is the predominant structure.

Topic 25 β- Pleated Sheets

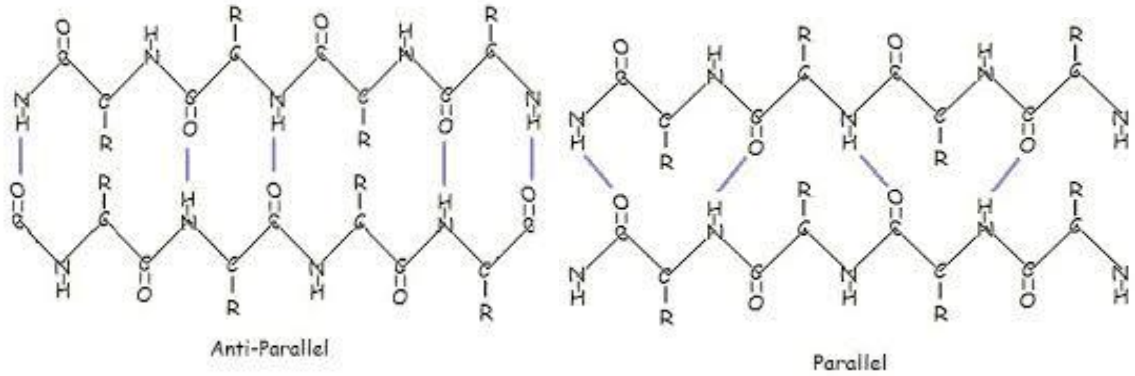
- Pauling and Corey predicted a second type of secondary structure which they called **β-sheets.**
- This is a more extended conformation of polypeptide chains.
- The backbone of the polypeptide chain is extended into a zigzag structure.
- The zigzag polypeptide chains are arranged side by side to form a structure resembling a series of pleats.



- The R groups of adjacent amino acids protrude from the zigzag structure in opposite directions.

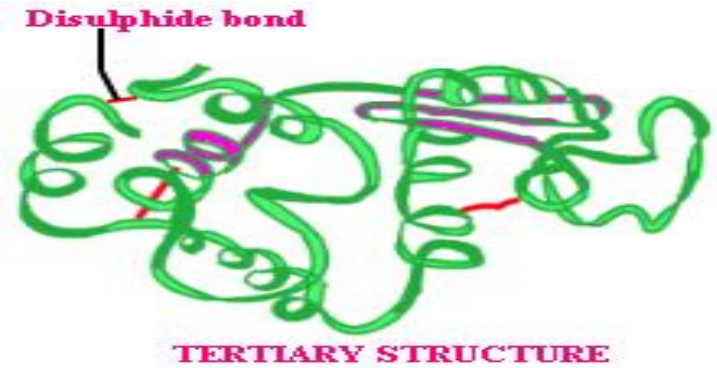


- Hydrogen bonds are formed between adjacent segments of polypeptide chain.
- The adjacent polypeptide chains in a sheet can be either parallel or antiparallel.



Topic 26 Tertiary Structure of Proteins

- The overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's **tertiary structure.**

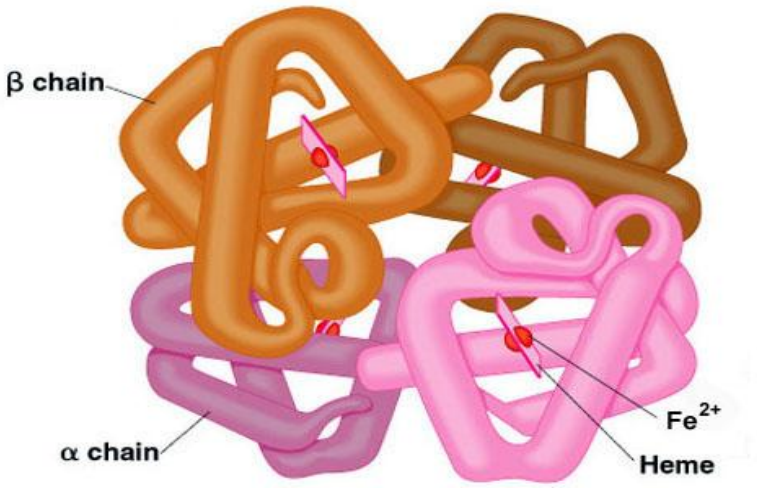


- It includes longer-range aspects of amino acid sequence.
- Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein.

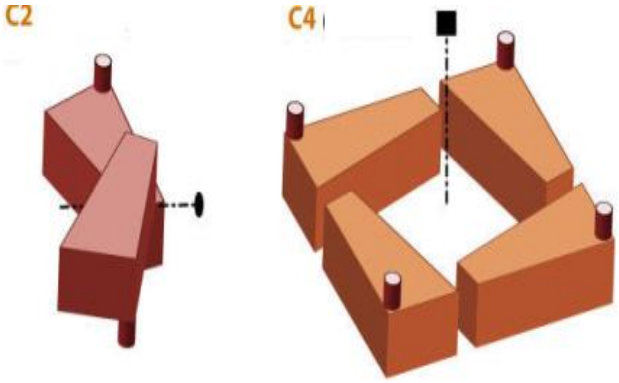
- Interacting segments of polypeptide chains are held in their characteristic tertiary positions by different kinds of weak interactions (and sometimes by covalent bonds) between the segments.
- Large polypeptide chains usually fold into two or more globular clusters known as **domains,** which often give these proteins a bi- or multilobal appearance.

Topic 27 Quaternary Structure of Proteins

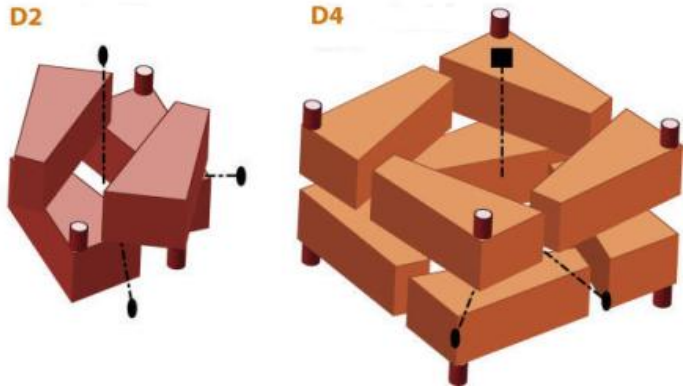
- Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different.
- The spatial arrangement of these subunits is known as a protein's quaternary structure.
- A multi-subunit protein is also referred to as a multimer.
- A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a protomer.



- Identical subunits of multimeric proteins are generally arranged in a symmetric patterns.
- Oligomers can have either rotational symmetry or helical symmetry.
- There are several forms of rotational symmetry. The simplest is cyclic symmetry, involving rotation about a single axis.

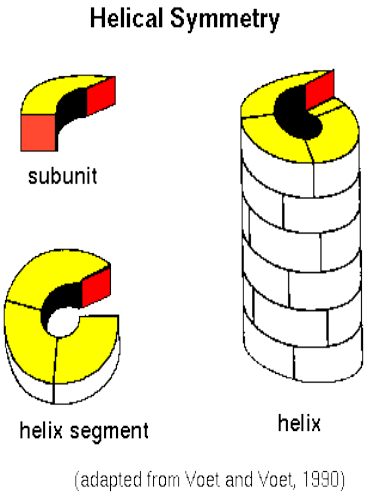


- A somewhat more complicated rotational symmetry is dihedral symmetry, in which a twofold rotational axis is present.



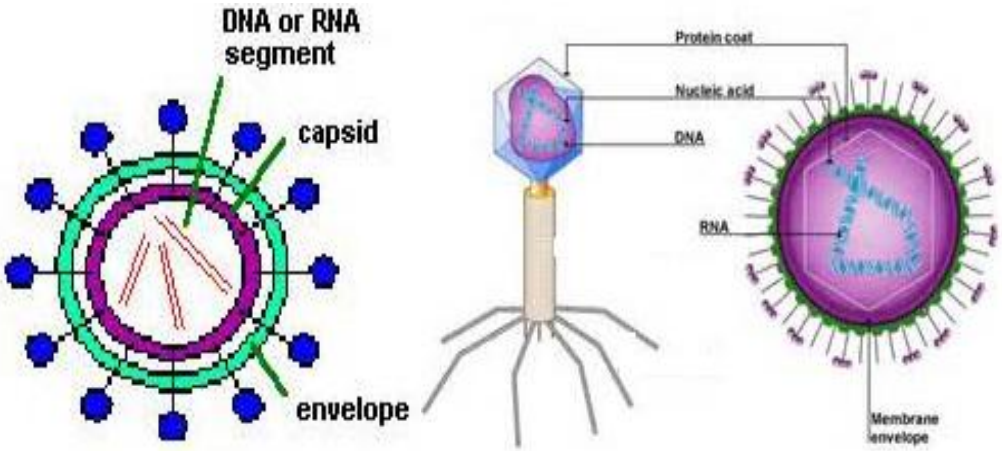
- More complex rotational symmetries include icosahedral symmetry.
- An icosahedron is a regular 12-cornered polyhedron having 20 triangular faces.

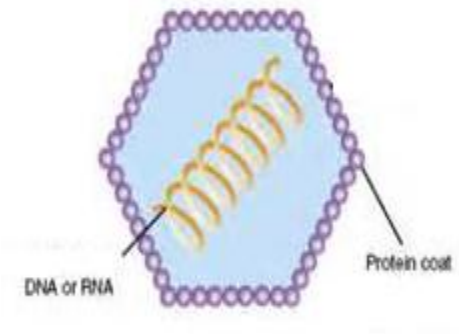
- The other major type of symmetry found in oligomers is helical symmetry.



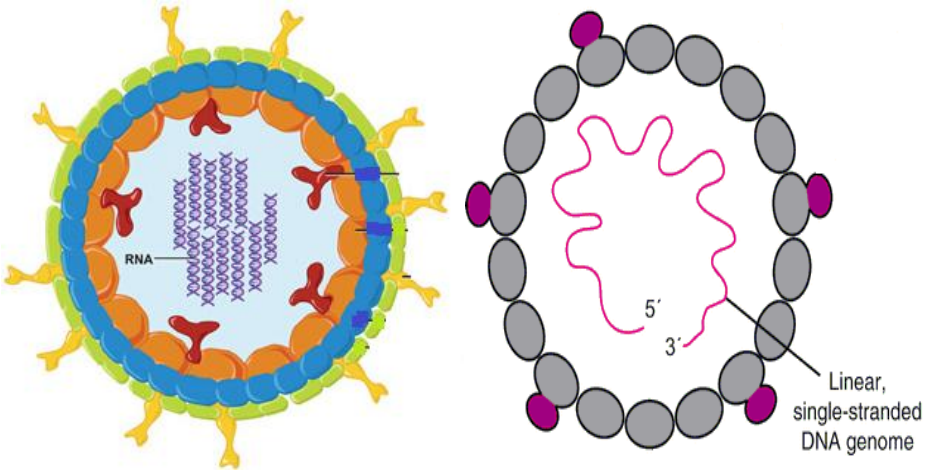
Topic 28 Genetic Materials in Viruses

- Viruses are exceptionally simple and extremely small microorganisms.
- They have a very simple structural organization consisting of a molecule of nucleic acid and a protein coat.

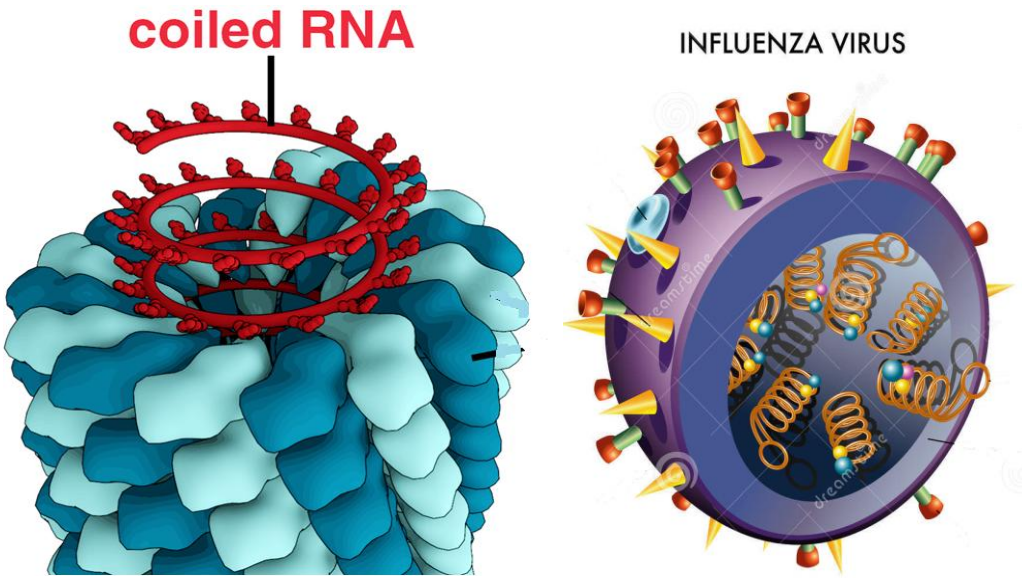




- Viruses can have either DNA or RNA as a genetic material but never both.
- The nucleic acid of a virus can be single-stranded or double-stranded.
- There are viruses with the familiar double-stranded DNA, with single-stranded DNA, with double-stranded RNA and with single-stranded RNA.



- The viral nucleic acid can be linear or circular.
- Some viruses may contain coiled RNA.
- In some viruses, the nucleic acid is in several separate segments.

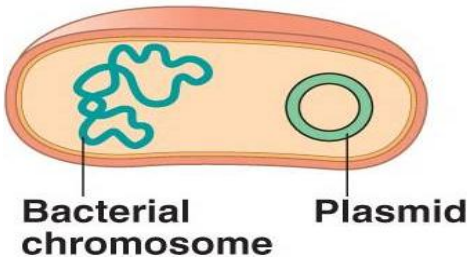


- The percentage of nucleic acid in relation to protein is about 1% for the influenza virus and about 50% for some bacteriophages.

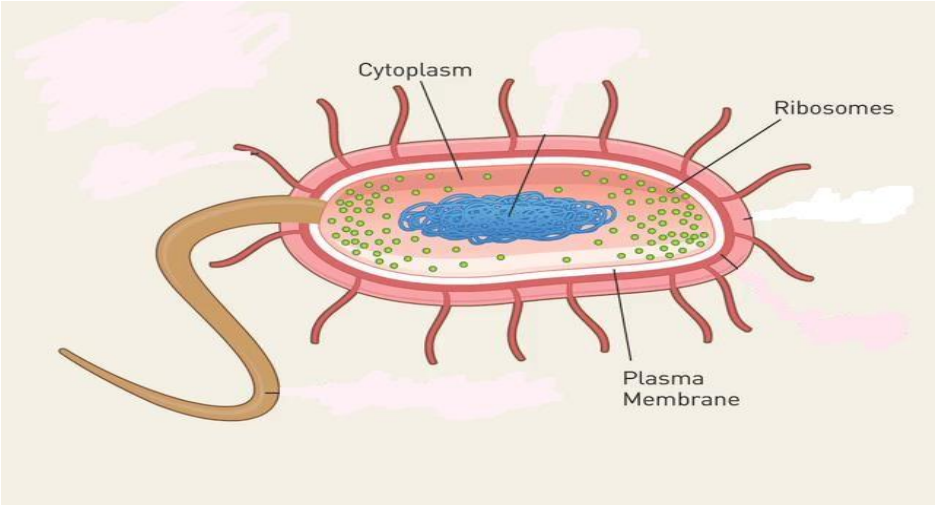
- The total amount of nucleic acid varies from a few thousand nucleotides to as many as 250,000 nucleotides.
- *E. coli's* chromosome consists of approx. 4 million nucleotide pairs.

Topic 29 and30 Organization of Genetic Material in Bacteria

- Bacteria typically have a single circular chromosome consisting of a single circular molecule of DNA with associated proteins.
- The bacterial chromosome is a very long (up to 1mm).
- It is looped and folded and attached at one or several points to the plasma membrane.

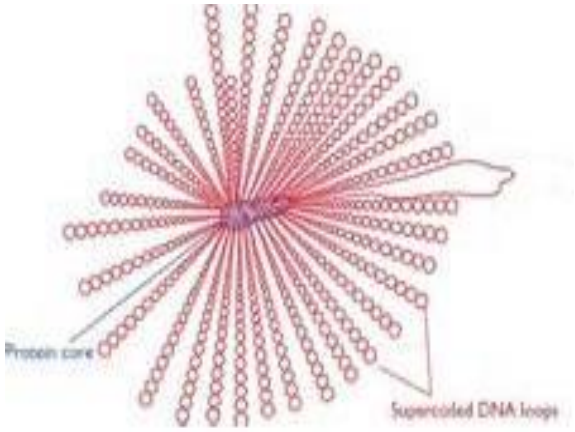


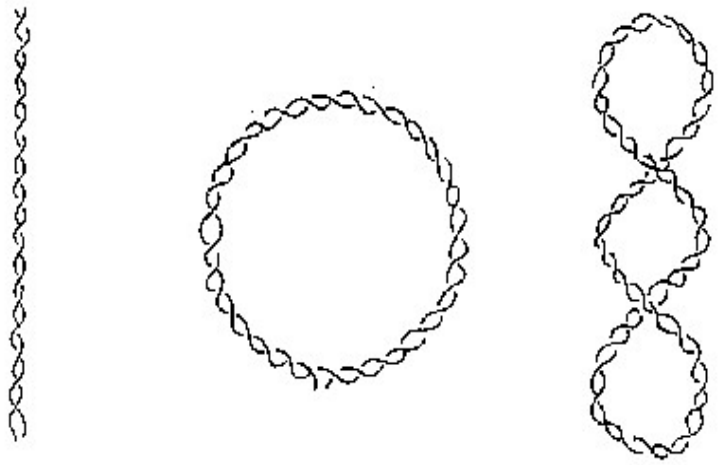
- Specific proteins interact with the bacterial DNA to form a highly condensed nucleoprotein complex called the **nucleoid.**



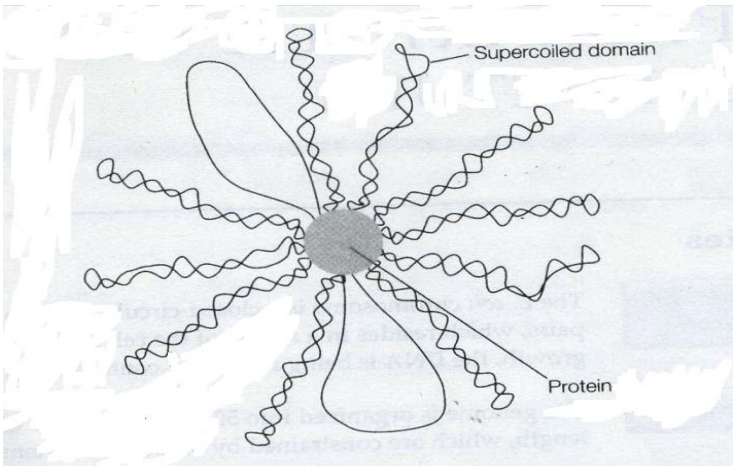
- Bacterial chromatin can be released from the cell by gentle lysis of the cell.
- Electron micrograph of the chromatin reveals that it consists of multiple loops which emerge from a central region of the chromatin.
- Some of the loops are super-coiled while some are relaxed.

- Relaxed loops are formed as a result of a nick introduced into super-coiled loops by a cellular DNase.





- If a super-coiled DNA molecule receives a nick, the strain of under-winding is immediately removed, and all the super-coiling is lost.
- Studies confirm that continued nuclease treatment increases number of relaxed loops.

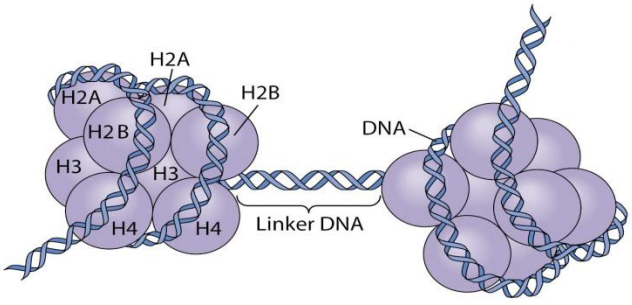


- The bacterial DNA is arranged in super-coiled loops that are fastened to a central protein matrix, so that each loop is topologically independent from all the others.
- So a nick that causes one super-coiled loop to relax would have no effect on other super-coiled loops.
- The super-coiled loops are dynamic structures which change during cell growth & division.
- An *E. coli* chromosome is estimated to have about 400 super-coiled loops.
- Each loop has an average length of about 10-20 kbp.
- The DNA compaction in a bacterial cell is contributed by super-coiling of loops, macromolecular crowding and DNA-binding proteins.

Topic 31 is not available yet.

Topic 32 Histone Proteins

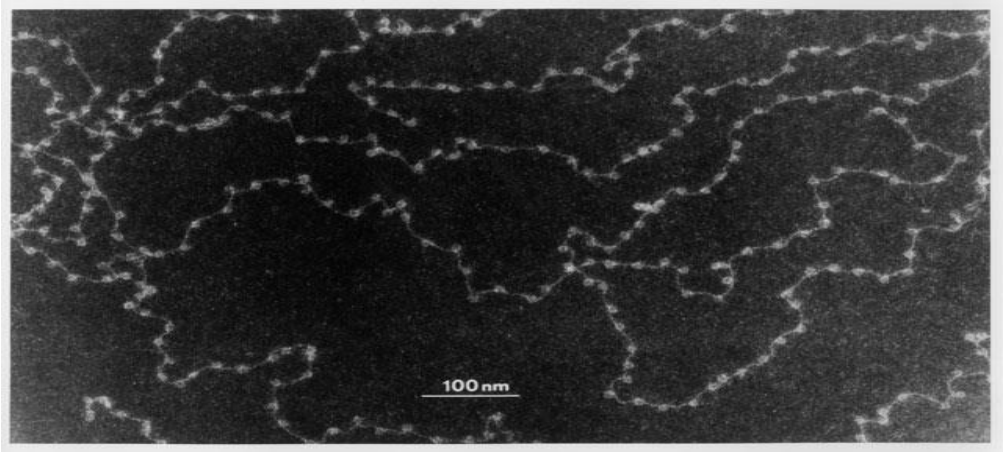
- Most abundant proteins in the chromatin are histones.
- There are nine types of histones including two each of H2A, H2B, H3 and H4 and one of H1.
- These histones fall in five major classes i.e., H1, H2A, H2B, H3 and H4.
- A typical human cell contains about 60 million copies of each kind of histone.



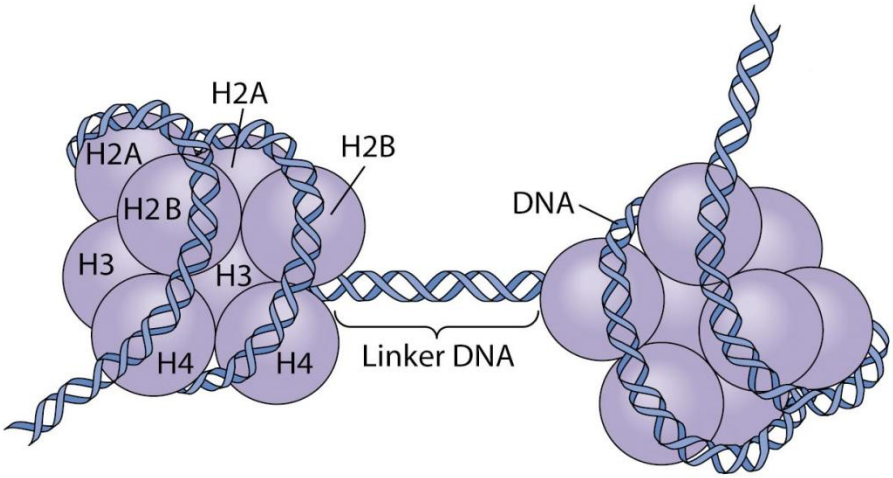
- All histones have a high percentage of arginine and lysine but the lysine-to-arginine ratio differs in each type of histone.
- The positively charged side chains of lysine and arginine enable histones to bind to the negatively charged phosphate groups of the DNA.
- The electrostatic attraction is an important stabilizing force in the chromatin.

Topic 33 : The Nucleosome

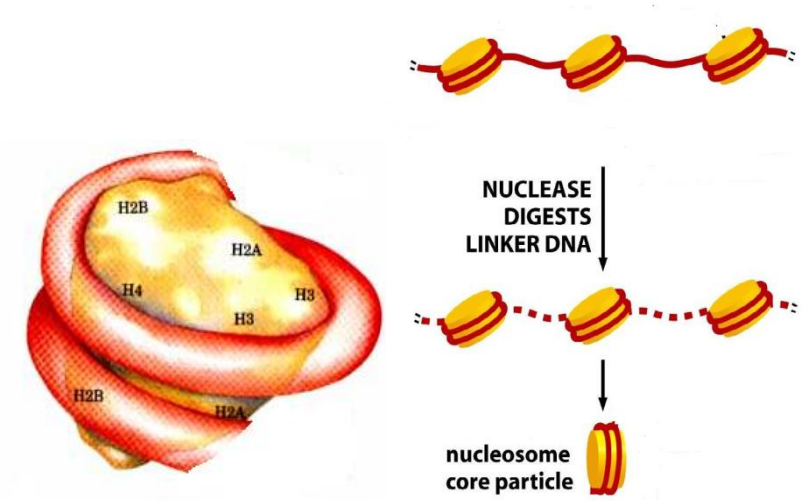
- The uncondensed chromatin resembles beads on a string when viewed under the electron microscope.
- Each bead is a nucleoprotein complex called nucleosome.



- Each nucleosome is formed by winding DNA fibre around a protein assembly consisting of eight histone molecules.
- The DNA connecting two nucleosomes is called **linker DNA.**

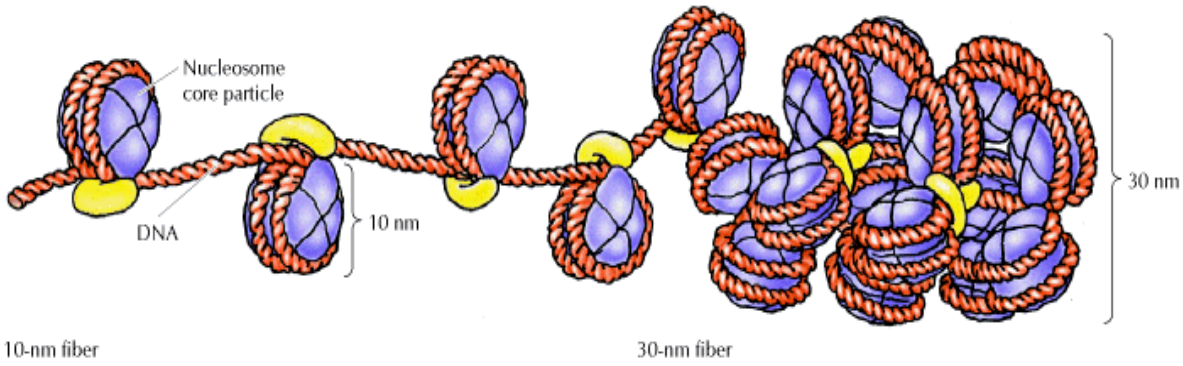


- The size of the linker DNA between the nucleosomes varies among different organisms and even different organs of the same organism.
- The length of DNA wrapped around nucleosomes also varies from one organism to the other ranging from about 170-240 bp.
- Prolonged nuclease digestion of chromatin cleaves additional nucleotides.
- The structure that remains is the nucleosome core particle.
- The nucleosome core particle consists of an octameric protein complex (two copies of each H2A, H2B, H3 & H4) with a 146 bp DNA fragment wound around it.

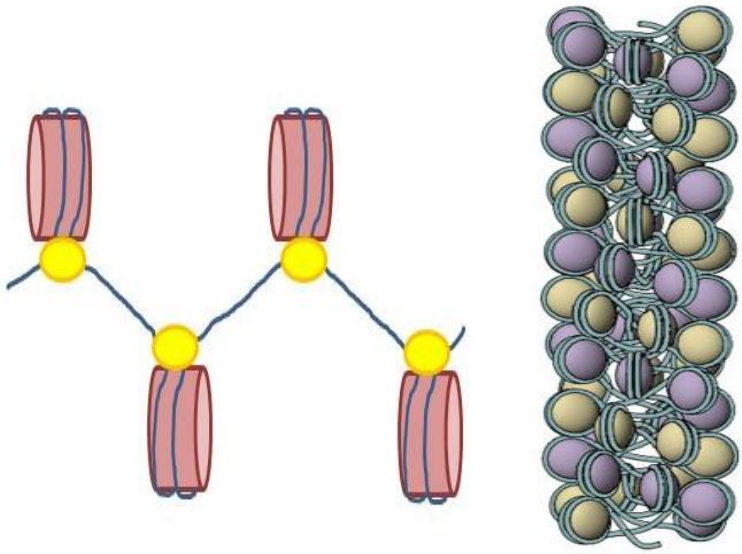


Topic 34 and 35 (The 30-nm Fiber)

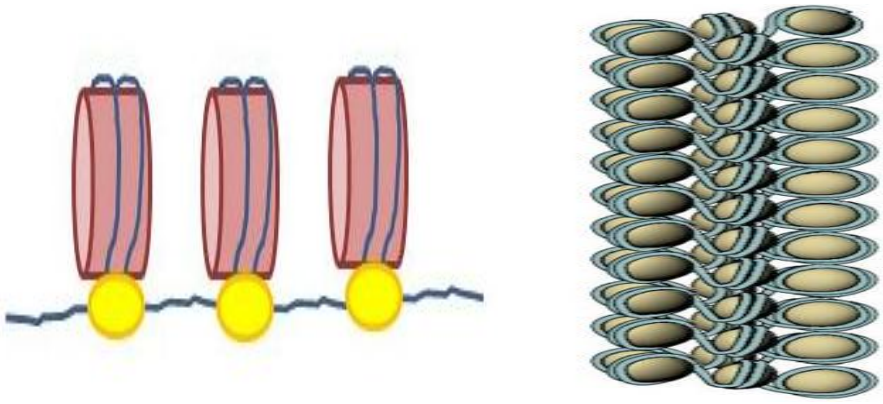
- It is still unclear that how a chain of nucleosomes folds into higher order structures.
- The next level of organization is a 30-nm fiber.



- Various models have been proposed to explain how nucleosomes fold to form the 30-nm fiber.
- However, two models gained the most support:-
    - Zigzag Model
    - Solenoid Model
- Zigzag model predicts that the linker DNA forms a straight path between successive nucleosomes.
- The nucleosomes lie on opposite sides of the fiber.



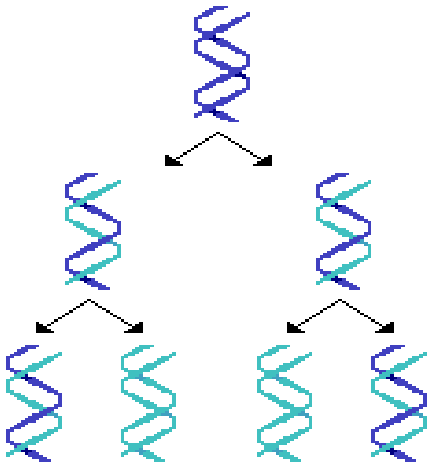
- The solenoid model predicts that the nucleosome chain forms a helical structure with about 6 nucleosomes per turn.
- Linker DNA is bent to connect neighbouring nucleosomes.



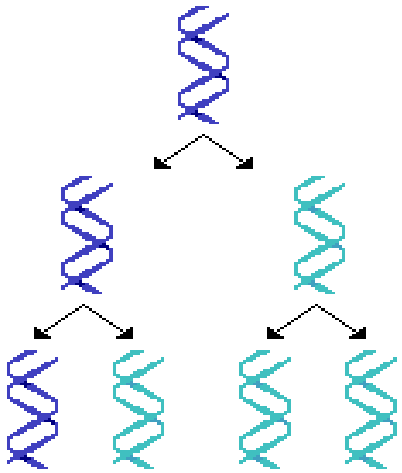
- Some of the reconstitution experiments appear to support Zigzag models while some others support Solenoid model.

Topic 36 Replication of DNA

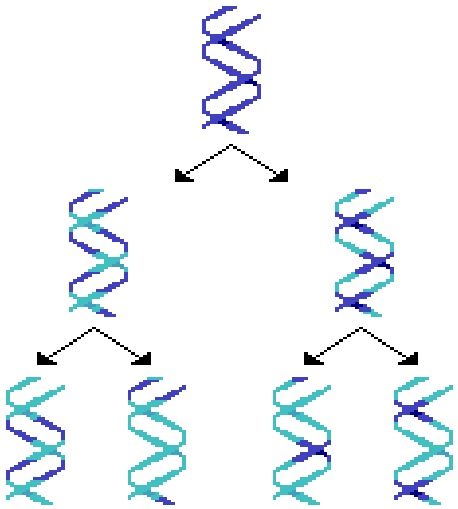
- The double-helical model for DNA includes the concept that the two strands are complementary.
- Thus, each strand can in principle serve as the template for making its own partner.
- A number of models were proposed to explain the mode of replication of DNA.
- But the semiconservative model for DNA replication is the correct one.
- The Watson–Crick model for DNA replication proposed that the two parental strands separate and that each then serves as a template for a new progeny strand.
- This is called semiconservative replication because each daughter duplex has one parental strand and one new strand which means that one of the parental strands is "conserved" in each daughter duplex.



- Another potential mechanism is **conservative replication,** in which the two parental strands stay together and somehow produce another daughter helix with two completely new strands.



- Yet another possibility is **dispersive replication,** in which the DNA becomes fragmented so that new and old DNAs coexist in the same strand after replication.

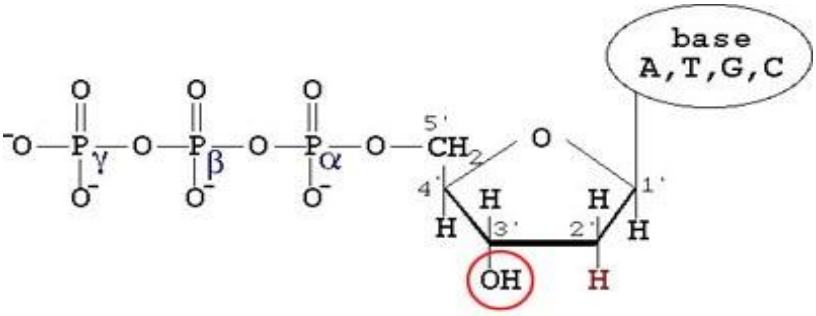


Topic 37 is not available yet.

Topic 38 and 39 Chemistry of DNA Synthesis

- Two key substrates are required for the synthesis of DNA to proceed.
    - Deoxynucleoside triphosphates
    - Primer:template junction
- Four deoxynucleoside triphospahtes namely dGTP, dCTP, dATP & dTTP are required.

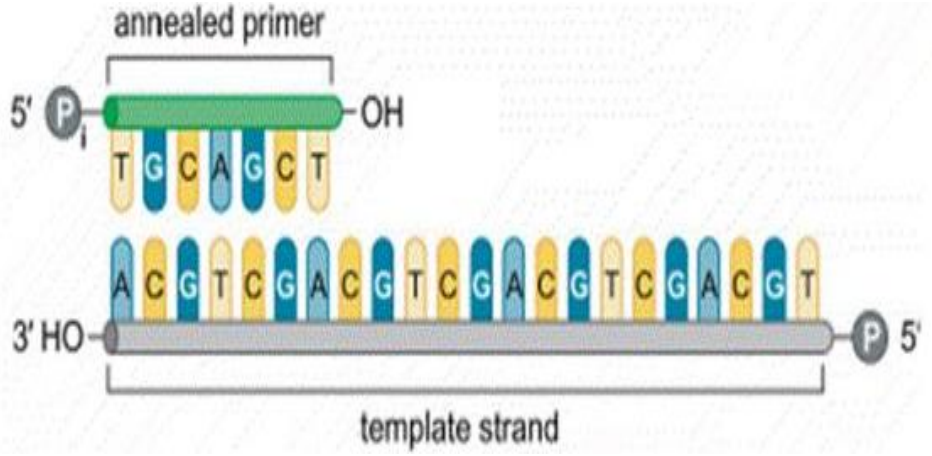
- Nucleoside triphosphates have three phosphoryl groups attached to the 5’ hydroxyl of deoxyribose.
- The innermost phosphoryl group is called the α-phosphate whereas the middle and outermost groups are called β- and ɣ- phosphates.



dNTP

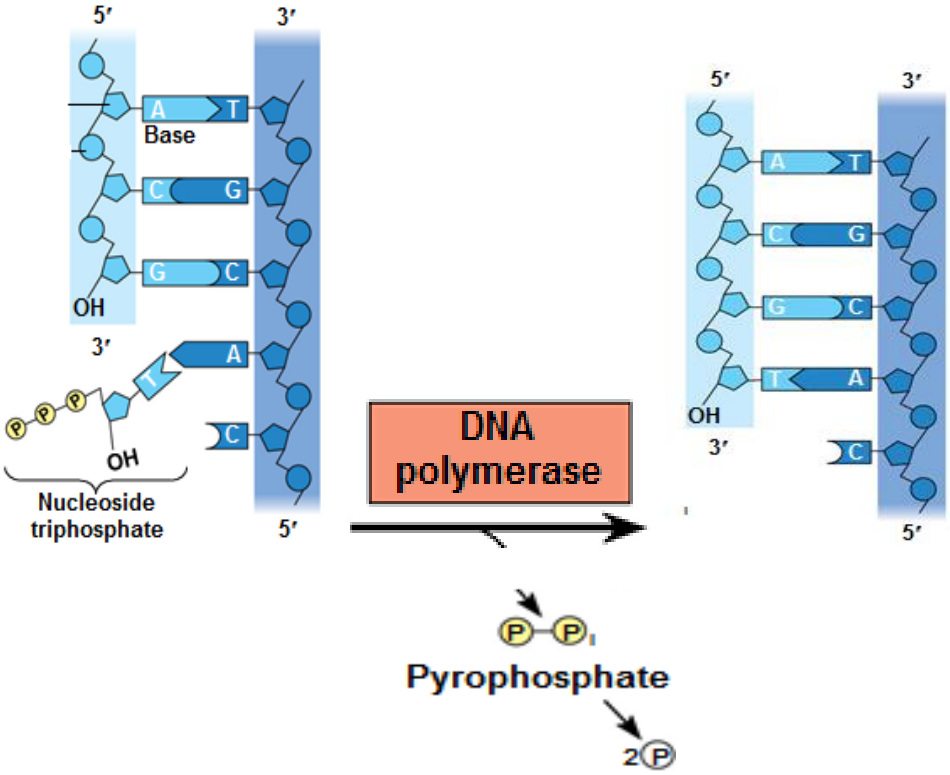
deoxyribonucleotide triphosphate

- The second important substrate for DNA synthesis is a particular arrangement of single stranded DNA (ssDNA) and double stranded DNA (dsDNA).
- This particular arrangement is called a primer:template junction.
- It has two components:-
    - The Template
    - The Primer



- The new chain of DNA grows by extending the 3’ end of the primer.
- The phosphodiester bond is formed in an S_N2 reaction.

- In this reaction, the hydroxyl group of the 3’ end of the primer attacks the α-phosphoryl group of the incoming nucleoside triphosphate.
- The leaving group of the reaction is pyrophosphate which arises from the release of β- and ɣ- phosphates of the nucleoside.

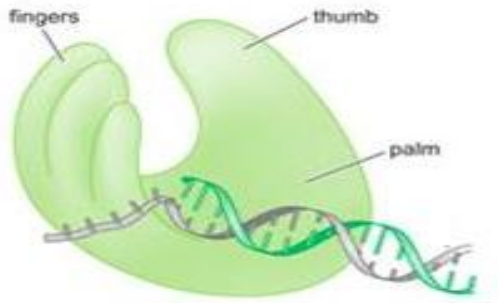


- The template strand directs which of the four nucleoside triphosphates is added.
- The incoming nucleoside triphosphate base pairs with the template strand.
- **What is the driving force for the addition of nucleotide to a growing polynucleotide chain?**
- The free energy for this reaction is provided by the rapid hydrolysis of the pyrophosphate into two phosphate groups by an enzyme known as pyrophosphatase.
- The net result of nucleotide addition and pyrophosphate hydrolysis is the simultaneous breaking of two high energy phosphate bonds.
- Therefore, DNA synthesis is a coupled process.
- This reaction is highly favourable with high value of K_{eq} which means that its an irreversible reaction.

Topic 40 is not available yet.

Topic 41 and 42 DNA Polymerases Resemble a Hand

- The structural studies on DNA polymerases reveal that the DNA substrate sits in a large cleft that resembles a partially closed right hand.
- Based on the hand analogy, the three domains of the DNA polymerase are called the thumb, fingers and palm.



- The palm domain is composed of a β-sheet and contains the primary elements of the catalytic site.
- This region of DNA polymerase binds two divalent metal ions (Mg^{2+} or Zn^{2+}).
- One metal ion reduces the affinity of the 3'-OH for its hydrogen.
- This generates a 3'-O_2 that is primed for the nucleophilic attack of the α-phosphate of the incoming dNTP.
- The second metal ion coordinates the negative charges of the β- and γ-phosphates of the dNTP and stabilizes the pyrophosphate produced by joining the primer and the incoming nucleotide.
- In addition to its role in catalysis, the palm domain also monitors the base pairing of the most recently added nucleotides.
- The fingers of the polymerase are also important for catalysis.
- Several residues located within the fingers bind to the incoming dNTP.
- More importantly, once a correct base pair is formed between the incoming dNTP and the template, the finger domain moves to enclose the dNTP.
- This closed form of the polymerase "hand" stimulates catalysis.

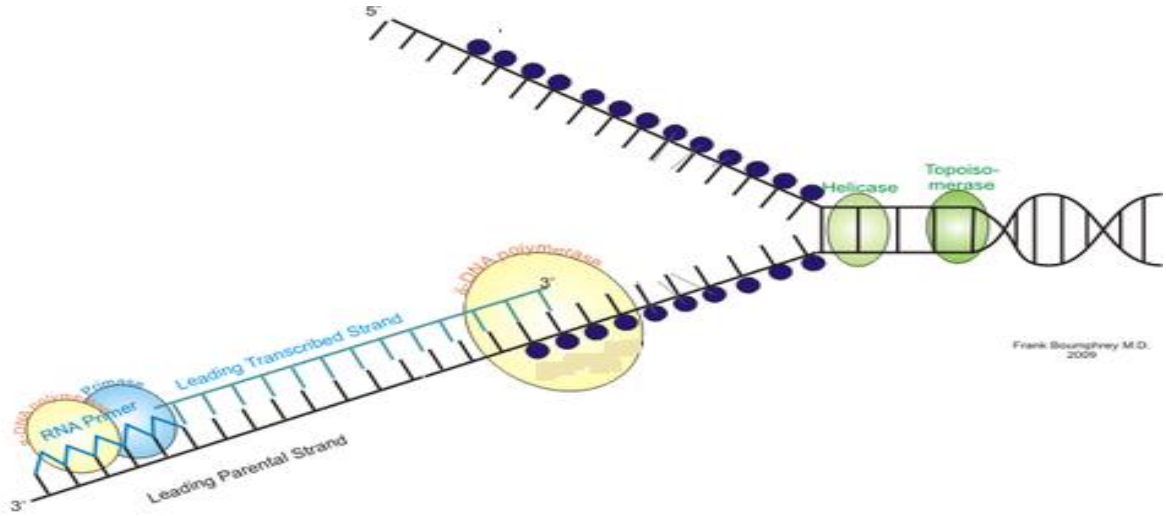
- In contrast to the fingers and the palm, the thumb domain is not intimately involved in catalysis.
- Instead, it interacts with the DNA that has been most recently synthesized.
- This serves two purposes:-
- First, it maintains the correct position of the primer and the active site.
- Second, the thumb helps to maintain a strong association between the DNA polymerase and its substrate.
- This association contributes to the ability of the DNA polymerase to add many dNTPs.

Topic 43 THE REPLICATION FORK

- In the cell, both strands of the DNA duplex are replicated at the same time.
- So it requires separation of the two strands of the double helix to create two template DNAs.
- The junction between the newly separated template strands and the unreplicated duplex DNA is known as the Replication Fork.

- The replication fork moves continuously towards the duplex region of unreplicated DNA.
- As the fork moves, it creates two DNA templates that each directs the synthesis of a complementary DNA strand.
- The antiparallel nature of DNA creates a complication for the simultaneous replication of the two exposed templates at the replication fork.
- Because DNA is synthesized only by elongating a 3’ end, only one of the two exposed templates can be replicated continuously as the replication fork moves.

- The newly synthesized DNA strand directed by this template is known as the leading strand.



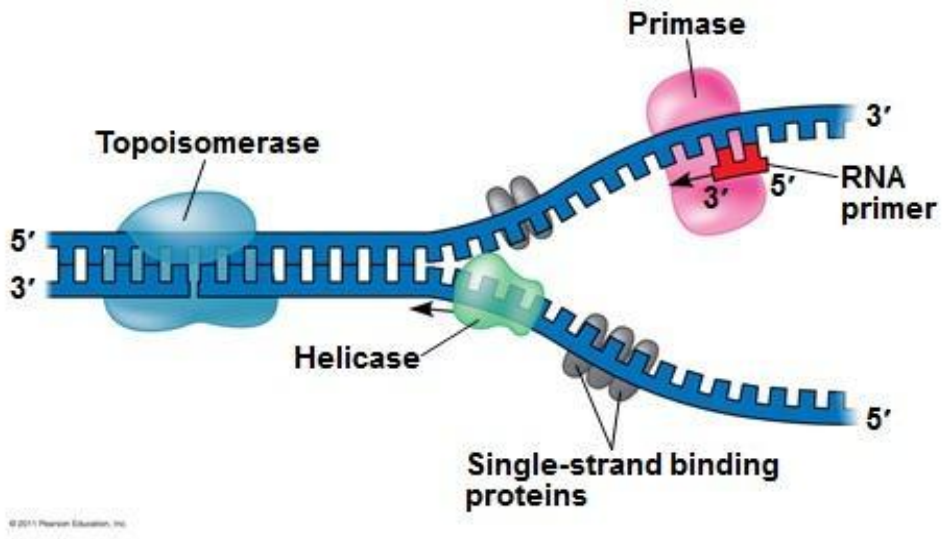
- Synthesis of the newDNA strand directed by the other ssDNA template is more complicated.
- This template directs the DNA polymerase to move in the opposite direction of the replication fork.
- The new DNA strand directed by this template is known as the lagging strand.
- This strand of DNA must be synthesized in a discontinuous fashion.
- Synthesis of the lagging strand must wait for movement of the replication fork to expose a substantial length of template before it can be replicated.
- Each time a substantial length of the template is exposed, DNA synthesis is initiated and continues until it reaches the 5’ end of the previous newly synthesized fragment of lagging strand DNA.
- The resulting short fragments of new DNA formed on the lagging strand are called Okazaki fragments.
- They vary in length from 1000 to 2000 nucleotides in bacteria and from 100 to 400 nucleotides in eukaryotes.
- Shortly after being synthesized, Okazaki fragments are covalently joined together to generate a continuous, intact strand of new DNA.

- Okazaki fragments are therefore transient intermediates in DNA replication.

Topic 44 THE RNA PRIMER

- All DNA polymerases require a primer with a free 3’-OH.
- They cannot initiate the synthesis of new DNA strand de novo.
- Then how?
- To accomplish this, the cell takes advantage of the ability of RNA polymerases to do what DNA polymerases cannot: start new RNA chains de novo.
- Primase is a specialized RNA polymerase dedicated to making short RNA primers (5–10 nucleotides long) on a ssDNA template.
- These primers are then extended by DNA polymerase.

Figure 16.13



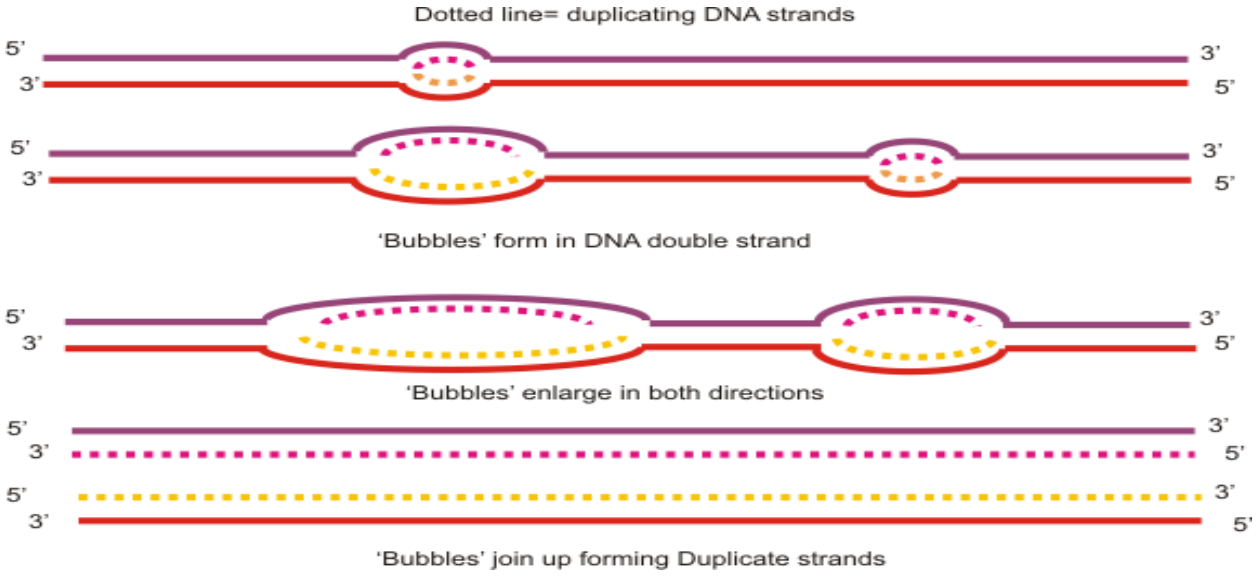
- Both the leading and lagging strands require primase to initiate DNA synthesis.
- Each leading strand requires only a single RNA primer.
- The discontinuous synthesis of the lagging strand means that new primers are needed for each Okazaki fragment.

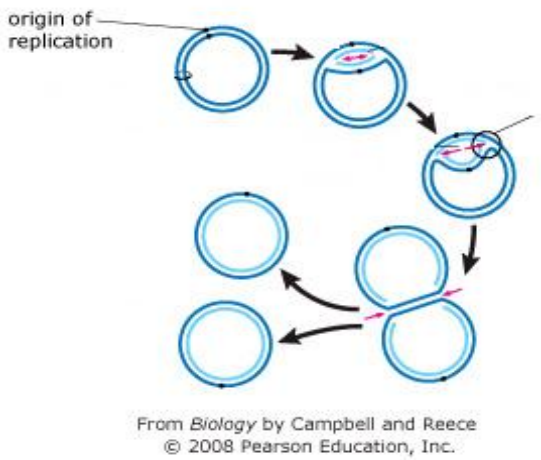
- Synthesis of the lagging strand can require hundreds of Okazaki fragments and their associated RNA primers.
- Primase activity is dramatically increased when it associates with another protein that acts at the replication fork called DNA Helicase.
- This protein unwinds the DNA at the replication fork, creating an ssDNA template that can be acted on by primase.

Topic 45 and 46 is not available yet.

Topic 47 INITIATION OF REPLICATION

- The initial formation of a replication fork requires the separation of the two strands of the DNA duplex to provide the ssDNA.
- ssDNA is required for DNA helicase binding and to act as a template for the synthesis of both the RNA primer and new DNA.
- Although DNA strand separation (DNA unwinding) is most easily accomplished at chromosome ends, but DNA synthesis generally initiates at internal regions.
- As the circular chromosomes lack the chromosome ends so it makes internal DNA unwinding essential for the replication initiation.
- The specific sites at which DNA unwinding and initiation of replication occur are called Origins of Replication.
- Depending on the organism, there may be as few as one or as many as thousands of origins per chromosome.





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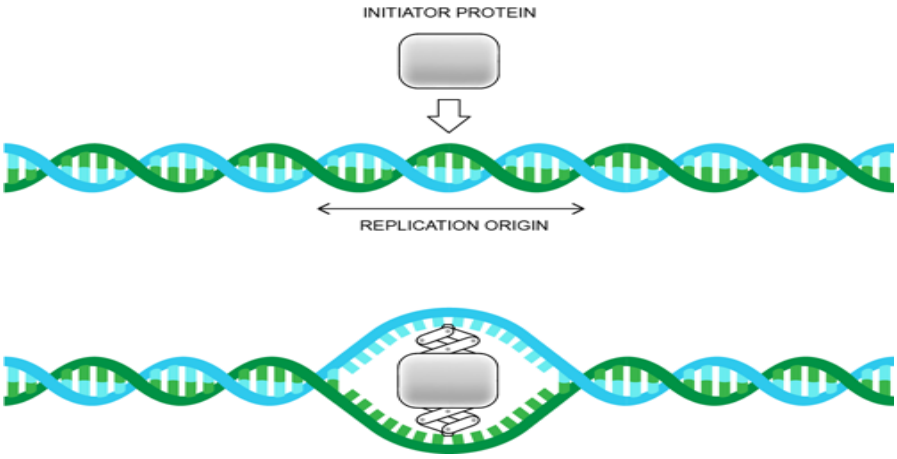
Topic 48 THE REPLICON MODEL

- How the initiation of replication takes place?
- It was explained by Francois Jacob, Sydney Brenner and Jacques Cuzin in 1963.
- This is called as The Replicon Model of replication Initiation.
- They defined all of the DNA replicated from a particular origin of replication as a replicon.
- As the single chromosome found in *E. coli* cells has only one origin of replication, the entire chromosome is a single replicon.
- In contrast, the presence of multiple origins of replication divides each eukaryotic chromosome into multiple replicons; one for each origin of replication.
- The replicon model proposed two components that controlled the initiation of replication; the replicator and the initiator.
- The replicator is defined as the cis-acting DNA sequences that are sufficient to direct the initiation of DNA replication.
- This is in contrast to the origin of replication, which is the physical site on the DNA where the DNA is unwound and DNA synthesis initiates.
- Although the origin of replication is always part of the replicator, sometimes the origin of replication is only a fraction of the DNA sequences required to direct the initiation of replication (the replicator).
- The second component of the replicon model is the initiator protein.

- This protein specifically recognizes a DNA element in the replicator and activates the initiation of replication.
- All initiator proteins select the sites that will become origins of replication.
- The initiator protein is the only sequence-specific DNA binding protein involved in the initiation of replication.

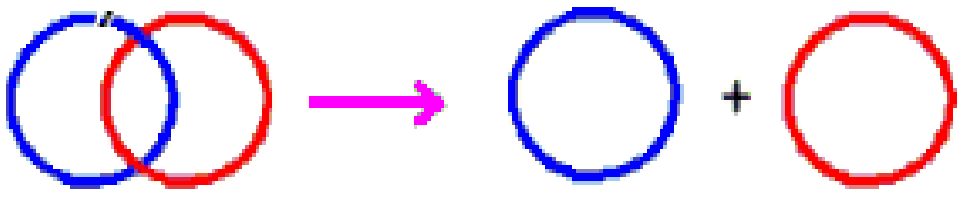


- All the remaining proteins other than initiator protein, required for replication initiation do not bind to a DNA sequence specifically.

Topic 49 is not available yet.

Topic 50 TYPE II TOPOISOMERASES

- Type II Topoisomerases are the enzymes which have the ability to break a dsDNA molecule and pass a second dsDNA molecule through this break.

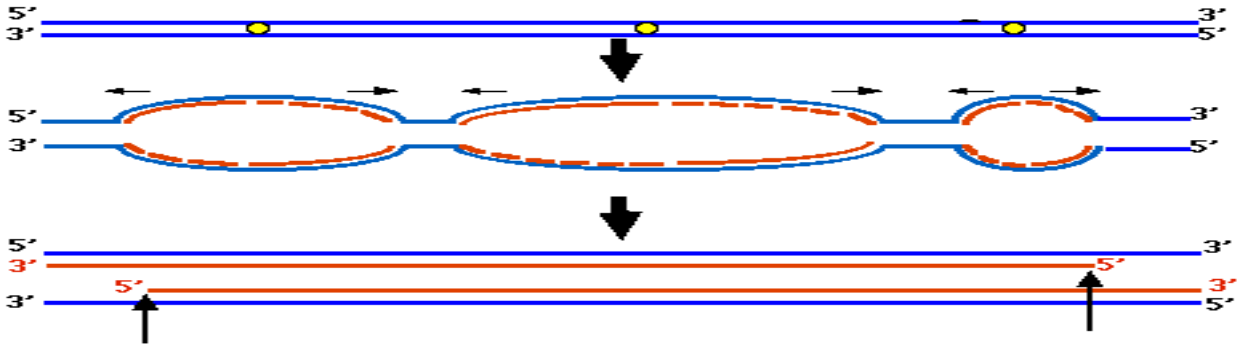


- So this reaction can easily decatenate the two circular daughter chromosomes and allow their segregation into separate cells.
- The activity of type II topoisomerases is also critical to the segregation of large linear molecules.

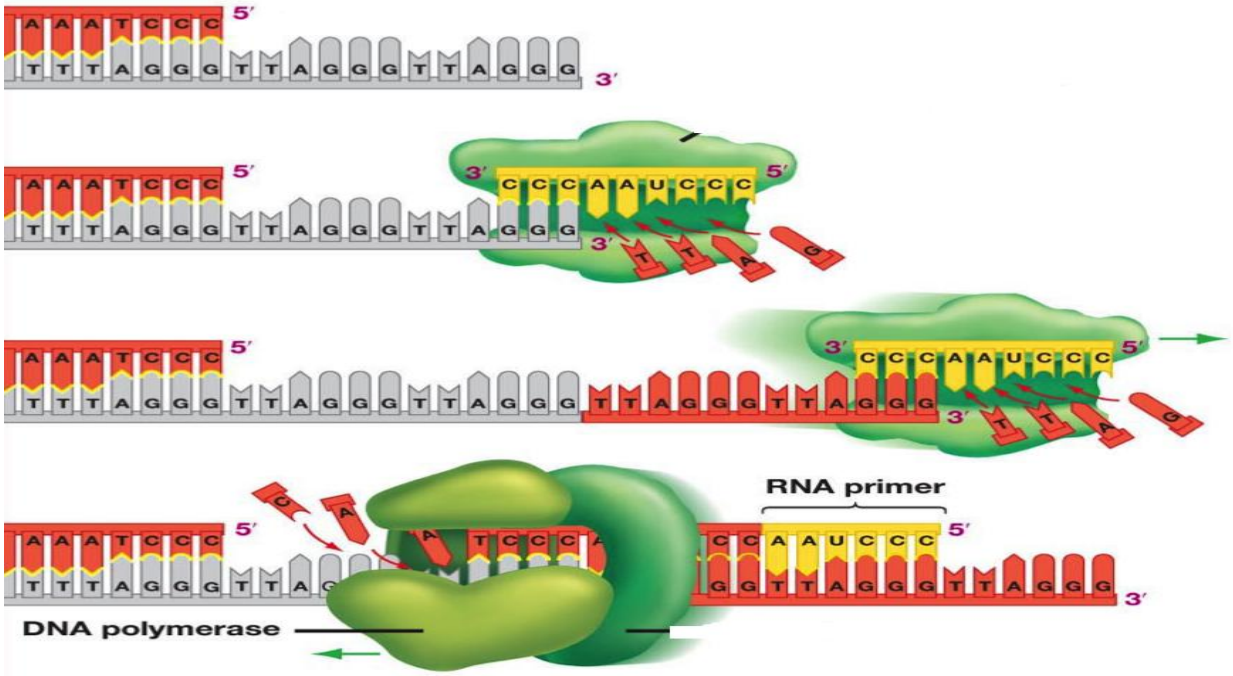
- Although there is no inherent topological linkage after the replication of a linear molecule, the large sized chromosomes necessitates the intricate folding of the DNA into loops which are attached to a protein scaffold.
- These attachments lead to many of the same problems that circular chromosomes have after replication.
- So type II topoisomerases allow these linked DNAs to be separated.
- So as in the case of circular chromosomes, type II topoisomerases also allow these linked DNAs to be separated.

Topic 51 TELOMERASE

- The requirement for an RNA primer to initiate all new DNA synthesis creates a dilemma for the replication of the ends of linear chromosomes.
- This is called as the end replication problem.
- This difficulty is not observed during the duplication of the leading-strand template.
- Because it requires only one RNA primer which completes the DNA synthesis up to extreme terminus of the strand.
- In contrast, the requirement for multiple primers to complete lagging-strand synthesis means that a complete copy of its template cannot be made.
- Even if the end of the last RNA primer for Okazaki fragment synthesis anneals to the final base of the lagging-strand template.
- Once this RNA molecule is removed, there will remain a short region (the size of the RNA primer) of un-replicated ssDNA at the end of the chromosome.
- Even if the end of the last RNA primer for Okazaki fragment synthesis anneals to the final base of the lagging-strand template.
- Once this RNA molecule is removed, there will remain a short region (the size of the RNA primer) of un-replicated ssDNA at the end of the chromosome.



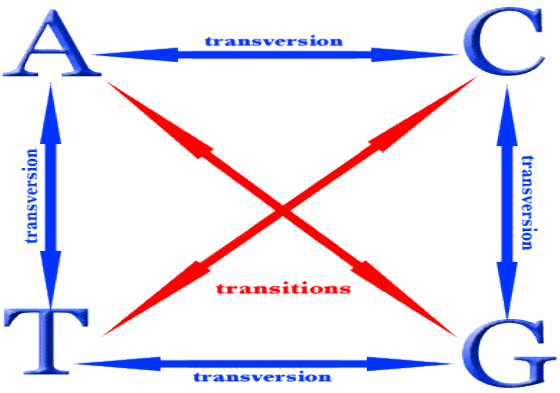
- Telomerase is a remarkable enzyme that includes multiple protein subunits and an RNA component.
- Like all other DNA polymerases, telomerase acts to extend the 3' end of its DNA substrate.
- But unlike most DNA polymerases, telomerase does not need an exogenous DNA template to direct the addition of new dNTPs.
- Instead, the RNA component of telomerase serves as the template for adding the telomeric sequence to the 3' terminus at the end of the chromosome
- Telomerase specifically elongates the 3'-OH of telomeric ssDNA sequences using its own RNA as a template.
- As a result, the newly synthesized DNA is single-stranded.
- So when telomerase acts on the 3' end of the telomere, it extends only one of the two strands of the chromosome.
- How is then 5' end extended?
- This is accomplished by the lagging-strand DNA replication machinery.
- By providing an extended 3' end, telomerase provides additional template for the lagging-strand replication machinery.
- By synthesizing and extending RNA primers using the telomerase extended 3' end as a template, the cell can effectively increase the length of the 5' end of the chromosome as well.


5′
3′
RNA primer
DNA polymerase

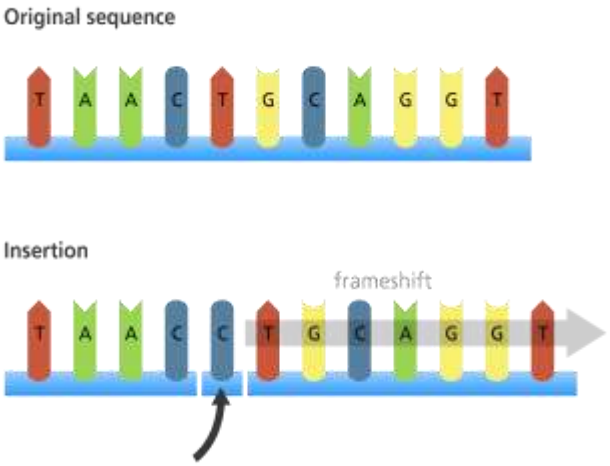
- **Topic 52 DNA MUTATIONS**
- DNA mutation can be defined as a permanent transmissible change in the genetic material (DNA/RNA).
- In other words, it is a permanent change in the nucleotide sequence of the genome of an organism.
- DNA can be easily damaged even under normal physiological conditions.
- Many different kinds of chemical and physical agents can damage DNA.
- Some of these agents are endogenous which are produced inside the cells as a result of normal metabolic pathways.
- While some others are exogenous agents which come from the surrounding environment.
- On one hand, DNA stability is required to ensure that the genetic information may pass accurately from one generation to the next
- It is also required for the correct functioning of thousands of genes.
- On the other hand the genetic variation is needed to drive evolution.
- If this variation would be lacking, the new species, including humans, would have not arisen.
- So the life and biodiversity depend on a happy balance between DNA damage (mutation) and its repair.

Topic 53 NATURE OF MUTATIONS

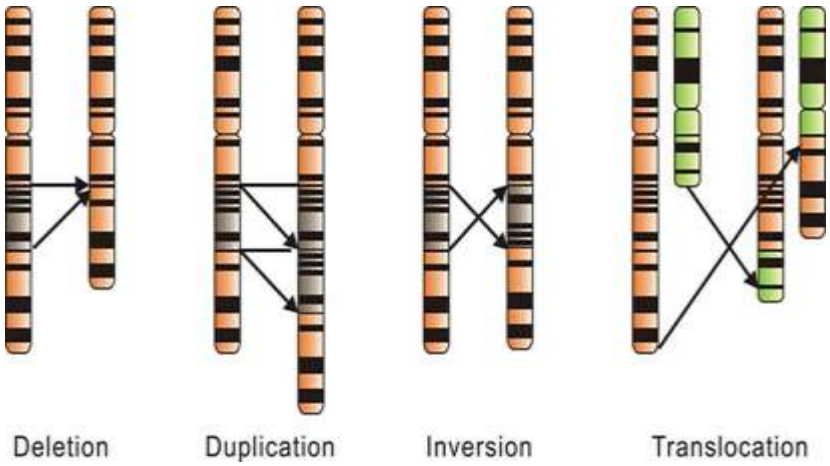
- DNA mutations may be very simple (single base change) or very complex and may include several thousands of nucleotides.
- The simplest mutations are switches of one base for another. There are two kinds of such mutations which include:-
    - Transitions
    - Transversions
- Transitions are pyrimidine-to-pyrimidine and purine-to-purine substitutions, such as thymine (T) to cytosine (C) and adenine (A) to guanine (G).
- Transversions are pyrimidine-to-purine and purine-to-pyrimidine substitutions, such as T to G or A and A to C or T.



- Other simple mutations are insertions or deletions of a nucleotide or a small number of nucleotides.



- All such mutations that alter a single nucleotide are called point mutations.
- Other kinds of mutations cause more drastic changes in DNA, such as extensive insertions and deletions and gross rearrangements of chromosome structure.
- Such changes might be caused, for example, by the insertion of a transposon, which typically places many thousands of nucleotides of foreign DNA in the coding or regulatory sequences of a gene.
- Another type of mutations which are more drastic occur at chromosomal levels.
- These are changes in appearance of the individual chromosomes through mutation-induced rearrangements.



Topic 54 REPLICATION ERRORS

- The replication machinery achieves a remarkably high degree of accuracy using a proofreading mechanism, which removes wrongly incorporated nucleotides.
- However, this proofreading is not foolproof.
- Some mis-incorporated nucleotides escape detection and become a mismatch between the newly synthesized strand and the template strand.
- If the mis incorporated nucleotide is not subsequently detected and replaced, the sequence change will become permanent in the genome.
- During a second round of replication, the mis-incorporated nucleotide will direct the incorporation of its complementary nucleotide into the newly synthesized strand.
- At this point, the mismatch will no longer exist; instead, it will have resulted in a permanent change (a mutation) in the DNA sequence.

Topic 55 RADIATION DAMAGE

- Cells are exposed to three types of high energy electromagnetic radiations which can damage their DNA.
- These radiations include:-

  Ultraviolet light (wavelength 100-400nm)

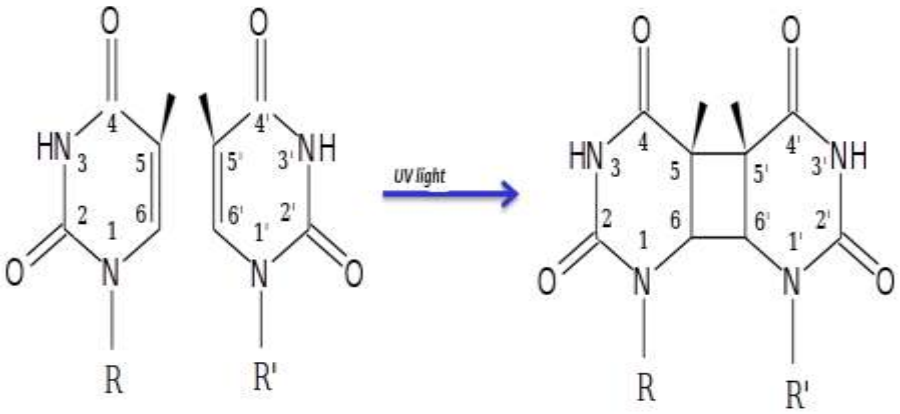
  X-rays (wavelength 0.01-100nm)

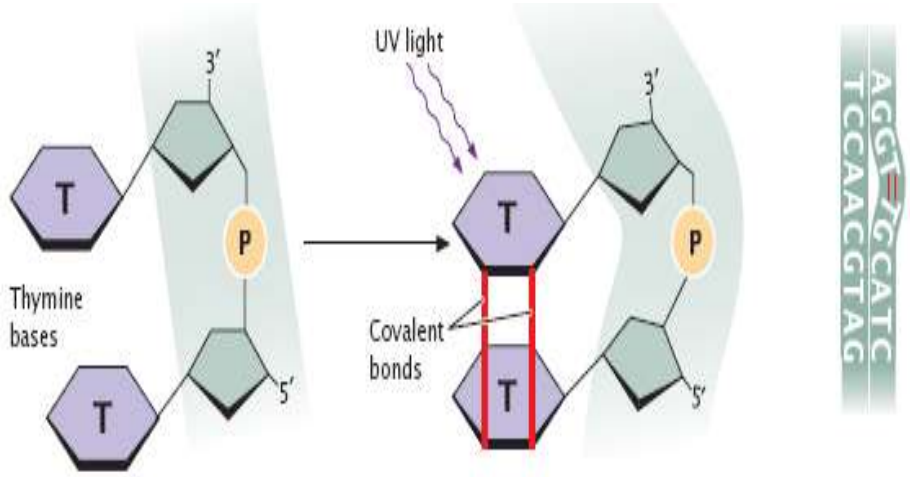
  Gamma rays (wavelength <0.01nm)
- Later two are ionizing radiations.

- Ultraviolet light is divided into three bands:
    - UV –A (321-400nm)
    - UV –B (296-320nm)
    - UV –C (100-295nm)
- The majority of UV light reaching on earth is UV – A.
- This is least energetic band and so does little damage to DNA.
- UV – B accounts for about 10% of the UV radiation reaching the earth's surface.
- It is responsible for most of the DNA damage in the skin.
- UV – C includes the wavelength of maximum DNA absorbance (260nm).
- So it would cause a great deal of DNA damage to exposed organisms if it were able to penetrate the earth's surface.
- Fortunately, very little UV - C reaches the earth's surface because the ozone layer prevents it from penetration.
- However, during lab studies, the germicidal lamps that produce UV - C light are used.

Topic 56 Cyclobutane pyrimidine Dimer

- Two major photoproducts account for nearly all of the UV - induced DNA damage.
- They involve the dimer formation between adjacent pyrimidine bases on the same DNA strand.
- The first pathway includes the formation of cyclobutane pyrimidine dimer (CPD) which accounts for about 75% of all the UV - induced damage.
- The cyclobutane ring is generated by forming one bond between C-5 atoms and another between C-6 atoms on adjacent pyrimidine rings.





- **Cyclobutane pyrimidine Dimer**
- The most common cyclobutane pyrimidine dimer is the thymine-thymine (T< >T) dimer.
- Cytosine-thymine (C< >T) and cytosine- cytosine (C< >C) dimers are also form but at slower rates.
- Structural studies show that:-

  1) B-DNA can accommodate a single T< >T dimer forcing the helical axis to bend by about 30˚ towards the major groove.

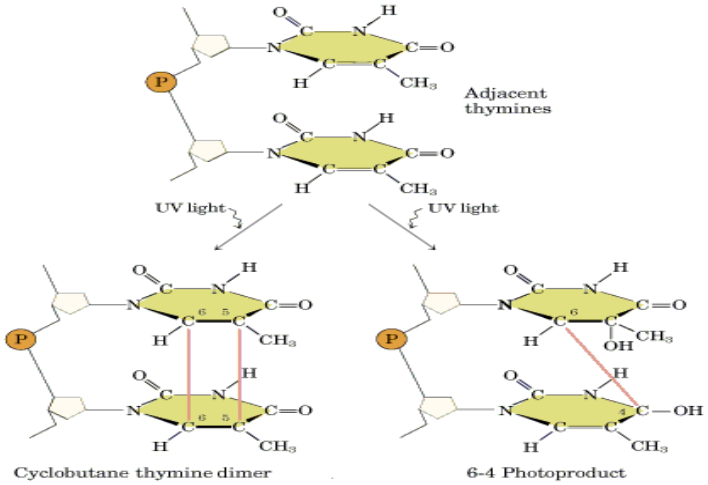
  2) The dimer's 3'-thymine can form a normal base pairing with its adenine partner on the complementary strand.

  3) The interaction between the 5'-thymine and its complementary adenine partner will be weaker than normal because a single hydrogen bond can be formed here.

- Thymine-thymine dimers are often used to study DNA repair systems because they are stable, easy to form and easy to detect.

Topic 57 (6-4) Photoproducts

- The second pathway, which accounts for most of the remaining UV-induced DNA damage, produces (6-4) photoproducts.
- A bond is formed between the C-6 atom of the 3'-pyrimidine (either thymine or cytosine) and the C-4 atom of the 5'-pyrimidine (usually cytosine).



- The (6-4) photoproduct causes a major distortion in B-DNA because the two pyrimidine rings are perpendicular to each other.
- If not removed, a pyrimidine dimer or (6-4) photoproduct can interfere with the normal operation of the replication and transcription machinery.
- This interference results in mutations and cell death.
- Even if the lesion is removed, the result can be a mutation.

Topic 58 X-rays & gamma rays damage

- Gamma radiations and X-rays (ionizing radiation) are particularly hazardous because they cause double-strand breaks in the DNA, which are difficult to repair.
- Ionizing radiations directly or indirectly generate many different kinds of DNA lesions.
- Direct damage takes place when DNA or water bound to it absorbs the radiation.
- Indirect damage takes place when water molecule or any other molecule surrounding the DNA absorb the radiation and form reactive species that then damage the DNA.
- The DNA lesions caused due to these radiations may be isolated or clustered.
- Many lesions within a few helical turns are called clustered lesions.

- One type of clustered lesion, the double-strand break, is generally thought to be the primary reason that ionization radiation is so lethal to cells.
- Double-strand breaks are also responsible for various chromosomal aberrations such as deletions, duplications, inversions and translocations.
- About 65% of the DNA damage caused by these radiations is due to indirect effects, primarily due to transfer of photons to water.
- The photon transfer activates the water and causing it to undergo two types of primary reactions.
- In the first of these reactions, the water molecule is ionized.

$H2O \rightarrow H2O^{\bullet+} + e^-$

- The $H2O^{\bullet+}$ that is formed readily dissociates and release a proton and a hydroxyl radical ($^{\bullet}OH$).

$H2O^{\bullet+} \rightarrow H^+ + {}^{\bullet}OH$

- The electron generated by the first reaction can combine with molecular oxygen to form a super-oxide radical ($^{\bullet}O2^-$).

  $e^- + O2 \rightarrow {}^{\bullet}O2^-$

- In the second type of primary reaction, excited water molecule (H2O*) splits into a hydrogen atom and a hydroxyl radical.

  $H2O \rightarrow H^{\bullet} + {}^{\bullet}OH$

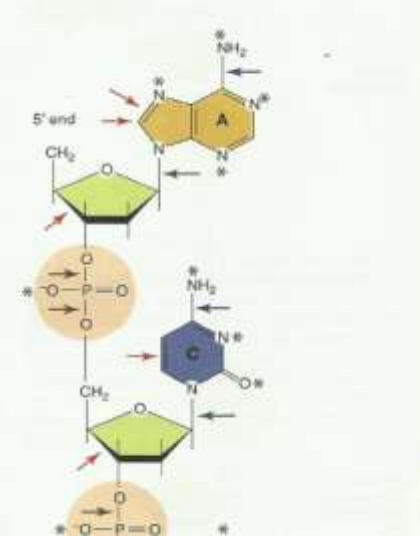
- So the three highly reactive chemical species i,e., $^{\bullet}H$, $^{\bullet}OH$ and $^{\bullet}O2^-$ are produced by the two primary pathways.
- Each of these attacks and damages whatever biomolecule they encounter.
- A wide variety of changes take place when that molecule happens to be DNA.

Topic 59 DNA INSTABILITY IN WATER

- DNA has three kinds of bonds with the potential for hydrolytic cleavage, namely:
  - Phosphodiester bond

- N-glycosyl bond
- Bonds linking exocyclic amine groups to bases

- **Hydrolytic Bond Cleavage**

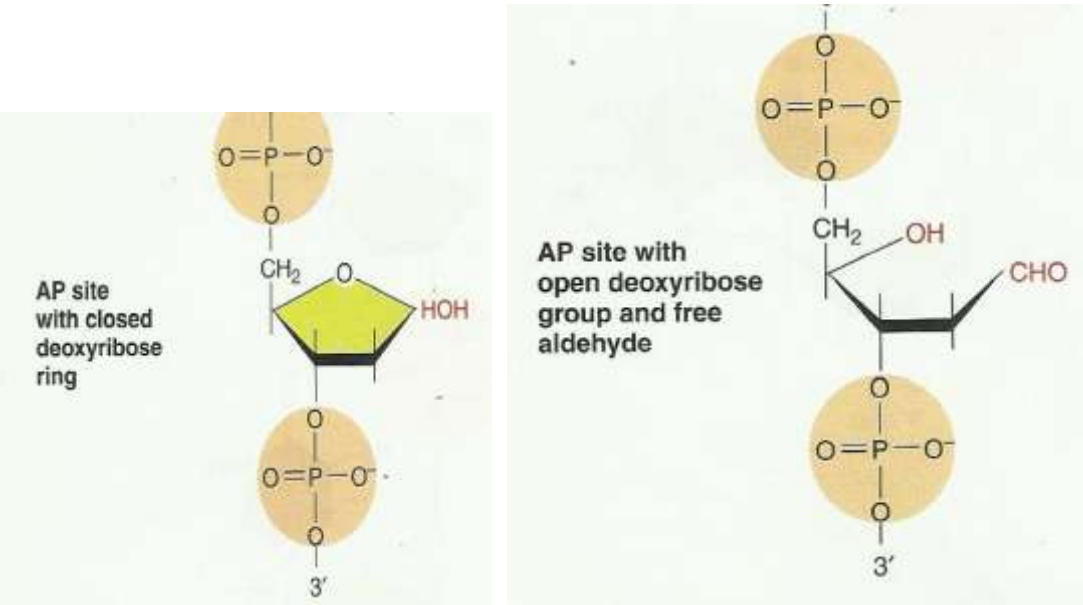


Phosphodiester Bond Cleavage

- Spontaneous phosphodiester bond cleavage, which introduces a nick in the DNA strand, is a very rare occurrence and doesn't make a significant contribution to DNA damage.

N-glycosyl Bond Cleavage

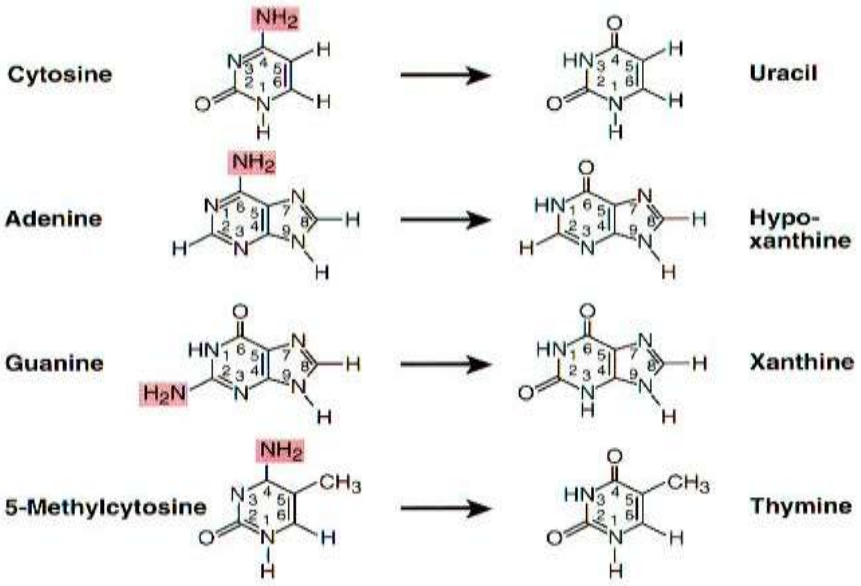
- N-glycosyl bond cleavage leads to the formation of an abasic site, which is also known as an AP site (AP for apurinic & apyrimidinic).



- According to current etimates, about 10,000 purine and 500 pyrimidine bases are lost from DNA in a mammalian cell nucleus each day.
- Experiments are also showing that purine N-glycosyl bonds are more easily hydrolyzed than pyrimidine N-glycosyl bonds.
- AP site formation sensitizes the neighbouring 3'-phosphodiester bond to cleavage which can be attributed to the formation of a free aldehyde group.
- A DNA strand with one or more AP sites makes a poor template because it lacks the information required to direct accurate replication and transcription.

Topic 60 and 61 Water-mediated Deamination

- Water-mediated deamination converts cytosine, guanine and adenine to uracil, xanthine and hypoxanthine, respectively.
- Hydrolytic deamination of cytosine is estimated to take place about 100 to 500 times a day in a mammalian cell.
- Whereas, combined guanine and adenine deaminations are estimated to occur at about 1 or 2% of that of cytosine deamination.



- A few cytosine bases in eukaryotic DNA are converted into modified base 5-methylcytosine.
- This modified base is concentrated in so called CpG islands.
- CpG islands are small segments of DNA often present in regulatory elements called promoters that are located just before the transcription unit they regulate.
- The frequency of spontaneous deamination of 5-methylcytosine bases in CpG islands is even greater than those of cytosine.
- The product of this deamination is thymine and not uracil.
- So it results in the conversion of a C – G base pair to a T – A base pair.
- Nitrous acid (HNO2) is formed from nitrites which are used as preservatives in processed meat.
- It reacts with amine groups attached to the ring structure in C, A and G and greatly increase their rate of deamination.
- Bisulfite ($HSO3^-$) used as an additive in fruit juices and dry fruits also greatly increases the rate of cytosine deamination but does'nt affect purine or 5-methylcytosine deamination.

Topic 62 and 63 Oxidative Damage to DNA

- The reactive oxygen species damage DNA.

- So the reactive oxygen species produced during cellular respiration (ETC) may have the potential to damage DNA.
- But it is unlikely that they do so because:-

  1) the respiratory chain doesn't normally release these reactive oxygen species.

  2) cells contain superoxide dismutase to convert superoxide radicals into molecular oxygen and hydrogen peroxide. Then catalase to convert hydrogen peroxide to oxygen and water.

  3) superoxides and hydroxyl radicals are so reactive that, if released from respiratory chain, they would react with the nearby biomolecules before they had chance to reach nuclear DNA.

- So the reactive oxygen species produced in this way do not damage the nuclear DNA under normal physiological conditions.
- The primary culprit appears to be the hydroxyl radical, which is produced by ionizing radiations.
- Hydroxyl radicals can also be produced chemically from hydrogen peroxide.
- Hydrogen peroxide is not as reactive as superoxide and hydroxyl radicals, so it has a much longer half-life in the cell, provided it escapes catalase and peroxidase.
- If it does escape, hydrogen peroxide can be converted into hydroxide radical by the following reaction:-

Fe^{2+} + H2O2

Fe^{3+} + •OH + OH^-

- This reaction is called the Fenton reaction and in this reaction copper, manganese and cobalt can replace iron.
- Hydroxyl radicals generated by any means are known to cause more than 80 different kinds of base damage.
- Two of the oxidized base products are 8-oxoguanine (oxoG) and thymine glycol.
- 8-Oxoguanine can base pair with adenine or cytosine and if uncorrected, this 8-oxoG – A base pair will be replicated to form a T – A base pair, thus causing a transversion mutation.
- On the other hand, thymine glycol inhibits DNA replication and is therefore cytotoxic.

- Hydroxyl radicals produced by the Fenton reaction are tend to be more widely dispersed than those produced by ionizing radiations and therefore, much less likely to produce double-stranded breaks.
- Cells can repair single-stranded breaks much more easily than they can repair double-stranded breaks.
- Reactive oxygen species can also convert various biomolecules into reactive species that can then damage DNA.
- For example, polyunsaturated fatty acid oxidation produces two aldehyde products, malondialdehyde and 4- hydroxynonenal, which contribute to base damage.

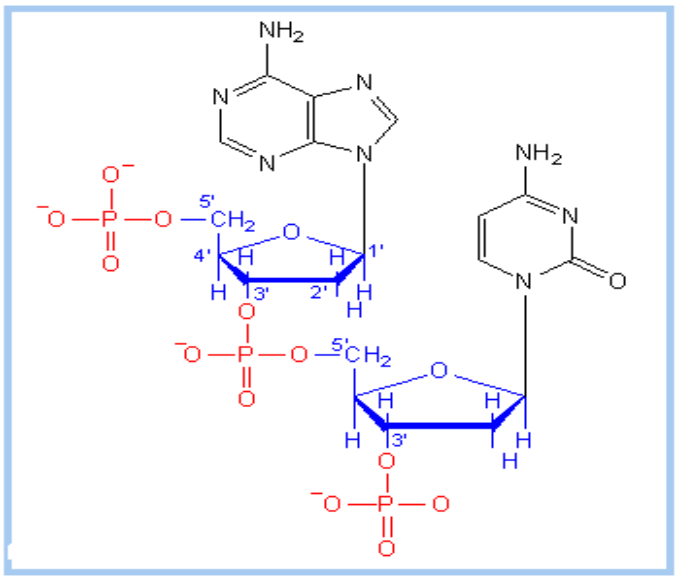
Topic 64 and 65 Alkylation Damage to DNA

- DNA has electron-rich atoms that are readily attacked by electron-seeking chemicals called electrophiles or electrophilic agents.
- Alkylating agents are a highly reactive group of electrophiles which transfer methyl, ethyl, or alkyl groups to the electron-rich atoms in the DNA and damage it.
- Alkylation in DNA takes place at:-

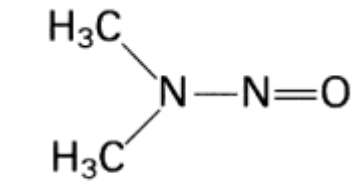
  A) nitrogen and oxygen atoms external to the base ring systems;

  B) non-bridging oxygen atoms in phosphate groups; and

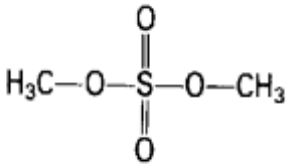
  C) nitrogen atoms in the base ring systems except those linked to deoxyribose.



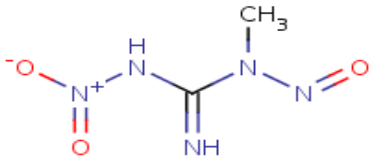
- Many different kinds of naturally occurring and synthetic chemical agents are known to transfer alkylating agents to DNA.

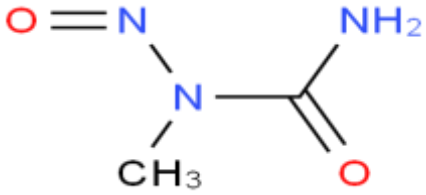


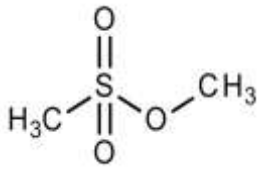
Dimethylnitrosamine



Dimethyl sulfate (DMS)







- The product formed by attaching a chemical group to DNA is called an adduct.
- If the chemical group attaches to a single site on the DNA then the product is termed as monoadduct.
- The exposure of DNA to dimethylnitrosamine leads to the production of a mono-adduct in which a single methyl group attaches to DNA.
- When DNA is exposed to methyl methane sulfonate (MMS) or N-methyl-N-nitrosourea (MNU), methylation takes place most frequently at:-

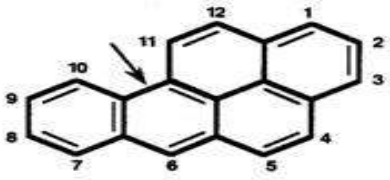
  i) N-7 position in guanine and

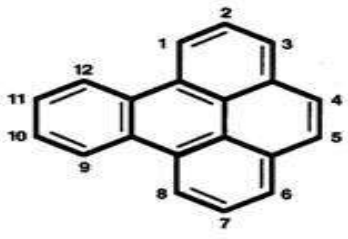
  ii) next most frequently at N-3 in adenine.
- N-methylguanine forms a base pair with cytosine, but it is readily removed from the DNA with the resultant formation of an abasic site.
- Methylation at N-3 in adenine is of great practical significance because N-3 methyl adenine formation blocks DNA replication but does not appear to lead to mutations.

- Therefore, a methylating agent that can transfer methyl group exclusively to N-3 position in adenine would have the potential to kill cancer cells without causing mutations.
- Methylation at O-6 in guanine and O-4 in thymine are much less frequent events than either of the above described methylations.
- O^6-methylguanine and O^4-methylthymine formation are quite important because the methylated bases mispair during DNA replication, resulting in transition mutations.
- The phosphate groups in DNA backbone can also be methylated.
- The resulting neutral phosphodiester is easily cleaved by water to produce single strand breaks.

Topic 66 DNA Damage by PAHs

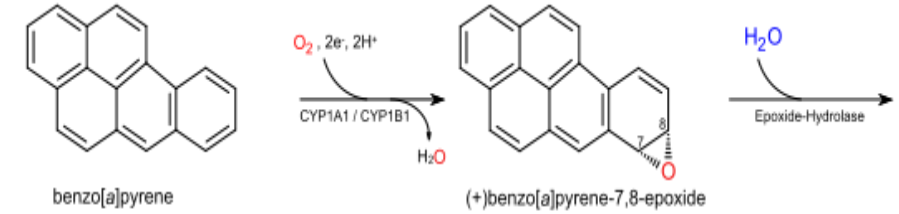
- Many environmental agents become active alkylating agents only after they are metabolized in the cells.
- One such agent is a mixture of polycyclic aromatic hydrocarbons (PAHs) formed by the incomplete combustion of the burning wood or coal used as fuel.
- Similar types of polycyclic aromatic hydrocarbons are also present in tobacco smoke and charbroiled meats.
- There are more than 100 different types of PAH compounds.

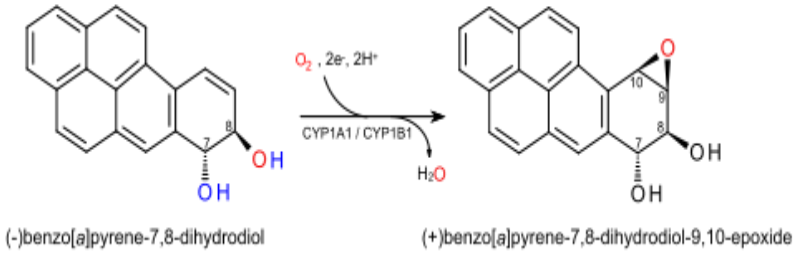




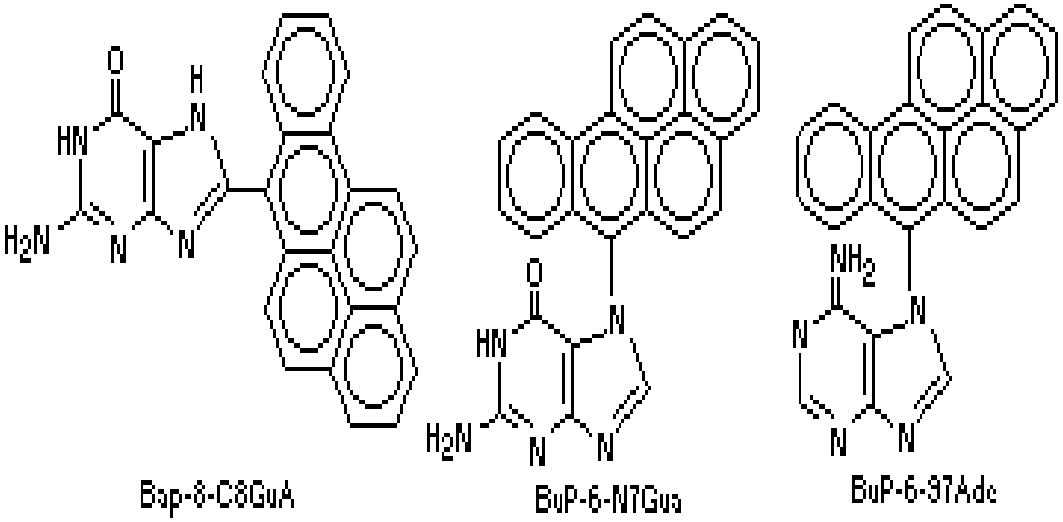
- The common structural feature in all PAHs is the presence of two or more fused aromatic rings.
- These are not able to damage DNA unless they are metabolically activated in the cell.

- One of the polycyclic aromatic hydrocarbon Benzo[a]pyrene is converted into an active epoxide alkylating agent through a metabolic pathway.





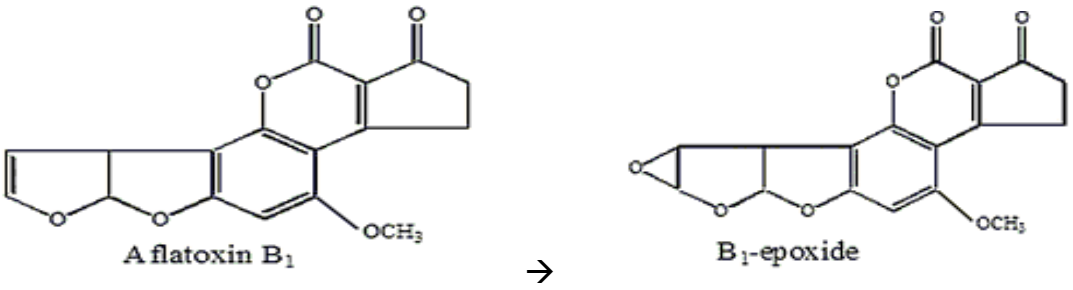
- Cytochrome P450 enzymes being not highly specific can act on PAHs such as benzo[a]pyrene and add oxygen atoms to form reactive three- membered epoxide rings.
- These epoxides then alkylate DNA, causing replication errors that result in mutations, which ultimately convert a normal cell into a cancer cell.



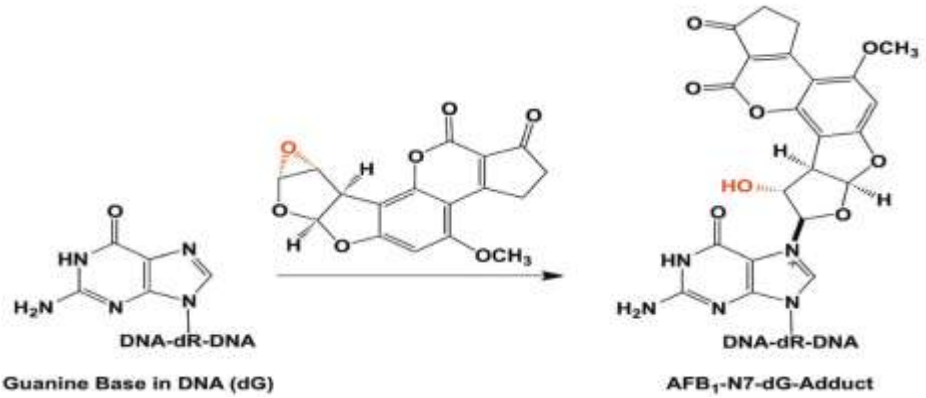
Topic 67 DNA Damage by Aflatoxins

- Another class of chemical carcinogens that must be activated before damaging DNA are called aflatoxins.
- They are produced from *Aspergillus flavus* and *A. parasiticus*, fungi that grow on peanut and other grains such as rice and corn.

- Animals feeding on contaminated peanuts or grains containing aflatoxins exhibit markedly increased rates of liver diseases including liver cancer.
- Aflatoxin B1, the most potent toxin produced by *A. flavus* presents a particularly serious health threat in the United States.
- Cytochrome P450 converts it into an epoxide derivative that damages DNA.



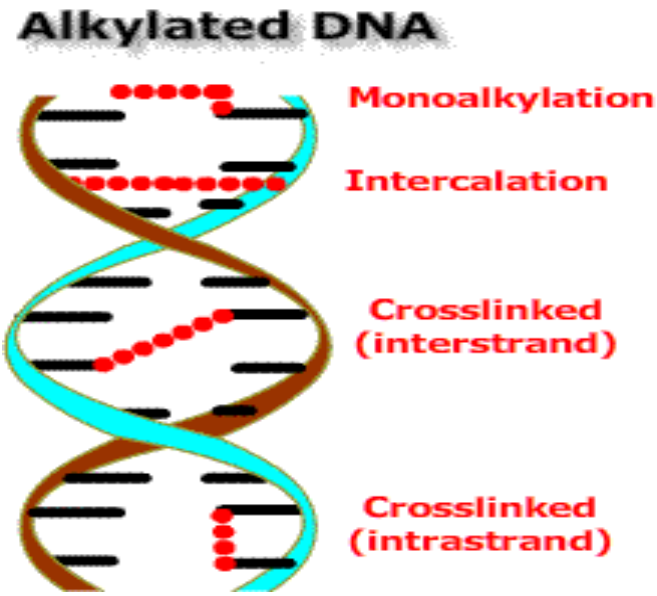
- Under ideal conditions, a small tripeptide glutathione will attack the epoxide ring, making the aflatoxin derivative soluble so that it can be excreted in the urine.
- If the reactive epoxide derivatives escape the attack of glutathione, they are free to attack guanine rings in DNA.



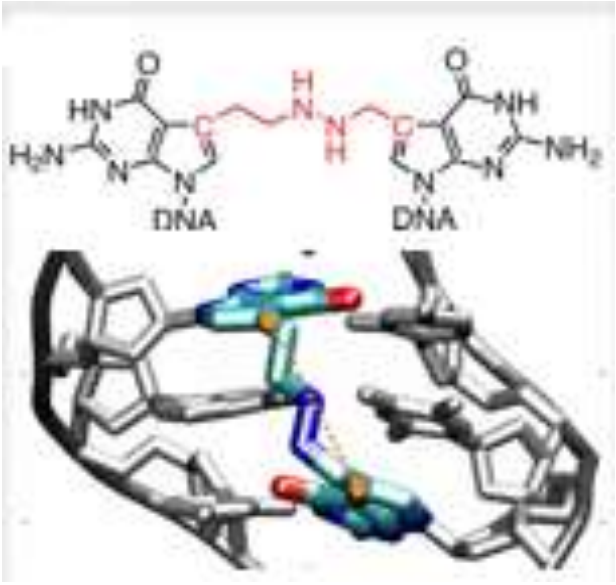
- The flat aflatoxin ring system inserts between DNA bases causing helical distortion that in turn leads to replication errors.

Topic 68 Chemical Cross-Linking Agents

- Many alkylating agents, having two reactive sites can form intra-strand or inter-strand cross-links in addition to forming mono-adducts.
- Inter-strand cross-links are of special interest because they prevent strand separation an, if not corrected, are lethal to the cells.



- One of the simplest cross-linking agents, nitrogen mustard gas (bis[2-chloroethyl] methylamine damages DNA by forming inter-strand cross-links.
- It does so by attacking N-7 on two guanines, which are on opposite strands of DNA double helix.

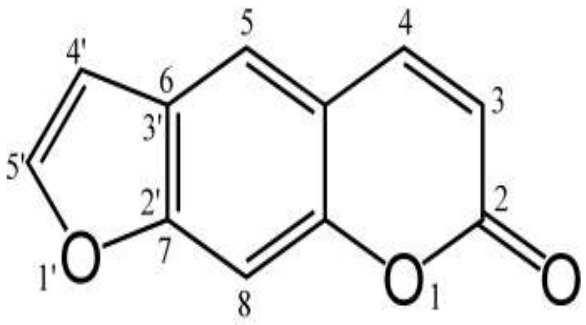


- Although a very toxic substance, nitrogen mustard gas has found clinical application as a chemotherapeutic agent for treating certain forms of cancers.

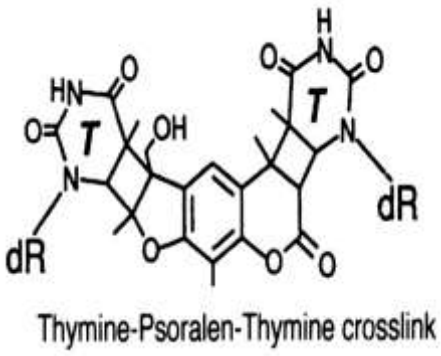
Topic 69 DNA damage by Psoralen

- Psoralen can form mono-adducts or cross-links.

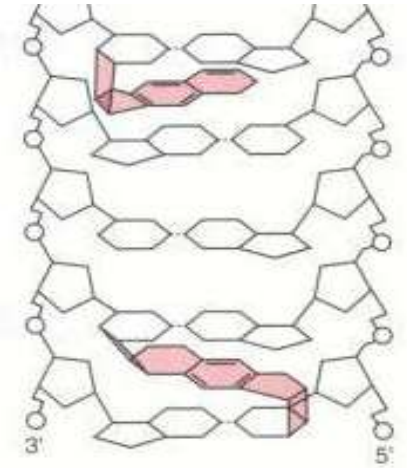
- It is a naturally occurring substance that can alkylate DNA if photo-activated.
- The planar psoralen molecule, which consist of a furan ring fused to a heterobicyclic ring system called coumarin, intercalates into the DNA molecule.



- Upon exposure to light with a wavelength of 400 to 450nm, the furan ring in psoralen becomes activated and adds across the 5,6 double bond in a pyrimidine base to form the 4',5', monoadduct.
- The planar tricyclic psoralen derivative in the monoadduct is in position to combine with a second pyrimidine base on the opposite DNA strand.
- But to do so, it must be activated by light with the wavelength of 320 to 400nm.
- The resulting photoproduct contain a cross-link between pyrimidine bases on opposite strands of the DNA duplex.



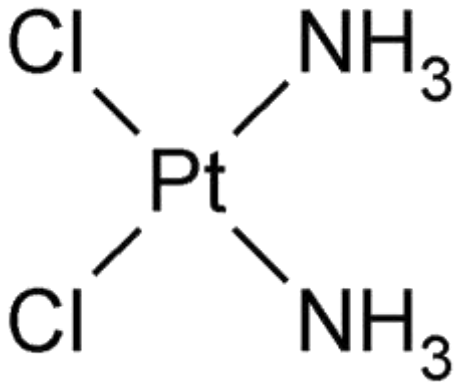
Thymine-Psoralen-Thymine crosslink



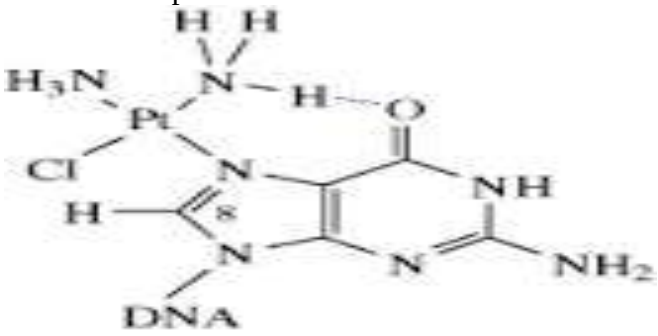
- If not properly repaired, psoralen damage causes mutations which are lethal to cells.

Topic 70 DNA damage by Cisplatin

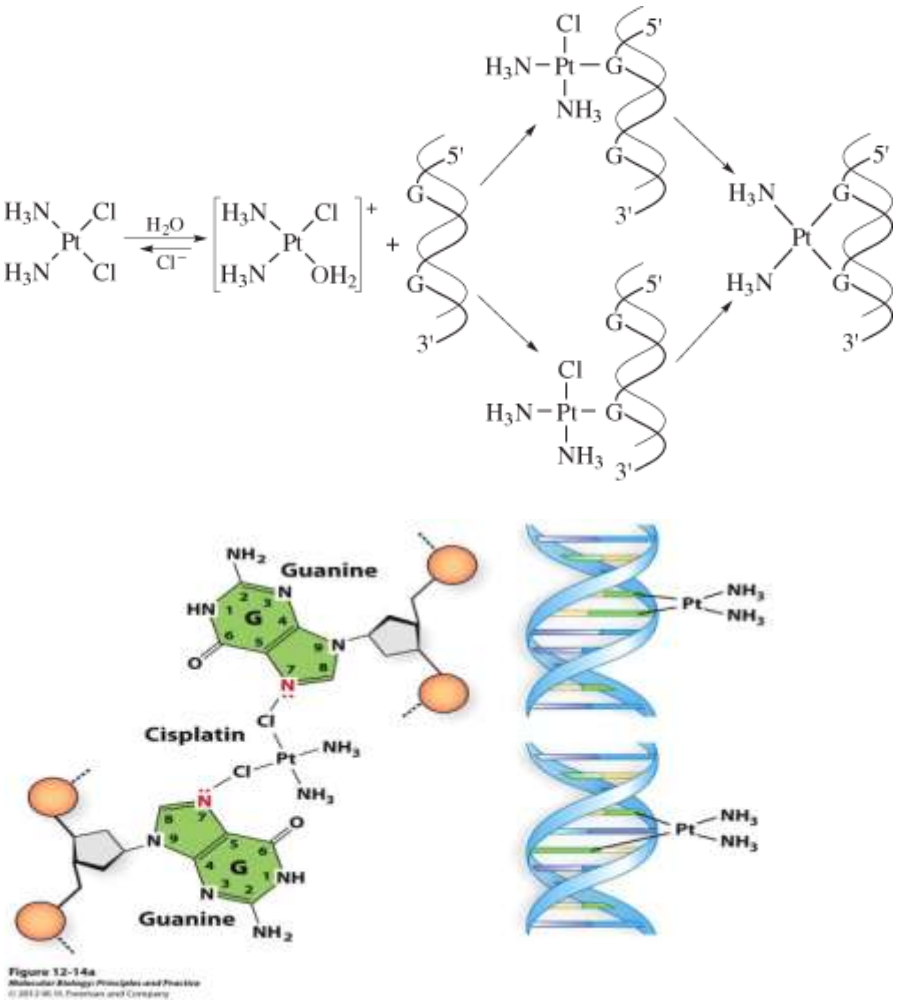
- While examining the effects of electric currents on *E. coli*, a new compound viz; *cis*-diamminedichloroplatinum, better known as Cisplatin was found to block cell division in *E. coli.*
- Efforts were made to see if cisplatin would also inhibit cell division in other kinds of cells.
- It was revealed that cisplatin blocks the division of tumor cells.



- After cisplatin enters the cell by passive diffusion or active transport, it undergoes hydrolysis to produce a highly reactive and charged complex ($Pt(NH3)2ClH2O^{+}$).
- This complex coordinates to the N-7 atom of either a guanine or adenine base in DNA.



- Then the remaining chloride ligand is displaced by hydrolysis , allowing the platinum to coordinate to a second purine base on the same or opposite strand of the double stranded DNA.



- Cisplatin's cytotoxic effects appear to be due to inter-strand cross-linking which blocks the replication and transcription machinery.
- Cisplatin is a very effective chemotherapeutic agent for treating cancers of the bladder, ovaries and testicles, but has a number of side effects.

Base Analogs and Intercalating Agents

- Mutations are also caused by compounds that substitute for normal bases called Base analogs or the compounds that slip between the bases called Intercalating agents and cause errors in replication.

Base Analogs and Intercalating Agents

- Base analogs are structurally similar to proper bases but differ in ways that make them treacherous to the cell.

Base Analogs and Intercalating Agents

- Thus, base analogs are similar enough to the proper bases to get taken up by cells, converted into nucleoside triphosphates, and incorporated into DNA during replication.

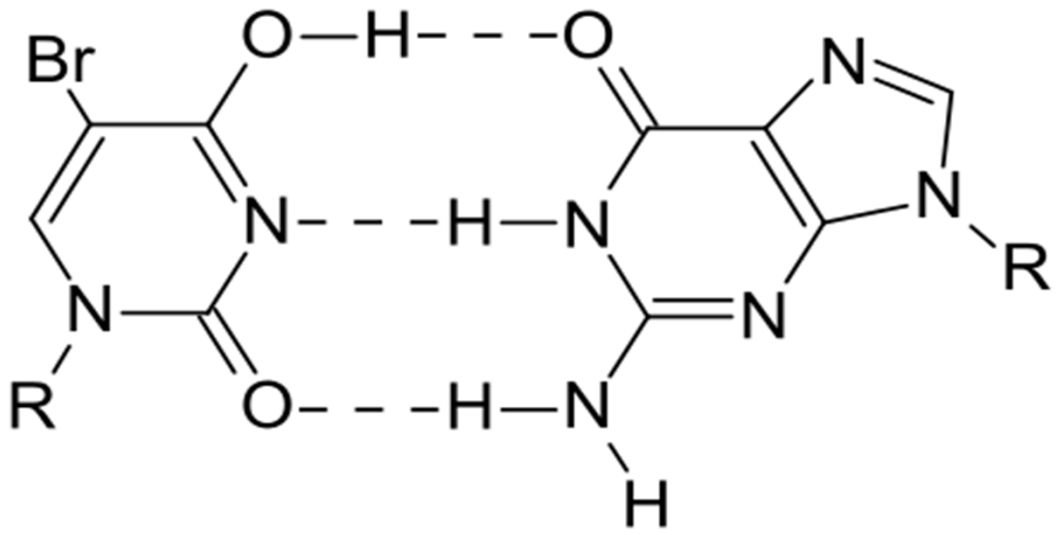
Base Analogs and Intercalating Agents

- But, because of the structural differences from the proper bases, the analogs base-pair inaccurately, leading to frequent mistakes during the replication process.

Base Analogs and Intercalating Agents

- One of the most mutagenic base analogs is 5-bromouracil, an analog of thymine. The presence of the bromo substituent allows the base to mispair with guanine.

Base Analogs and Intercalating Agents



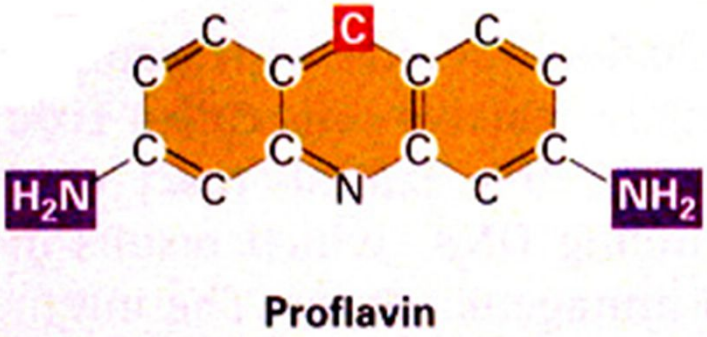
5-BrU

Guanine

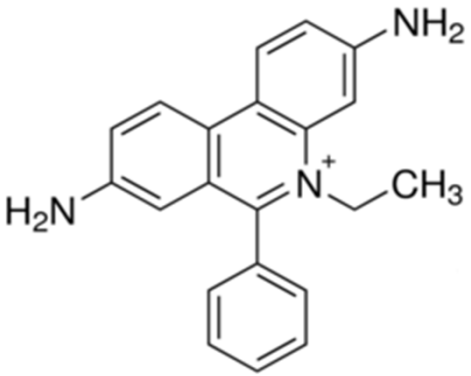
Base Analogs and Intercalating Agents

- Intercalating agents are flat molecules containing several polycyclic rings that bind to the purine or pyrimidine bases of DNA, just as the bases bind or stack with each other in the double helix.

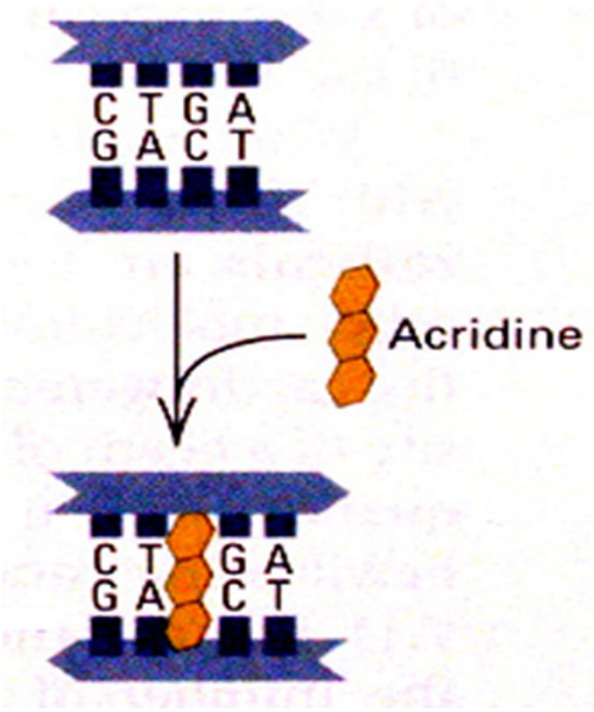
Base Analogs and Intercalating Agents



Proflavin



Ethidium



Base Analogs and Intercalating Agents

- Intercalating agents, such as proflavin, acridine, and ethidium, cause the deletion or addition of a base pair or even a few base pairs.

Base Analogs and Intercalating Agents

- These intercalating agents also cause short insertions or deletions.
- But how do these agents cause short insertions and deletions?

Base Analogs and Intercalating Agents

- One possibility in the case of insertions is that, by slipping between the bases in the template strand, these mutagens cause the DNA polymerase to insert an extra nucleotide opposite the intercalated molecule.

Base Analogs and Intercalating Agents

- Conversely, in the case of deletions, the distortion to the template caused by the presence of an intercalated molecule might cause the polymerase to skip a single or few nucleotides.

END

Direct Reversal of DNA Damage

- The first clue to the existence of an enzyme that catalyzes the direct reversal of DNA damage was reported by Albert Kelner in 1949.

Direct Reversal of DNA Damage

- In his experiments, Kelner first irradiated bacteria with UV light at doses that killed most of the bacteria.
- Then he tested the survivors to isolate the desired mutants.

Direct Reversal of DNA Damage

- Even though Kelner was very careful in his experimentation, he noticed a great deal of variation in the number of survivors from one experiment to the other.

Direct Reversal of DNA Damage

- Finally, he found that cells placed in dark after UV treatment had a very low survival rate whereas those placed in light had a high survival rate.

Direct Reversal of DNA Damage

- Exposure to light thus reversed the UV light's bactericidal effects.
- Similar phenomenon was also observed by Renato Dulbecco while studying UV-irradiated phage T2.

Direct Reversal of DNA Damage

- Dulbecco prepared multiple plates each containing the same number of UV-irradiated phages and sensitive bacteria, and placed the plates in a stack under a florescent bulb in the lab.

Direct Reversal of DNA Damage

- Each of the stacked plates should have about the same number of plaques.
- However, the plaque number decreased dramatically, going from top to bottom of the stack.

Direct Reversal of DNA Damage

- Dulbecco explained this by proposing that the plates on the top of the stack were exposed to more light from the bulb as compared to the plates on the bottom of the stack.

Direct Reversal of DNA Damage

- He tested this hypothesis by exposing some of the plates to more fluorescent light while keeping others in dark.

Direct Reversal of DNA Damage

- As expected, the number of plaques on plates exposed to light was much higher than those left in the dark.

Direct Reversal of DNA Damage

- The bacteria were somehow using the visible light to repair the UV damaged DNA in the phage.

Direct Reversal of DNA Damage

- The chemical basis for this light-dependant phenomenon, which Dulbecco called photoreactivation, remained to be elucidated.

Photoreactivation

- Claud S. Rupert and co-workers devised an in vitro photoreactivation system in 1957, taking a major step toward determining the chemical mechanism of photoreactivation.

Photoreactivation

- They used a straightforward approach in which they isolated DNA from the gram-negative bacteria *Haemophilus influenzae*.

Photoreactivation

- They irradiated this DNA with UV light to inactivate its transforming ability.
- Then they demonstrated that a cell-free *E. coli* extract restored the transforming activity in the presence of light.

Photoreactivation

- Although, this study had the potential to open the way for the purification and characterization of photoreactivation enzyme, investigators still needed to establish the chemical nature of this repair.

CPD Photolyase

- The problem of the chemical nature of photoreactivation was resolved over next few years when investigators came to know that UV irradiation induces the formation of cyclobutan pyrimidine dimer in DNA.

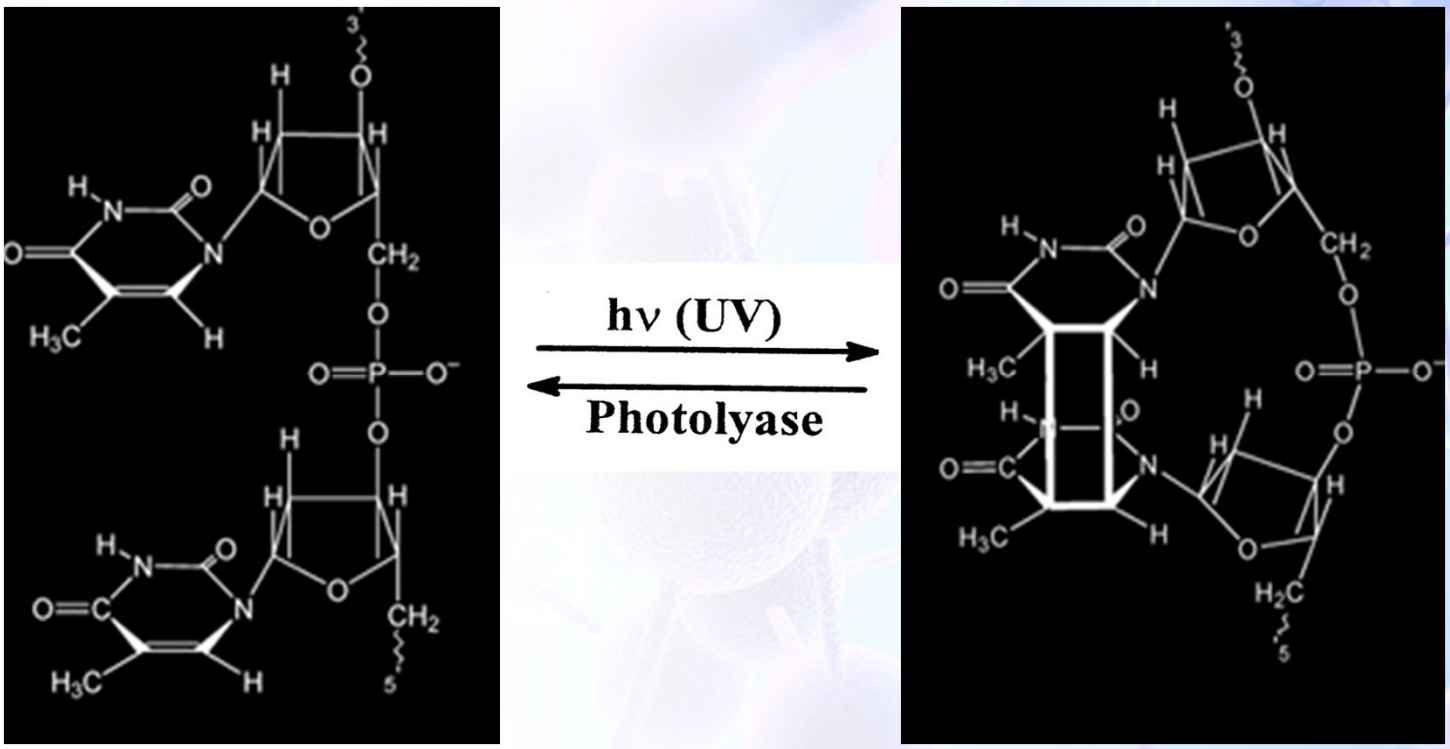
CPD Photolyase

- Further studies showed that the photoreactivation enzyme reverses the UV induced damage by using the energy provided by blue light (350-450nm) to drive cyclobutane ring disruption.

CPD Photolyase

- When it was recognized that the photoreactivation enzyme catalyzes the disruption of carbon – carbon bonds, it was given a more descriptive name of CPD photolyase.

CPD Photolyase



CPD Photolyase

- The bacteria that lack CPD photolyase can’t repaire cyclobutane pyrimidine lesions through photoreaction thus can’t reverse the UV induced damage in DNA.

CPD Photolyase

- CPD photolyases are present in a wide variety of organisms including bacteria, archaea, plants, and animals but not in humans and other placental mammals.

CPD Photolyase

- These are monomeric proteins ranging in size from about 450-550 amino acid residues.
- All CPD photolyases have two domains, designated as N – and C – terminal domains.

CPD Photolyase

- A light absorbing pigment, or chromophore pigment binds to each domain through non-covalent bonding.

CPD Photolyase

- This chromophore factor acts as a photoantenna to capture light with wavelengths that would not otherwise be available.

Mechanism of CPD Photolyase

- CPD photolyase can bind to DNA in the dark by recognizing the altered DNA structure caused by a CPD formation rather than a specific nucleotide sequence.

Mechanism of CPD Photolyase

- This binding is about 10^5 tighter when a DNA segment contains a CPD than when it does not.
- Half of the binding energy appears to come from interactions between enzyme and the DNA back bone.

Mechanism of CPD Photolyase

- The other half of energy comes from interactions between the $FADH^-$ at the active site and the CPD.
- However, the light harvesting antenna pigment does not influence binding.

Mechanism of CPD Photolyase

- Once the enzyme – DNA complex is formed, the CPD is flipped out of the DNA double helix and into the enzyme's active site.

Mechanism of CPD Photolyase

- After the CPD flips into the enzyme's active site, the energy of an absorbed photon is transferred from the light harvesting antenna pigment to the $FADH^-$.

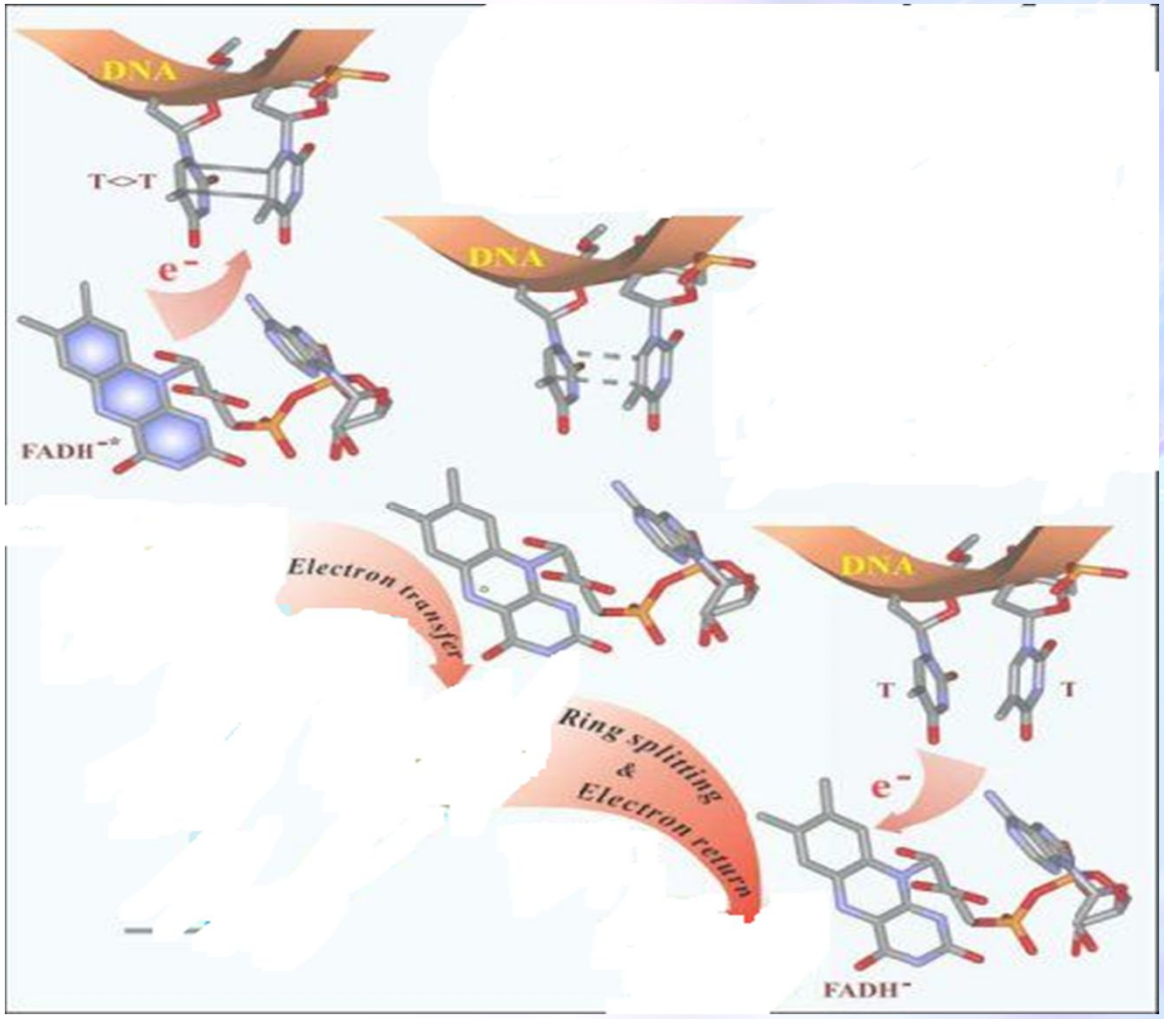
Mechanism of CPD Photolyase

- $FADH^-$ then transfers an electron to the CPD to induce cyclobutane ring cleavage.

Mechanism of CPD Photolyase

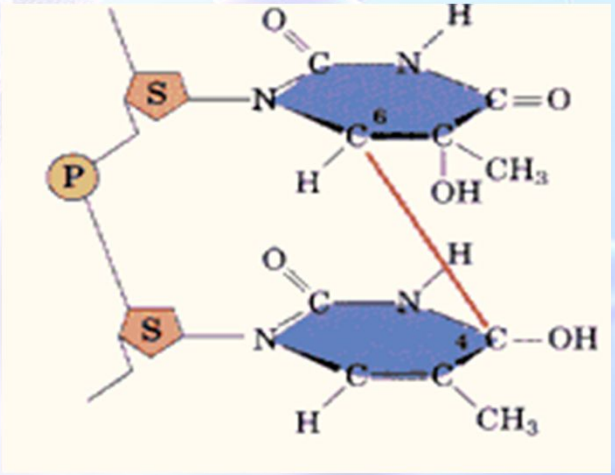
- The catalytic cycle is completed when the electron is transferred back from the repaired thymine to the FADH cofactor.

Mechanism of CPD Photolyase



(6-4) Photolyase

- UV irradiation also induces the formation of another type of pyrimidine dimer, the (6-4) photoproduct.



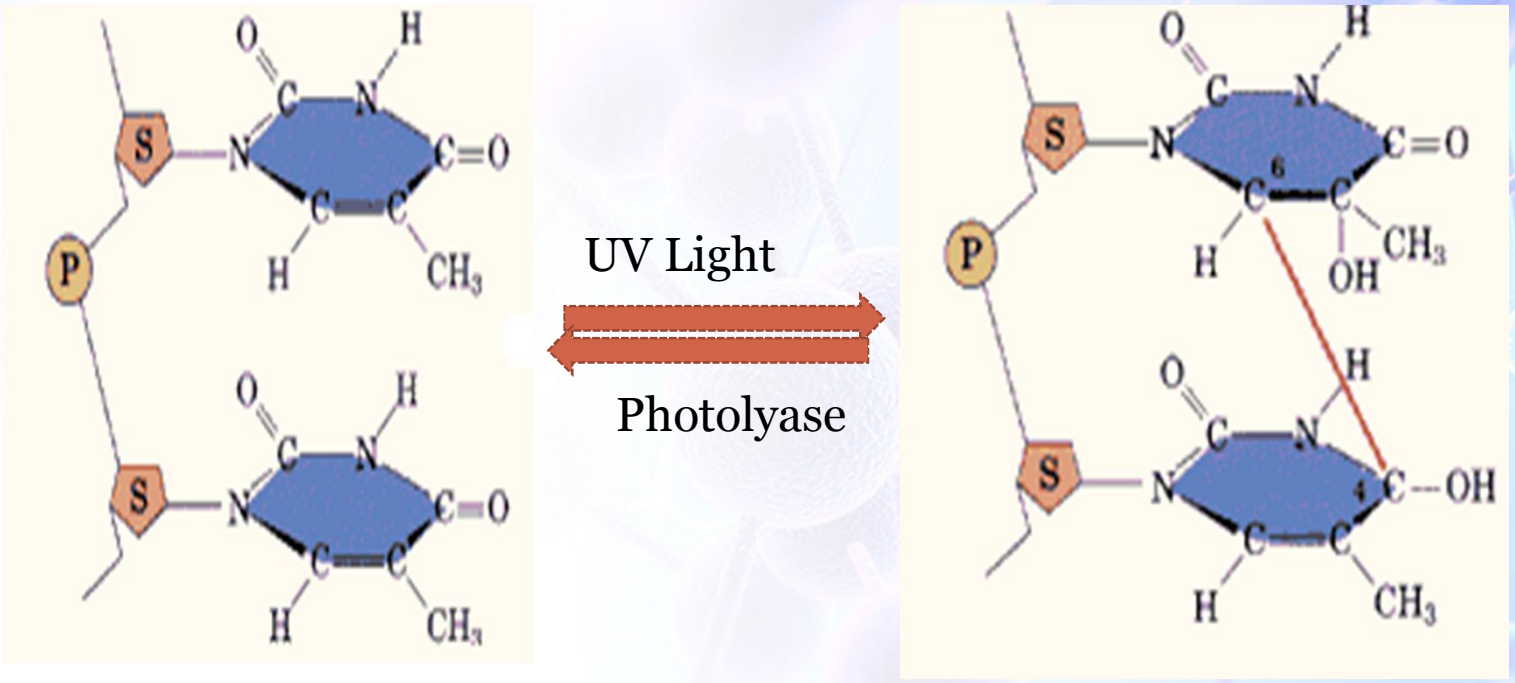
(6-4) Photolyase

- Takeshi Todo, Taisei Nomura and coworkrs reported in 1993 that *Drosophila melanogaster* has a photolyase that reverses (6-4) photoproduct lesions in the DNA.

(6-4) Photolyase

- This photolyase was designated as (6 – 4) photolyase.
- It is widely distributed in plants and animals, but has not been detected in bacteria and mammals.

(6-4) Photolyase



(6-4) Photolyase

- Considerably less information is known about the mechanism of action of (6-4) photolyase as compared to CPD photolyase; however, the two enzymes seem to work in a similar way.

(6-4) Photolyase

- The (6-4) photolyase binds to damaged DNA, causing the (6-4) photoproduct to flip out of the DNA and into the active site of the enzyme.

(6-4) Photolyase

- In the active site of the enzyme, it undergoes a rearrangement to produce a product that receives an electron from an excited $FADH^-$ molecule.

(6-4) Photolyase

- The final outcome is that the organisms containing (6-4) photolyase can use light energy to convert (6-4) photoproducts back to normal pyrimidine rings.

(6-4) Photolyase

- Organisms can also repair dimer lesions introduced by UV light by excising damaged nucleotides and replacing them with normal nucleotides.

(6-4) Photolyase

- This type of excision repair is the major pathay for repairing UV-induced damage to DNA in organisms such as humans that lack both the types of photolyases.

Damage Reversal by Dealkylation

- Another means of direct reversal of DNA damage is by dealkylation.
- Direct dealkylation reactions have probably been most extensively studied in *E. coli*.

Damage Reversal by Dealkylation

- Three different proteins can catalyze the direct removal of alkyl groups attached to oxygen atoms in DNA.
- These include:-

Damage Reversal by Dealkylation

- 1) O^6-alkylguanine DNA alkyltransferase I
- 2) O^6-alkylguanine DNA alkyltransferase II
- 3) Alkylguanine DNA alkyltransferase

Damage Reversal by Dealkylation

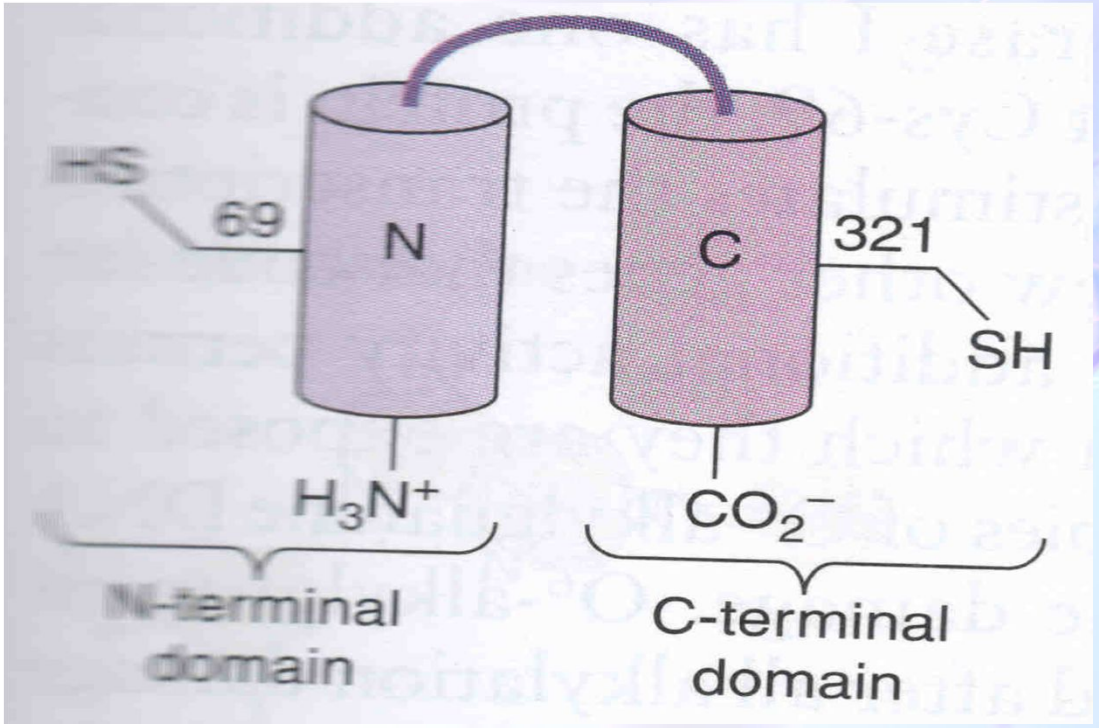
O^6-alkylguanine DNA alkyltransferase I

- O^6-alkylguanine DNA alkyltransferase I can remove methyl & other alkyl groups attached to O-6 in guanine.
- It can also remove alkyl groups attached to O-4 of thymine and to phosphotriesters.

Damage Reversal by Dealkylation

- This enzyme is a monomer that has a flexible linker connecting its N- and C- terminal domains.
- Each domain has an active site that performs a specific function.

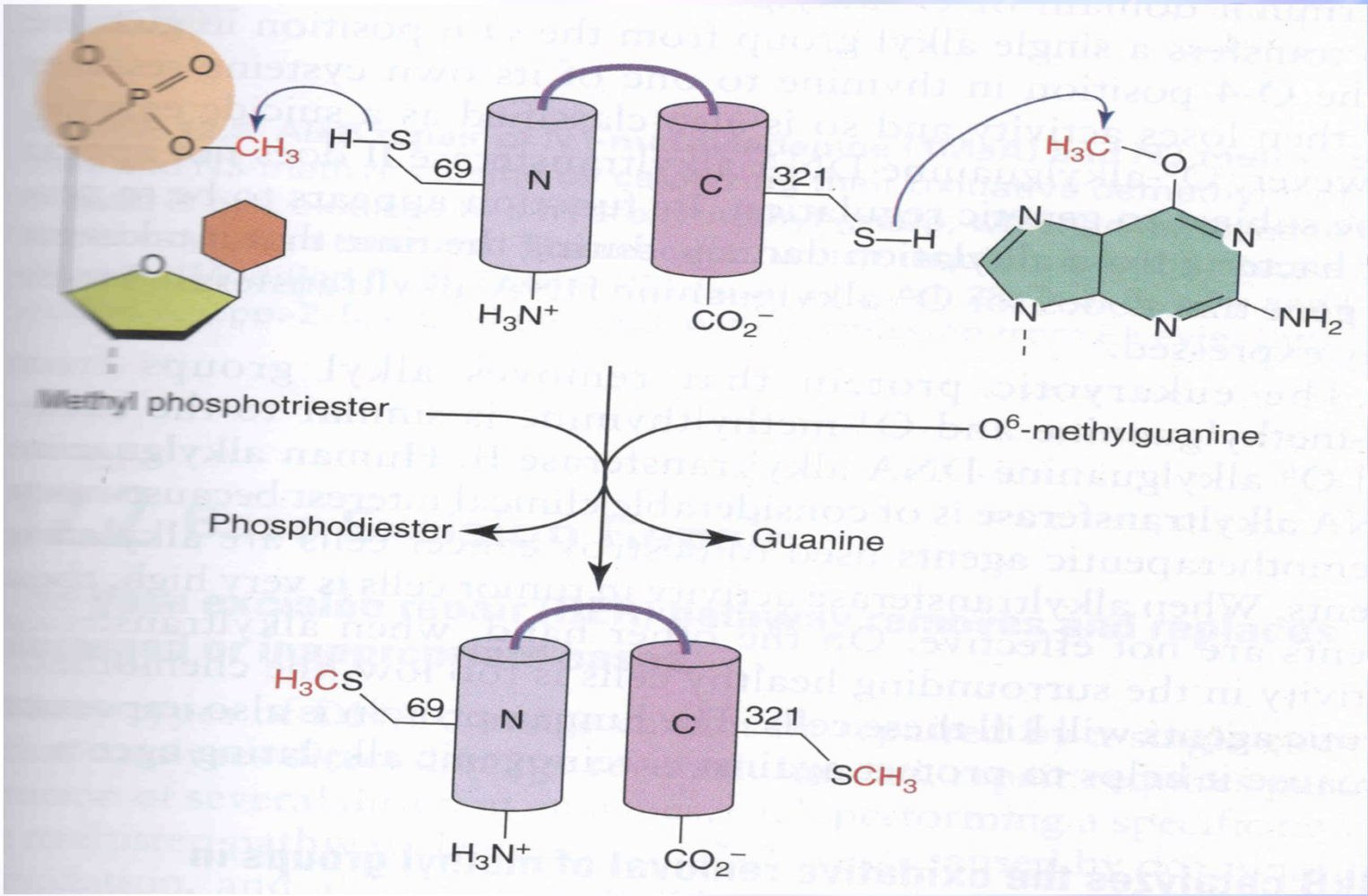
Damage Reversal by Dealkylation



Damage Reversal by Dealkylation

- The N- terminal domain transfers an alkyl group from an S_p phosphotriester to one of its own cysteine residues, Cys-69.

Damage Reversal by Dealkylation



Damage Reversal by Dealkylation

- The C- terminal domain transfers an alkyl group from either O^6-alkylguanine or O^4-alkylthymine to one of its own cysteine residues, Cys-321.

Damage Reversal by Dealkylation

- Once alkylated, the protein can't be regenerated and therefore behaves more like an alkyl transferring agent than an enzyme.

Damage Reversal by Dealkylation

- Such proteins as O^6-alkylguanine DNA alkyltransferase I, which lose their activity after acting one time, are called suicide enzymes.

Damage Reversal by Dealkylation

- O^6-alkylguanine DNA alkyltransferase I, has one additional remarkable function.
- After methylation at Cys-69, this protein is converted to a transcriptional activator.

Damage Reversal by Dealkylation

- This activator stimulates the transcription of the gene that codes for it as well as some other genes that code for the proteins that repair DNA damage.

Damage Reversal by Dealkylation

- This additional activity permits the bacteria to adapt to environments in which they are exposed to alkylating agents by synthesizing more copies of these enzymes which can repair damaged DNA.

Damage Reversal by Dealkylation

- O^6-alkylguanine DNA alkyltransferase I that is synthesized after all the alkylation damage has been repaired will remain unmethylated.

Damage Reversal by Dealkylation

- This unmethylated form of the enzyme blocks transcription of the same genes that were activated by the methylated proteins.

Dealkylation Enzymes

O^6-alkylguanine DNA alkyltransferase II

- O^6-alkylguanine DNA alkyltransferase II is another E. coli's alkyl transferring enzyme.
- It has properties very similar to that of C-terminal domain of O^6-alkylguanine DNA alkyltransferase I.

Dealkylation Enzymes

- It also transfers a single alkyl group from the O – 6 position in guanine or the O-4 position in thymine to one of its own cysteine residues.

Dealkylation Enzymes

- After this transfer, the enzyme loses its activity and so is also classified as a suicide enzyme.
- However, this enzyme doesn’t appear to be subjected to genetic regulation.

Dealkylation Enzymes

- Its function appears to be to protect bacteria from alkylation damage during the time that it takes for the gene of O^6-alkylguanine DNA alkyltransferase I to be fully expresed.

Dealkylation Enzymes

Alkylguanine DNA alkyltransferase

- The eukaryotic protein that removes alkyl group from O^6-methylguanine or O^4-methylthymine is similar to the bacterial O^6-alkylguanine DNA alkyltransferase II.

Dealkylation Enzymes

- Human alkylguanine DNA alkyltransferase is of considerable clinical interest because many chemotherapeutic agents used to destroy cancer cells are alkylating agents.

Dealkylation Enzymes

- When alkyltransferase activity is very high in the tumor cells, these agents are not effective.

Dealkylation Enzymes

- On the other hand, when alkyltransferase activity in the surrounding healthy cells is too low, the chemotherapeutic agents will kill these cells.

Dealkylation Enzymes

- The human protein/enzyme is also important because it helps to protect the cells against carcinogenic alkylating agents.

Dealkylation Enzymes

AlkB

- An entirely different type of alkylation damage repair activity was found in *E. Coli* in 2002 which is carried out by a protein AlkB.

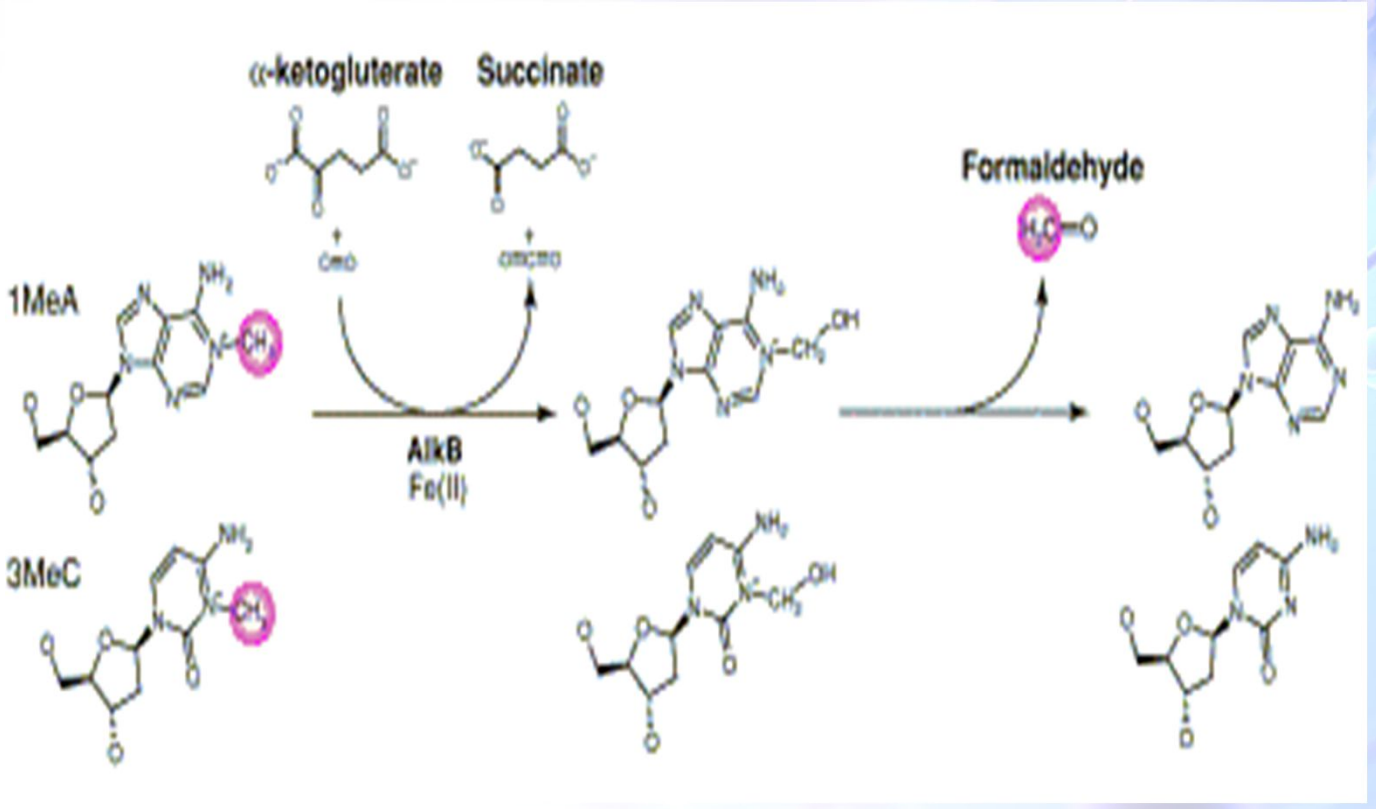
Dealkylation Enzymes

- AlkB catalyzes the direct conversion of 1-methyladenine, 1-methylguanine, 3-methylcytosine and 3-methylthymine to adenine, guanine, cytosine and thymine, respectively.

Dealkylation Enzymes

- The AlkB catalyzed reaction requires, $Fe2^{+}$, molecular oxygen, and α - ketoglutarate.
- A similar type of enzyme has been found in other organisms including human & mammals.

Dealkylation Enzymes



Base Excision Repair

- Many types of DNA damages can’t be repaired by a single enzyme that catalyzes direct damage reversal.

Base Excision Repair

- Instead, repair requires the participation of several different enzymes, each performing a specific task in multistep pathway.

Base Excision Repair

- Damage to DNA bases caused by deamination, oxidation, and alkylation is mainly repaired by one such multistep pathway which is called, Base Excision Repair (BER).

Base Excision Repair

- BER pathway is same in all organisms.
- Enzymes involved in the base excision repair pathway also participate in the repair of single-strand break in DNA.

Base Excision Repair

- Base excision repair derives its name from the first step in the pathway, N-glycosyl bond cleavage.
- This cleavage excises the damaged or inappropriate base from the DNA to form an abasic site.

Base Excision Repair

- Because no single enzyme can distinguish the four bases present in DNA from a wide variety of altered bases, cells must use many different enzymes to perform this function.

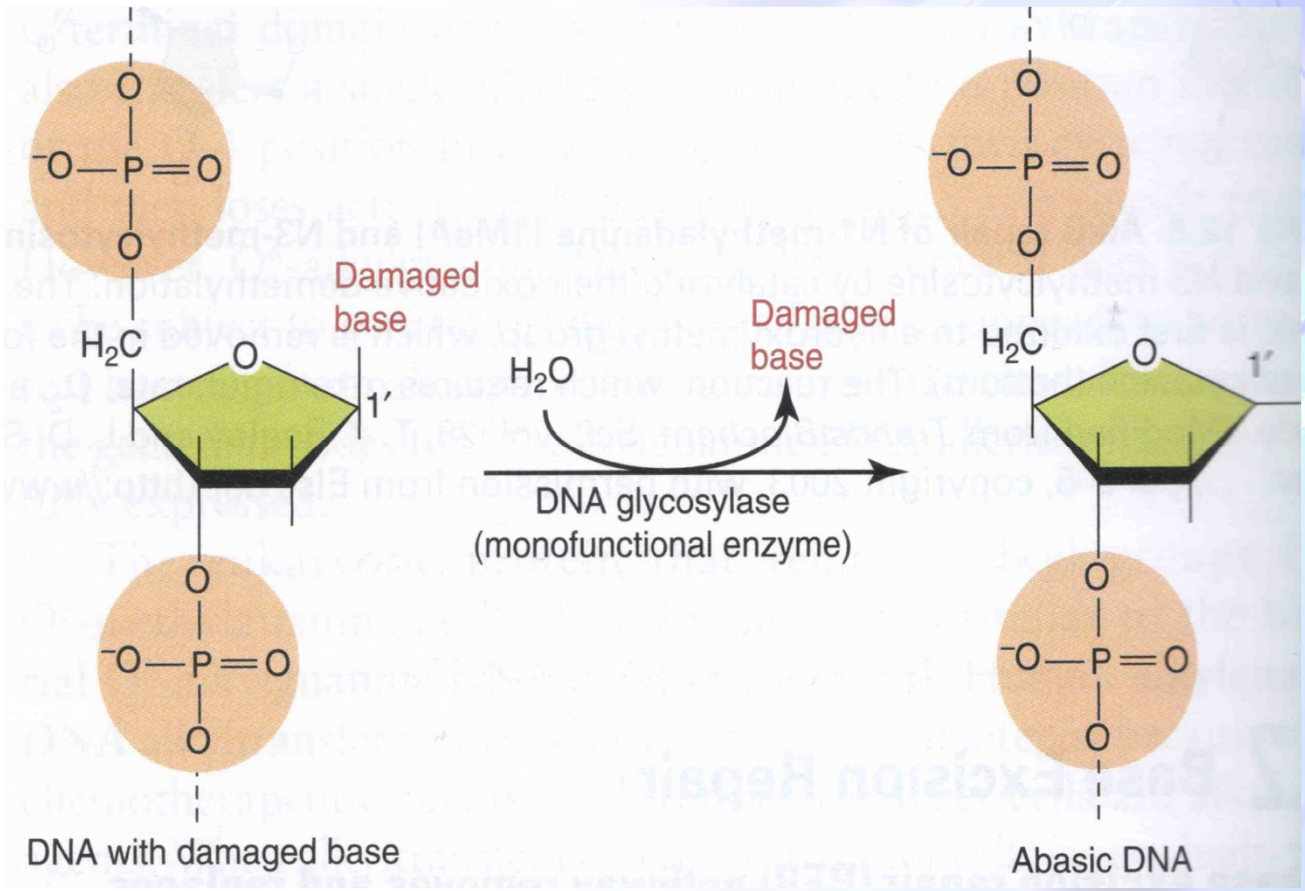
Base Excision Repair

- Some N-glycosylases are monofunctional enzymes with only the ability to excise a damaged base.
- Such enzymes are called DNA glycosylases.

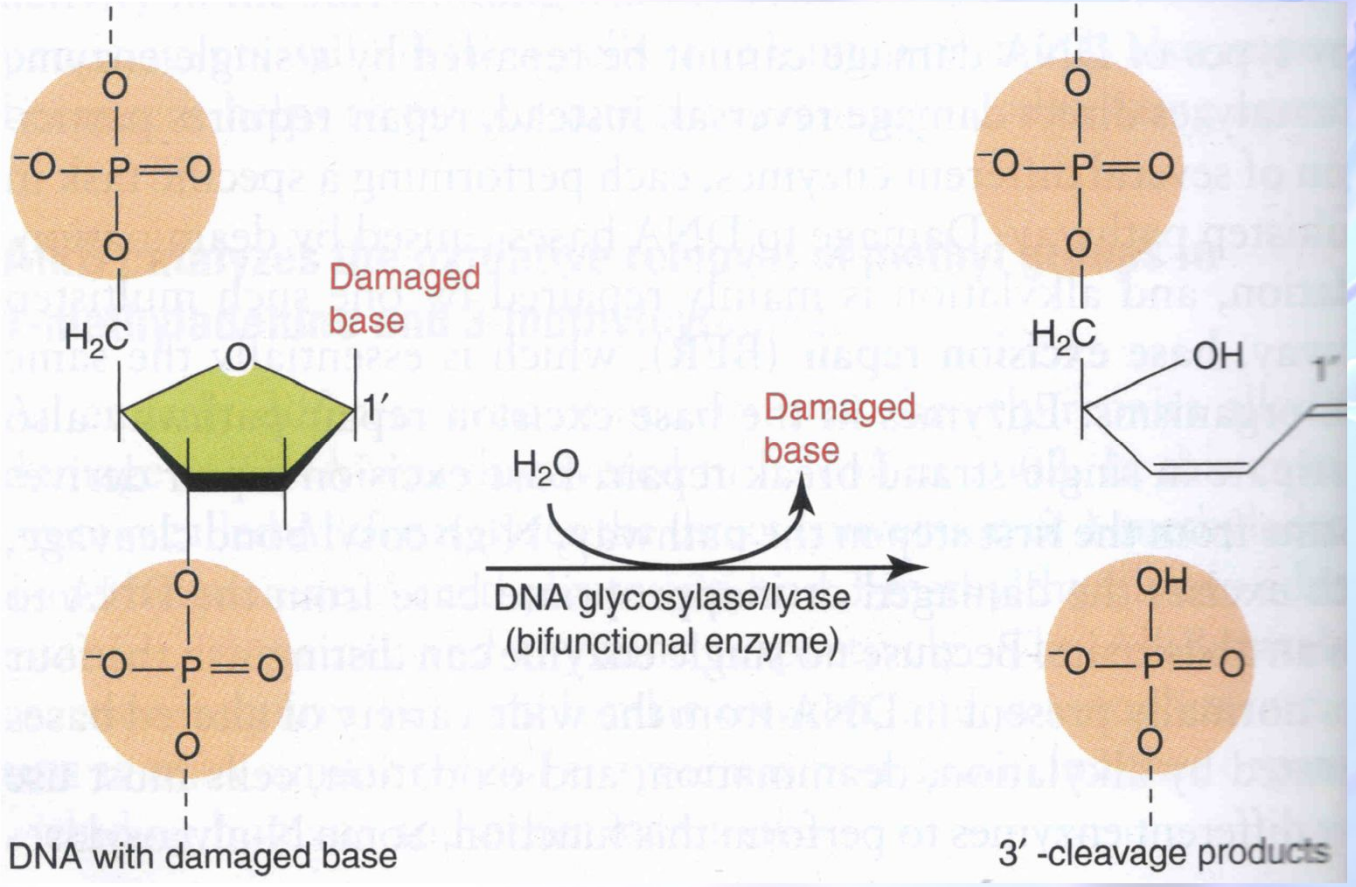
Base Excision Repair

- Other N-glycosylases have an AP lyase activity that cleaves the bond between sugar and the phosphate 3’ to the damaged site.
- This enzymes is designated as DNA glycosylase/lyase.

Base Excision Repair



Base Excision Repair



Base Excision Repair

- Both types of enzymes detect the damaged base, flip it out of the DNA helix into an active site pocket, and then cleave the N-glycosyl bond.

Base Excision Repair

- Although some glycosylases excise a specific base, most have a somewhat broader specificity.
- The *E.coli* enzyme Uracil N-glycosylase (Ung) is specific for uracil.

Base Excision Repair

- Whereas another *E. coli* N-glycosylase, 3-methyladenine DNA glycosylase (AlkA), acts on 3- or 7-methyl purines, 3- or 7-ethylpurine, ethenoadenine, and O^2-methyl pyrimidine.

Base Excision Repair

- A null mutation in a gene that codes for a DNA glycosylase or DNA glycosylase/lyase is not lethal which probably reflects overlapping functional abilities among the glycosylases.

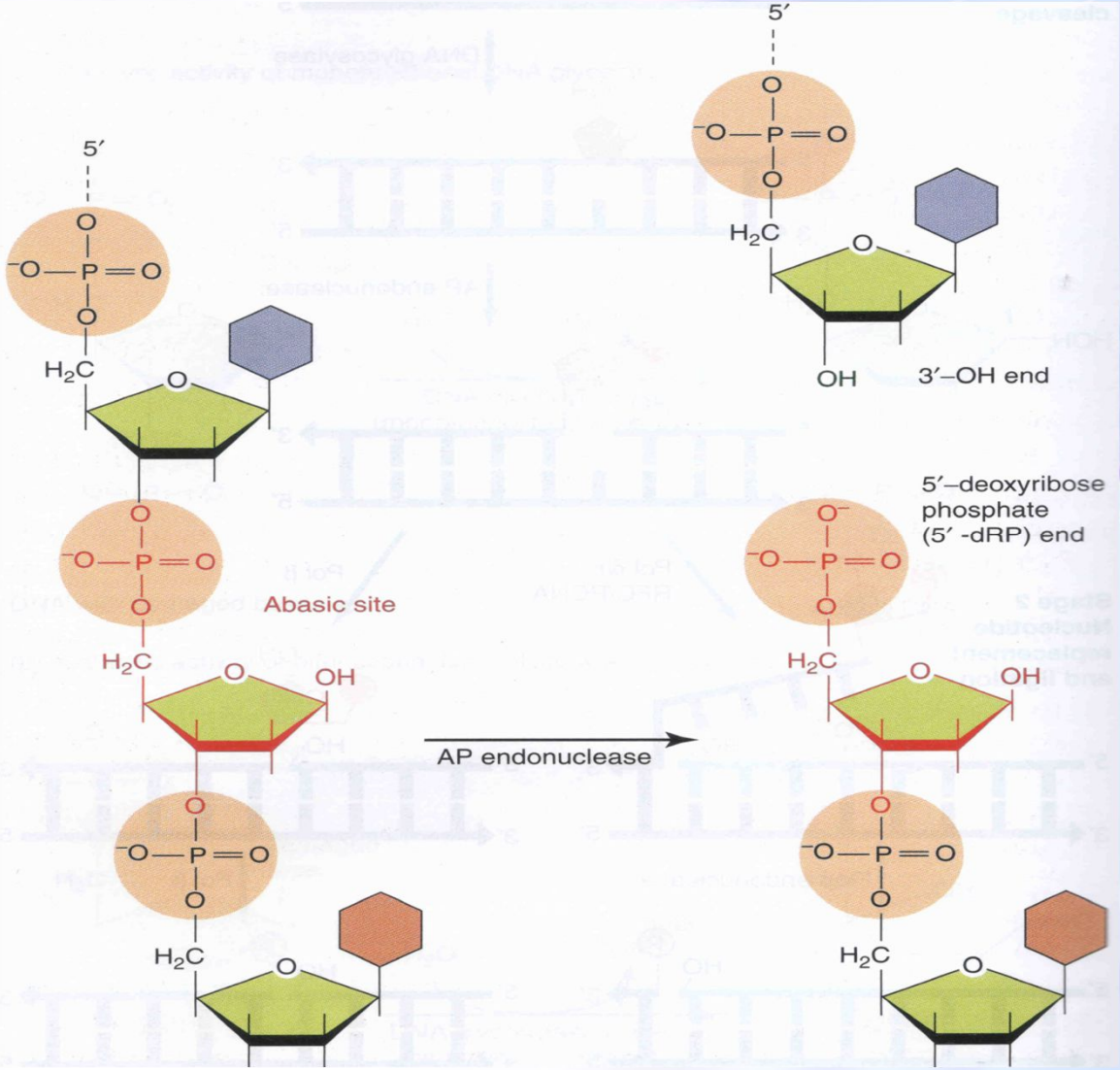
Base Excision Repair Pathway

- The DNA glycosylase catalyzes base excision to produce an AP (apurinic and apyrimidininc) site.

Base Excision Repair Pathway

- The next enzyme in the pathway, AP endonuclease, hydrolyzes the phosphodiester bond 5’ to the AP site to generate a nick.

Base Excision Repair Pathway



Base Excision Repair Pathway

- *E.coli* has two well-characterized AP endonucleases:-
  - Exonuclease III (Xth)
  - Endonuclease IV (Nfo)
- Exonuclease III (Xth) despite its name accounts for most of the bacterial AP endonuclease activity.

Base Excision Repair Pathway

- Both are multifunctional enzymes that have 3'-phosphate and 3'-repair phosphodiesterase activities.
- The former activity removes phosphate groups from the 3' end of a DNA strand.

Base Excision Repair Pathway

- The latter activity i.e., 3’-repair phosphodiesterase activity removes the 3’-unsaturated aldehydic group produced by DNA glycosylase/lyase action.

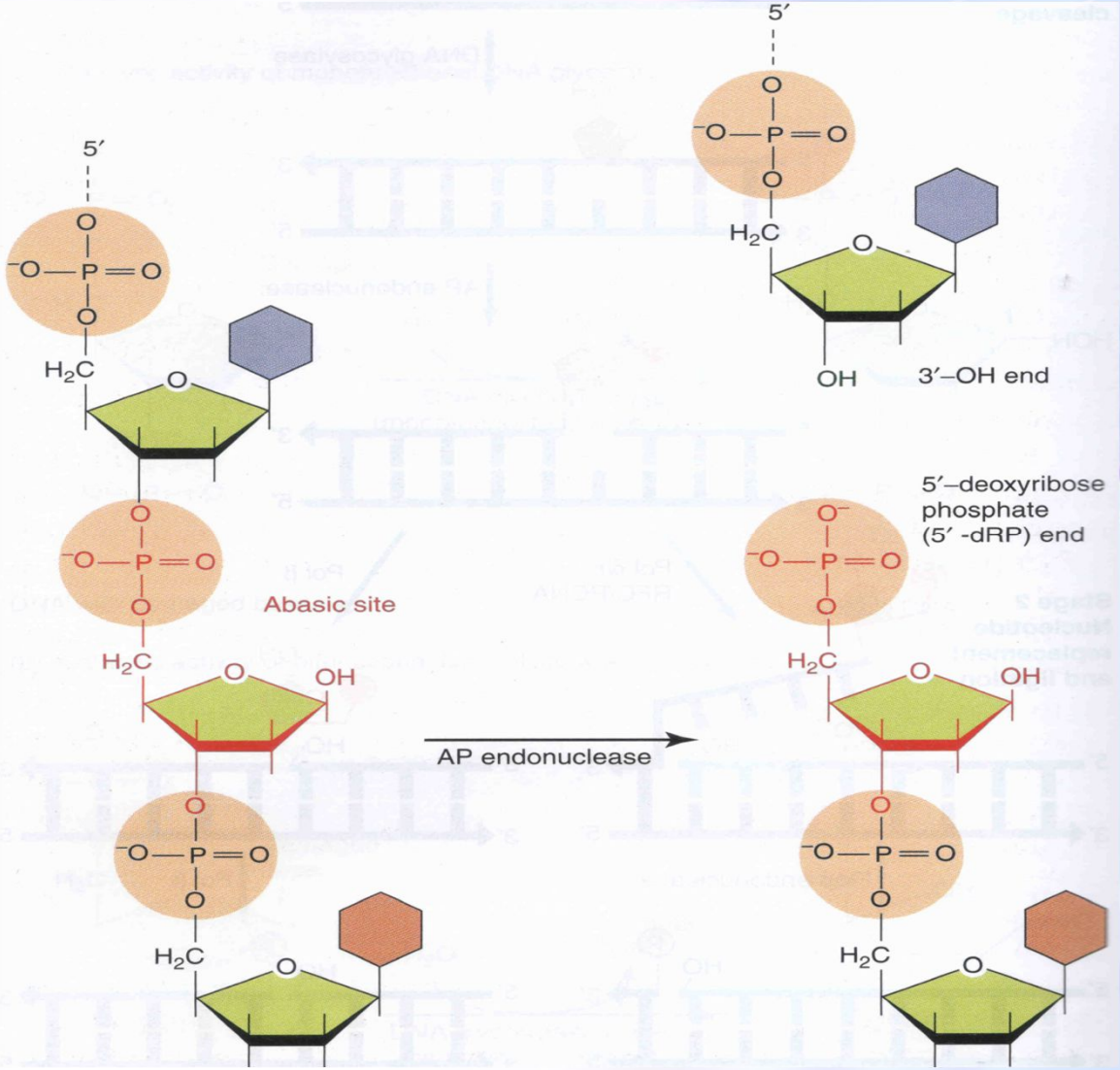
Base Excision Repair Pathway

- These activities are important because DNA polymerase can't attach new nucleotides to a blocked 3'-end.
- The mammalian AP endonuclease, APE1 is homologous to *E. coli* exonuclease III.

Short Patch Repair

- Base excision by DNA glycosylase and strand cleavage by AP endonuclease introduces a gap with a 5’-deoxyribose phosphate (5’-dRP) on one side and a 3’-OH on the other.

Short Patch Repair



Short Patch Repair

- Additional enzymes are required to fill in the gap and remove the 5’-deoxyribose phosphate.
- Cells can repair the damage by two different pathways viz. short patch repair & long patch repair.

Short Patch Repair

- In the short patch repair, only a single nucleotide is replaced.
- This repair pathway involves the following enzyme catalyzed reactions:-

Short Patch Repair

- (1) DNA polymerase adds a deoxyribonucleotide to the 3’-OH end.
- (2) Deoxyribose phosphate lyase (dRPase) removes 5’-deoxyribose phosphate from the 5’-end; and

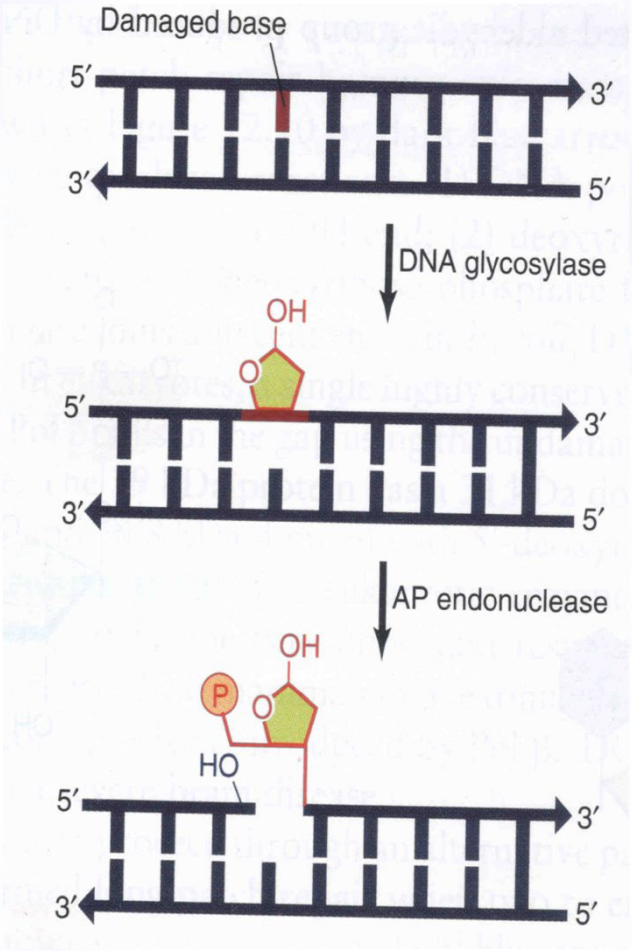
Short Patch Repair

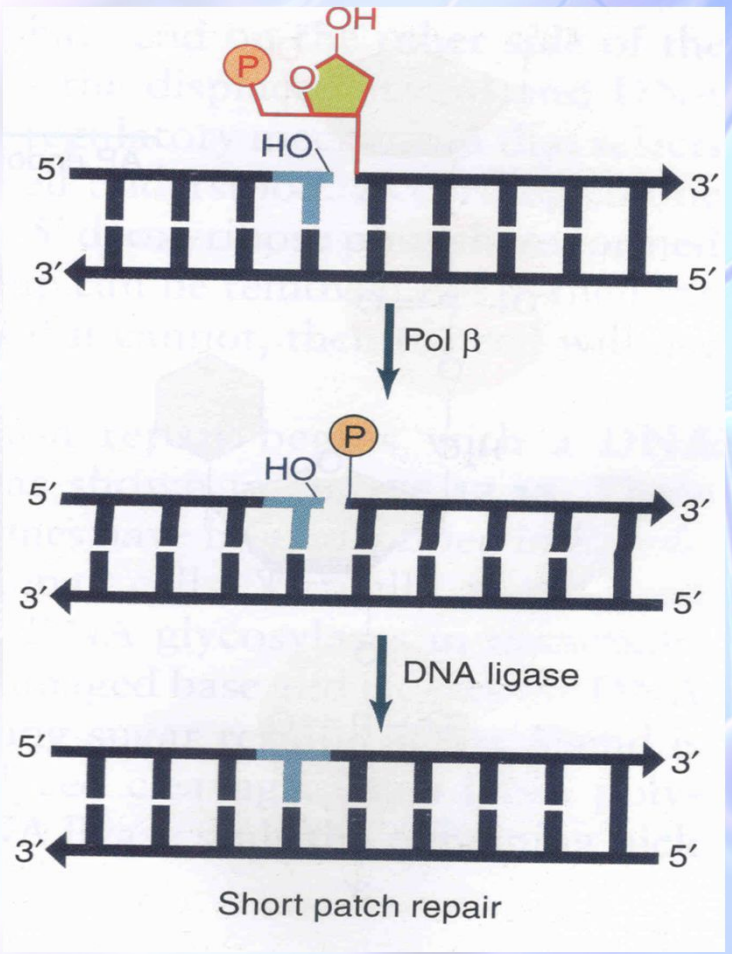
- (3) DNA ligase joins the adjacent ends.
- In *E. coli*, DNA polymerase I fills in the gap.

Short Patch Repair

- While in eukaryotes, a single highly conserved enzyme, DNA polymerase β (Pol β) fills in the gap using the undamaged DNA strand as the template.

Short Patch Repair





Short Patch Repair

- The eukaryotic enzyme differs from its bacterial counterpart in one very important respect, i.e., it lacks 3’ → 5’ proofreading activity.

Short Patch Repair

- Two mammalian 3’ exonucleases, TREX1 and TREX2 may correct the errors introduced by Pol β.

Long Patch Repair

- The alternative pathway to repair the damage in mammalian cells is termed as long patch repair.
- In this pathway, 2 – 8 nucleotides are replaced.

Long Patch Repair

- In this case, DNA polymerase δ or Ɛ catalyzes chain extension with the assistance of the RFC (replication factor C) clamp loader and the PCNA sliding clamp.

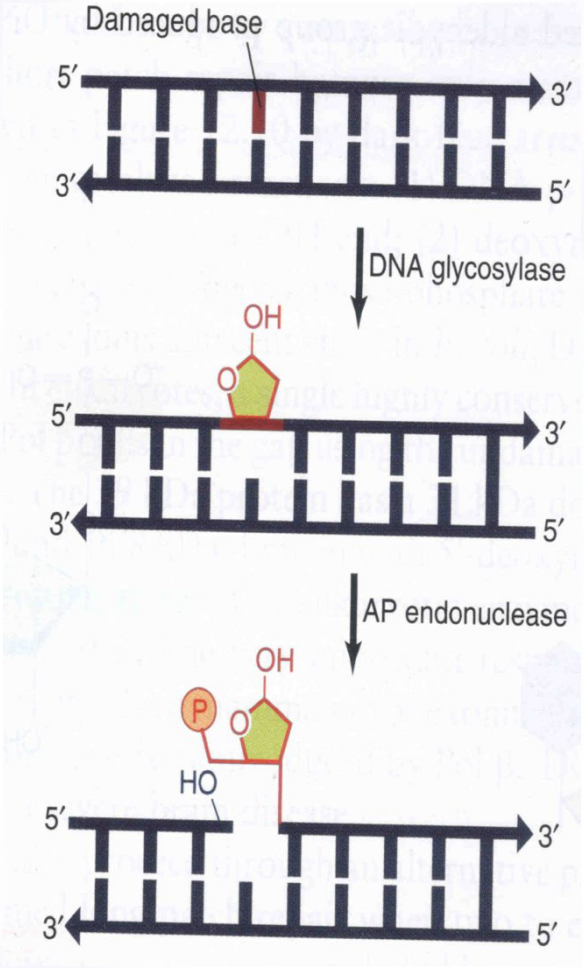
Long Patch Repair

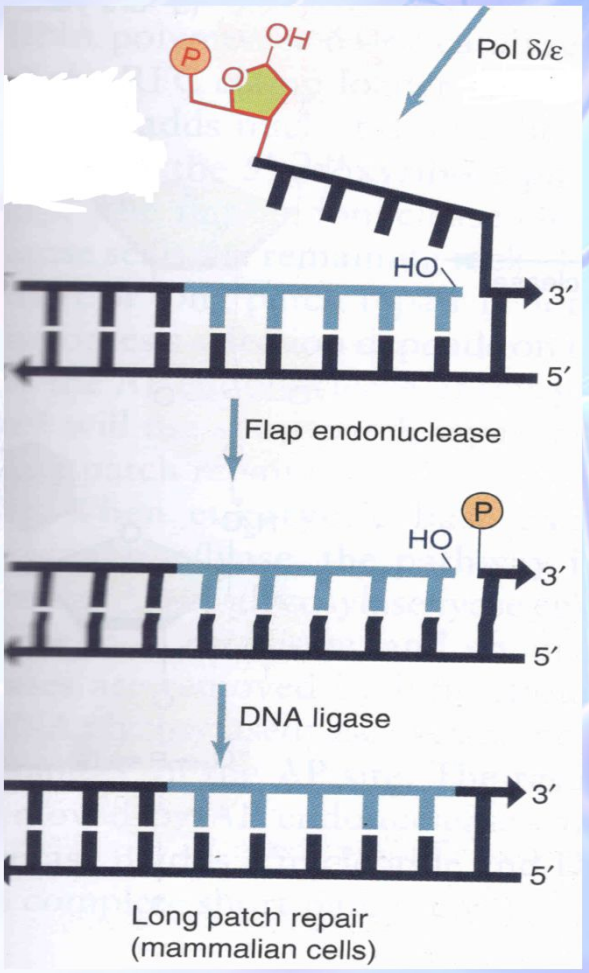
- As the polymerase adds nucleotides to the 3’-OH end on one side of the gap, it displaces the 5’-deoxyribose phosphate end on the other side of the gap.

Long Patch Repair

- The flap endonuclease cleaves the displaced strand and DNA ligase seals the remaining nick.
- The regulatory mechanism that selects long or short patch repair is not well understood.

Long Patch Repair

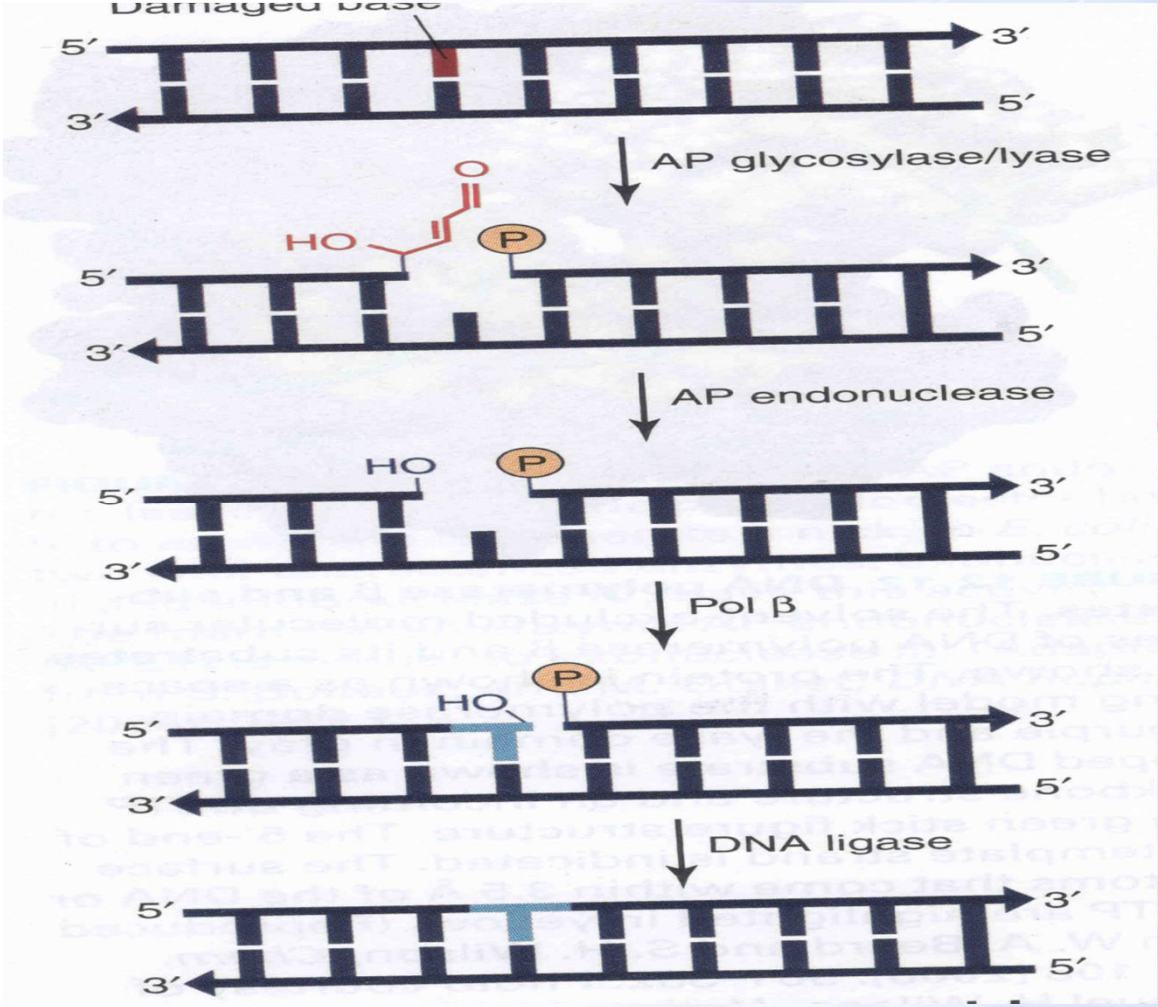




Long Patch Repair

- When eukaryotic base excision repair begins with a DNA glycosylase/lyase, the pathway is shown below:-

Long Patch Repair



Long Patch Repair

- Three distinct DNA glycosylase/lyase enzymes have been identified in *E. coli*, three in *S. cerevisiae*, and six in human cells.

Long Patch Repair

- Virtually all oxidized bases are removed by bifunctional DNA glycosylases in mammals.
- DNA glycosylase/ lyase excises the damaged base and cleaves the DNA strand 3’ of the AP site.

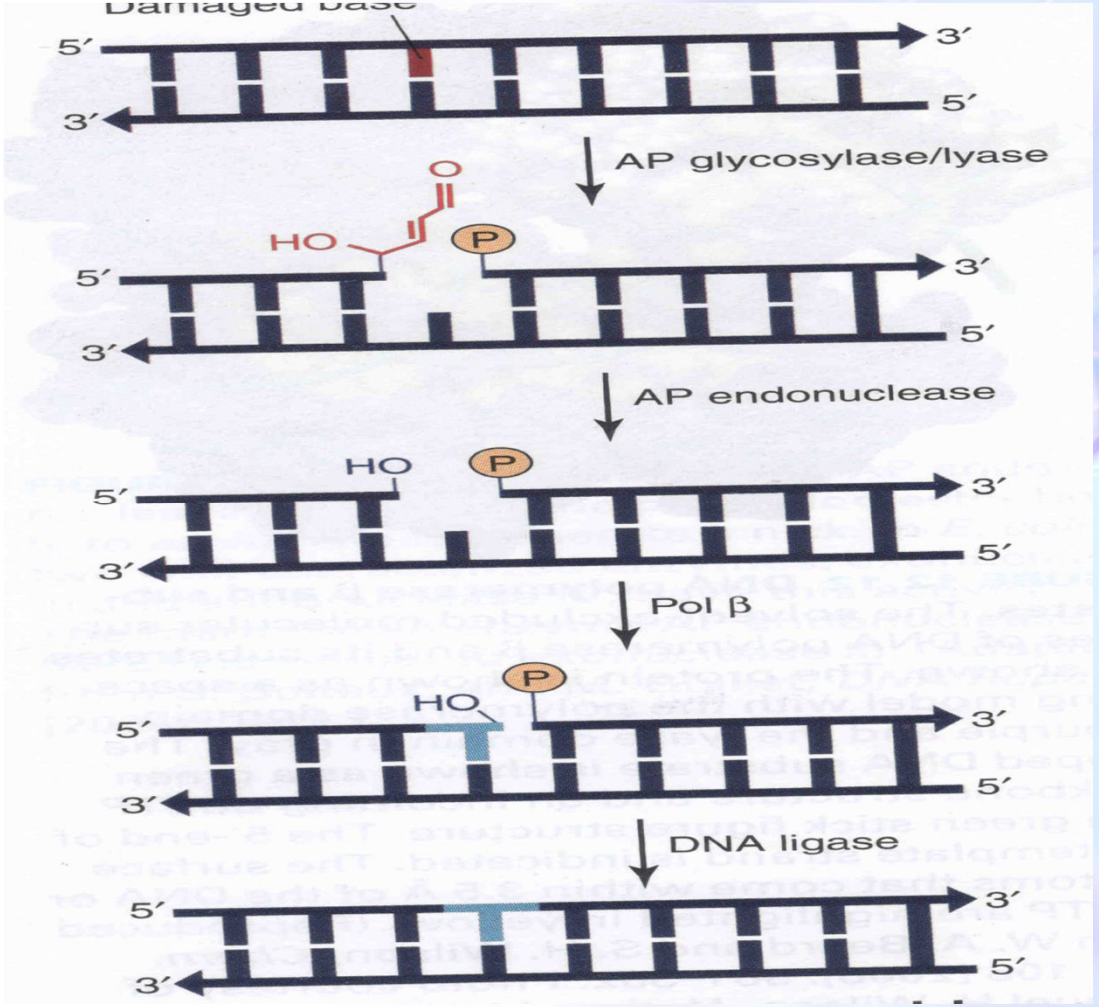
Long Patch Repair

- The resulting sugar residue at the 3’-end is removed by the AP endonuclease catalyzed cleavage.

Long Patch Repair

- Then DNA polymerase β adds a nucleotide and DNA ligase seals the remaining nick to complete short patch repair.

Long Patch Repair



Nucleotide Excision Repair

- Nucleotide excision repair (NER) pathway removes bulky adducts from DNA by excising an oligonucleotide bearing the lesion and replacing it with new DNA.

Nucleotide Excision Repair

- This repair mechanism excises UV-induced cyclobutane pyrimidine dimers, (6-4) photoproducts, damaged bases formed by alkylating agents and certain types of cross-links.

Nucleotide Excision Repair

- The efficiency of repair for different kinds of lesions can vary a lot.
- In general, there is a direct relationship between the amount of helical distortions produced by lesion and the efficiency of this repair.

Nucleotide Excision Repair

- The basic nucleotide excision repair pathway is same in all the organisms.
- It involves the following steps:-
- 1. damage recognition

Nucleotide Excision Repair

- 2. an incision in the damaged DNA strand on each side of the lesion,
- 3. excision of the oligonucleotide created by the incision,

Nucleotide Excision Repair

- 4. synthesis of new DNA to replace the excised DNA segment using the undamaged DNA strand as a template, and
- 5. ligation of the remaining nick.

Nucleotide Excision Repair

- Although the basic nucleotide excision repair pathway is similar in all the organisms, there are considerable differences in the proteins that carry out the various steps.

Nucleotide Excision Repair of UV-induced Damage

- UV-irradiated *E. coli* can regain their ability to survive after incubation in dark.
- However, they recover more slowly than when incubated in the light.

Nucleotide Excision Repair of UV-induced Damage

- This observation suggests that the bacteria use some process other than photoreactivation to repair the UV-induced DNA damage.

Nucleotide Excision Repair of UV-induced Damage

- Richard Setlow & William Carrier and, working independantly, Richard Boyce & Paul Howard-Flanders, used a similar approach to investigate this alternative process in 1964.

Nucleotide Excision Repair of UV-induced Damage

- Both groups cultured *E. coli* in the presence of [^{3}H]thymine to label the DNA and then irradiated the cells with UV light to induce the formation of cyclobutane thymine dimer.

Nucleotide Excision Repair of UV-induced Damage

- Then they:-
- 1) incubated the UV-irradiated cells in the dark so that the photoreactivation could not take place;
- 2) removed samples after various incubation times and added TCA to them;

Nucleotide Excision Repair of UV-induced Damage

- 3) separated acid-insoluble DNA from acid soluble nucleotides;
- 4) digested the DNA and oligonucleotides to release intact thymine cyclobutane dimers; and

Nucleotide Excision Repair of UV-induced Damage

- 5) detected the released dimers by chromatography.
- The experiments revealed that as the incubation time in the dark increases,

Nucleotide Excision Repair of UV-induced Damage

- Cyclobutane thymine dimers disappear from the acid-insoluble DNA and appear in the acid soluble oligonucleotide fraction.

Nucleotide Excision Repair of UV-induced Damage

- These results were correctly interpreted to mean that bacteria can excise an oligonucleotide containing a lesion and replace the excised oligonucleotides with newly synthesized DNA.

Nucleotide Excision Repair of UV-induced Damage

- Subsequent studies showed that eukaryotes and the archaea also have nucleotide excision repair pathways.

UvrA, UvrB, and UvrC Proteins

- Genetic studies revealed that three *E. coli* genes viz., *uvrA, uvrB*, and *uvrC*, code for proteins that are essential for damage recognition, incision, and excision.

UvrA, UvrB, and UvrC Proteins

- All three genes have been cloned and the proteins that they encode (UvrA, UvrB, and UvrC) have been purified and characterized.

UvrA, UvrB, and UvrC Proteins

- Under normal physiological conditions, *E. coli* has about 25 molecules of UvrA, 250 molecules of UvrB, and 10 molecules of UvrC.

UvrA, UvrB, and UvrC Proteins

- After UV damage of DNA, UvrA and UvrB levels increase ten- and four folds, but the UvrC level remains the same.

UvrA, UvrB, and UvrC Proteins

- Although UvrA, UvrB, and UvrC do not combine to form a stable ternary complex, the polypeptides nevertheless are said to be part of a UvrABC damage-specific endonuclease.

UvrA, UvrB, and UvrC Proteins

- The UvrABC damage-specific endonuclease is called as UvrABC endonuclease in short but some investigators preferably term them as UvrABC excinuclease.

UvrA, UvrB, and UvrC Proteins

- UvrABC excinuclease is termed so because the proteins participate in incision and excision reactions.

UvrA, UvrB, and UvrC Proteins

- The three polypeptides work in the order suggested by their names, that is their order of action is UvrA, UvrB, and the UvrC.

The NER Pathway

- The crystal structure of UvrA•DNA complex reveals that UvrA does not make direct contact with the modified thymine but does bind to DNA regions on either side of the lesion.

The NER Pathway

- Based on this information, it appears that UvrA makes an important contribution towards recognition of DNA damaged lesion.
- This UvrA•DNA complex is formed in vitro in the absence of UvrB.

The NER Pathway

- The recognition process appears to be more complicated in vivo where UvrA is a part of a $(UvrA)_2 \bullet UvrB$ complex.

The NER Pathway

- UvrB plays a central role in nucleotide excision repair by interacting with both the UvrA and UvrB, although not at the same time.
- UvrB has at least two catalytic sites that are essential for its function.

The NER Pathway

- UvrB combines with UvrA to form a $(UvrA)_2 \bullet UvrB$ complex either in solution or on DNA.
- Initially, this protein complex binds to DNA at some distance from the damaged site.

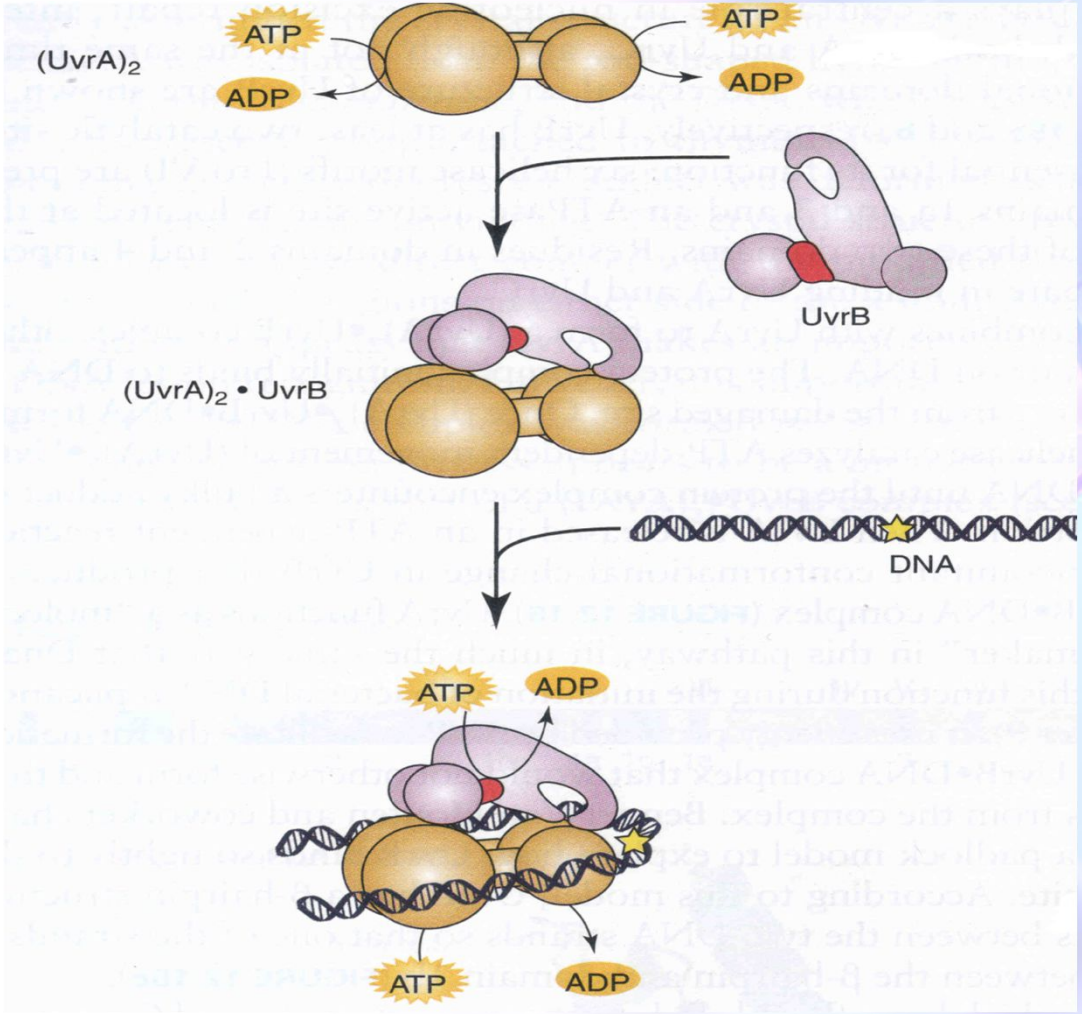
The NER Pathway

- Once $(UvrA)_2$•UvrB • DNA complex is formed, the UvrB helicase catalyzes ATP-dependant movement of $(UvrA)_2$•UvrB along the DNA until the protein complex encounters a bulky adduct or helix distortion.

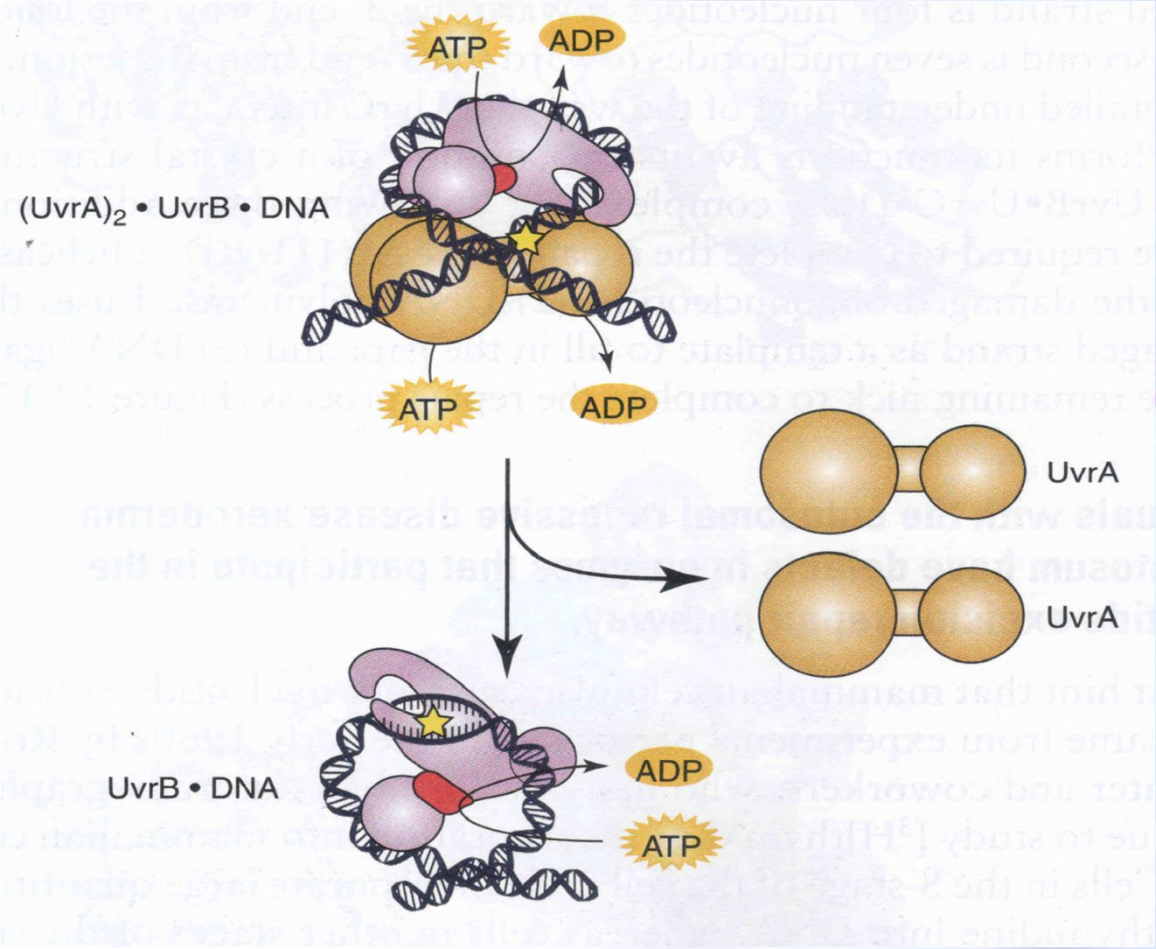
The NER Pathway

- The UvrA is released in an ATP-dependant reaction with a concomitant conformational change in the UvrB that produces a stable UvrB•DNA complex.

The NER Pathway



The NER Pathway



The NER Pathway

- UvrA functions as a “molecular match maker” in this pathway.
- It uses energy of ATP to facilitate the formation of UvrB•DNA complex and then dissociates from the complex.

The NER Pathway

- UvrC, which has a flexible linker that connects its N- and C-terminal domains, binds to the UvrB•DNA complex and makes two incisions, one on each side of the lesion.

The NER Pathway

- The first incision of the damaged strand is four nucleotides towards the 3’-end from the lesion and the second is seven nucleotides toward the 5’-end from the lesion.

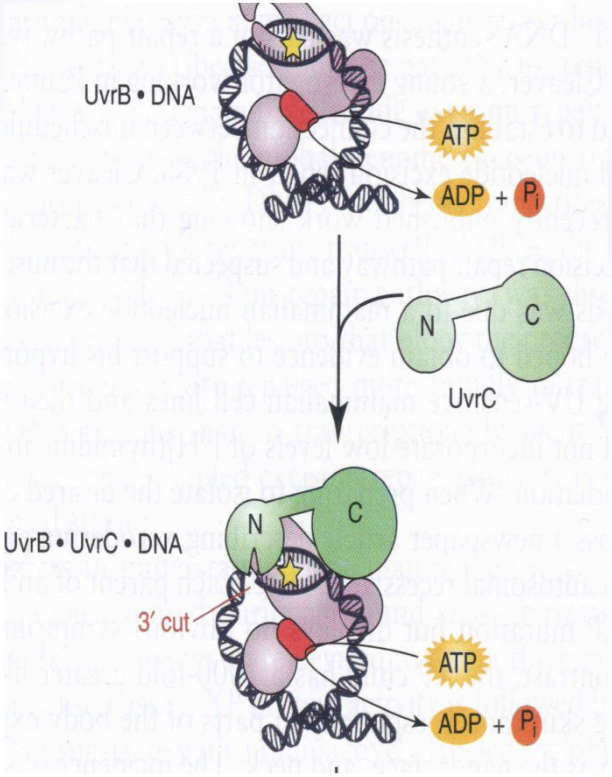
The NER Pathway

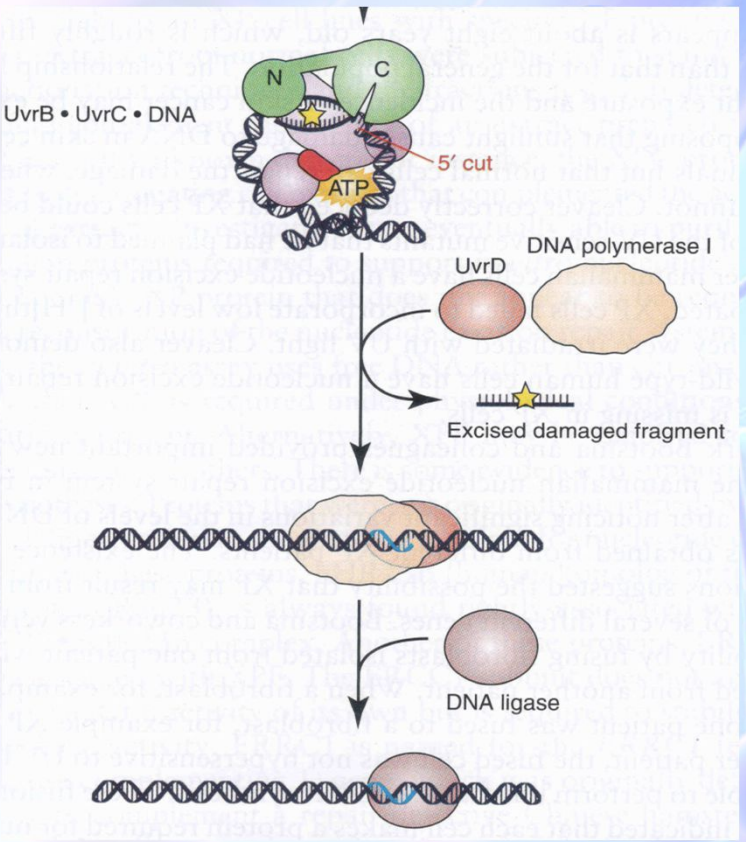
- How UvrC interacts with UvrB and performs its function is still to be described.
- Following three more steps are required to complete the repair process:-

The NER Pathway

- 1). UvrD, a helicase, excises the damaged oligonucleotide;
- 2). DNA polymerase I uses the undamaged strand as a template to fill the gap;
- 3). DNA ligase seals the remaining nick to complete the repair process.

The NER Pathway





Mismatch Repair

- The mismatch repair system corrects rare base pair mismatches and short insertions or deletions that appear in DNA following replication.

Mismatch Repair

- DNA polymerases introduce about one mispaired nucleotide per 10^5 nucleotides.
- However, the 3’ → 5’ proofreading exonuclease increases replication fidelity by 100- fold by removing mispaired nucleotides.

Mismatch Repair

- Although an error frequency of one nucleotide in 10^7 may seem extremely low, it would result in a high mutation rate.

Mismatch Repair

- The second type of error that occur during replication is the short insertions and deletions, which result from the fact that repeated-sequence motifs sometimes dissociate and then re-anneal incorrectly.

Mismatch Repair

- As a result, the newly synthesized strand will have a different number of repeats than the template strand.
- Introduction of an insertion or deletion into the newly synthesized DNA is likely to produce a mutation.

Mismatch Repair

- Cells with a non-functional mismatch repair system have a high rate of mutation due to their inability to efficiently repair base pair mismatches, short insertions or deletions that arise during replication.

Mismatch Repair System in *E. coli*

- Let us begin the examination of mismatch repair by considering the *E. coli* mismatch repair system because this system has been the most extensively studied.

Mismatch Repair System in *E. coli*

- Although this system provides valueable informations for studying mismatch repair in other organisms, it differs from the mismatch repair systems of gram-positive bacteria & eukaryotes in one important respect.

Mismatch Repair System in *E. coli*

- The *E. coli* mismatch repair system can distinguish a newly synthesized strand from a parental strand because only the latter has methyl groups attached to sites with the sequence GATC.

Mismatch Repair System in *E. coli*

- *E. coli* has a deoxy-adenosine methylase that transfers methyl groups from S-adenosylmethionine molecules to deoxyadenosines in GATC sequences.

Mismatch Repair System in *E. coli*

- The time of methylation by deoxyadenosine methylase, however, lags behind that of nucleotide addition at the replication fork about two minutes, so the newly synthesized strand is transiently unmethylated.

Mismatch Repair System in *E. coli*

- The *E. coli* mismatch repair system exploits this period of transient unmethylation to identify and cut GATC sites in a newly synthesized strand with a mismatch.

MutS, MutL, & MutH Proteins

- Genetic and biochemical studies have demonstrated that three *E. coli* proteins viz., MutS, MutL, and MutH are dedicated to mismatch repair.

MutS, MutL, & MutH Proteins

- Although these proteins are essential for mismatch repair, they are not sufficient.
- Several additional enzymes and protein factors also make important contributions.

MutS, MutL, & MutH Proteins

- Among these enzymes and protein factors are:-
- i. DNA helicase II (UvrD),
- ii. Single-stranded DNA binding protein (SSB)
- iii. 5’ $\longrightarrow$ 3’ exonucleases (ExoVII or RecJ)

MutS, MutL, & MutH Proteins

- iv. 3’ → 5’ exonucleases (ExoI, ExoVII, or ExoX),
- v. DNA polymerase III holoenzyme,
- vi. DNA ligase, and
- vii. Deoxyadenosine methylase.

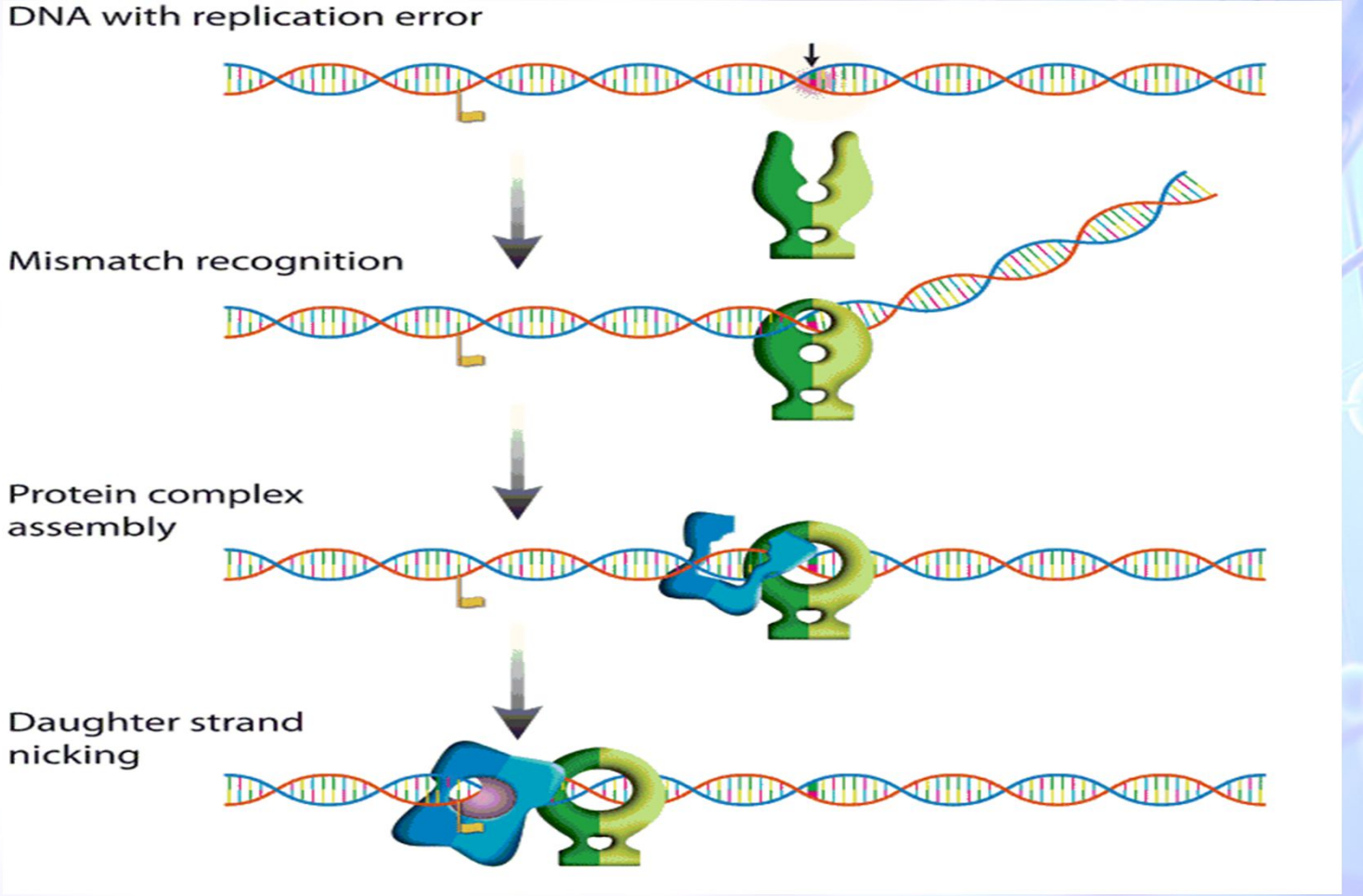
MutS, MutL, & MutH Proteins

- The process of repair begins when MutS (a homodimer or homotatramer) binds to the mismatch.
- MutS recruits MutL (a homodimer) in an ATP-dependant fashion.

MutS, MutL, & MutH Proteins

- Then the MutS•MutL complex activates MutH, which makes an incision at the nearest unmethylated GATC site, either 5’ or 3’ to the mismatch, in the newly synthesized strand.

MutS, MutL, & MutH Proteins



MutS, MutL, & MutH Proteins

- MutH shares sequence homology with the type II restriction endonuclease, Sau3AI.
- Both enzymes recognize and cleave GATC sequences.

MutS, MutL, & MutH Proteins

- MutH does not bind or cleave fully methylated GATC sites, whereas Sau3AI cleaves fully, hemi and unmethylated GATC sites.

MutS, MutL, & MutH Proteins

- Next in the process is that helicaseII (UvrD) unwinds the DNA and SSB binds to the resulting single strands.

MutS, MutL, & MutH Proteins

- When the incision is 5’ to the mismatch, ExoVII or RecJ exonucleases hydrolyze the nicked strand in a 5’ $\longrightarrow$ 3’ direction.

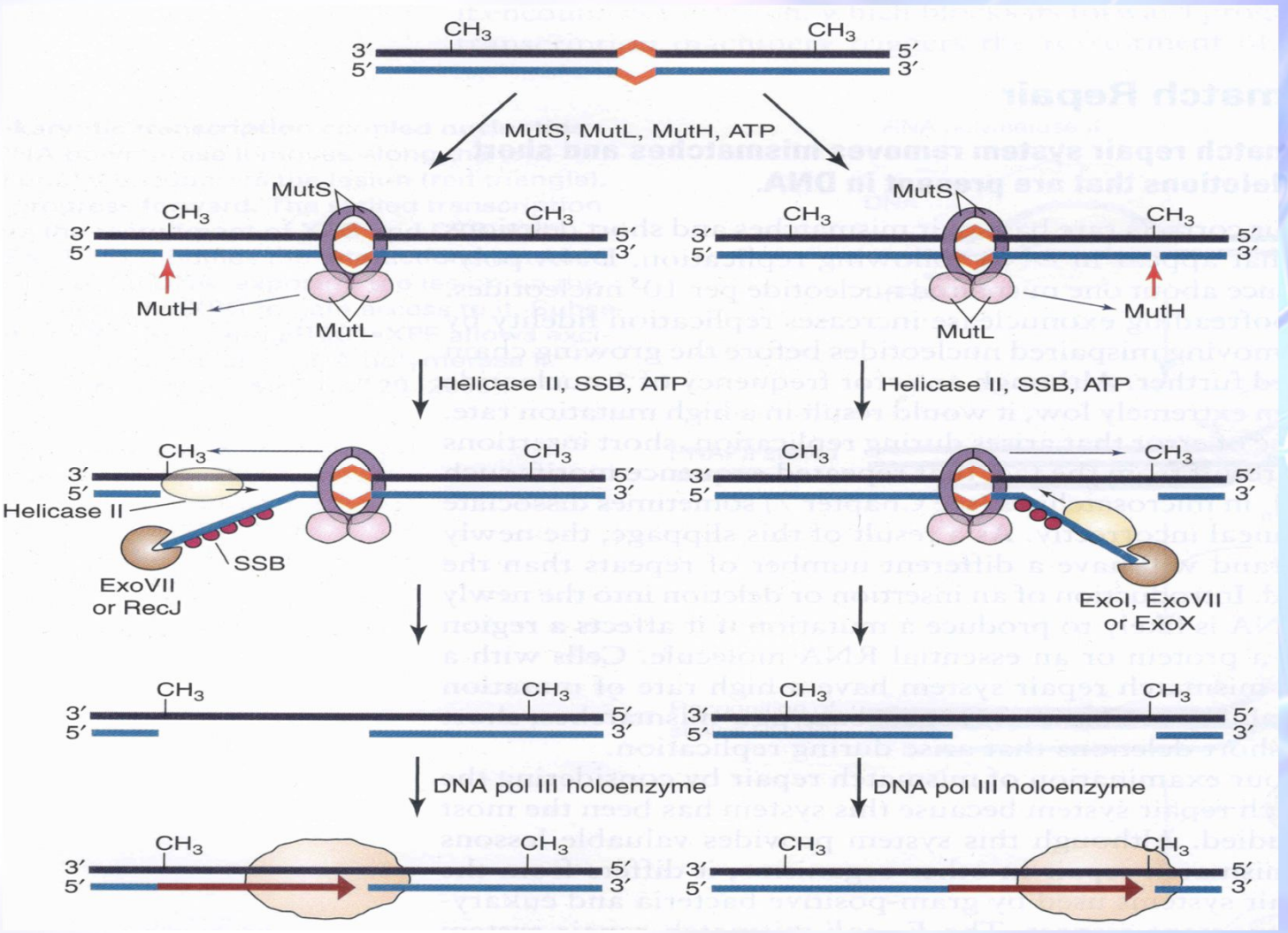
MutS, MutL, & MutH Proteins

- When the incision is 3’ to the mismatch, ExoI, ExoVII or ExoX exonucleases hydrolyze the nicked strand in a 3’ $\longrightarrow$ 5’ direction.

MutS, MutL, & MutH Proteins

- DNA polymerase III holoenzyme fills the gap with new DNA.
- DNA ligase seals the remaining nick and deoxyadenosine methylase adds a methyl group to the GATC site.

MutS, MutL, & MutH Proteins



MutS, MutL, & MutH Proteins

- All organisms that have a mismatch repair system have MutS and MutL homologs.
- However, MutH is present only in *E. coli* and some other gram-negative bacteria.

Mismatch Repair in Eukaryotes

- Eukaryotes have proteins that are homologous to MutS and MutL but lack homologs to MutH.

Mismatch Repair in Eukaryotes

- There are three human MutS homologs designated as MSH2, MSH3, and MSH6 which participate in mismatch repair.

Mismatch Repair in Eukaryotes

- MSH2 and MSH6 combine to form a hetrodimer called MutSα, and MSH2 and MSH3 combine to form a second heterodimer called MutSβ.

Mismatch Repair in Eukaryotes

- The structures of MutSα and MutSβ are thought to be similar to the MutS homodimer in bacteria.

Mismatch Repair in Eukaryotes

- MutSα initiates mismatch repair at single mismatches and small insertions/deletion loops, whereas MutSβ only initiates mismatch repair at insertion/deletion loops of various sizes.

Mismatch Repair in Eukaryotes

- Mammalian homologs of the bacterial MutL protein that participate in mismatch repair are designated as MLH1 and PMS2 and the heterodimer containing these two subunits is called MutLα.

Mismatch Repair in Eukaryotes

- Paul Modrich and coworkers have reconstituted the human mismatch repair system *in vitro*.
- A strand break on either side of the mismatch is sufficient to direct repair.

Human Mismatch Repair System

- Paul Modrich and coworkers have reconstituted the human mismatch repair system *in vitro*.
- A strand break on either side of the mismatch is sufficient to direct repair.

Human Mismatch Repair System

- Purified human MutSα, MutLα, ExoI (a 5’ → 3’ exonuclease), RPA, PCNA, RFC, and DNA ploymeraseδ are required for bidirectional repair.

Human Mismatch Repair System

- MutSα, ExoI, and RPA are adequate to excise a mismatch when the nick is on the 5’ side of the mismatch but MutLα, PCNA, and RFC are also required when the nick is on the 3’ side of the mismatch.

Human Mismatch Repair System

- The observation that the mismatch repair system can degrade newly synthesized strands with a nick on the 3'-side of the mismatch was very puzzling because ExoI degrades DNA in a 5' → 3' direction.

Human Mismatch Repair System

- Modrich and co-workers solved the puzzle by demonstrating that MutLα is a latent endonuclease that is activated in a mismatch.

Human Mismatch Repair System

- Once activated, MutLα preferentially makes incisions in the strand that already has a nick, that is, the discontinuous strand during replication.

Human Mismatch Repair System

- The endonuclease activity appears to require an amino acid motif present in the PMS2 but not the MLH1 subunit.

Human Mismatch Repair System

- In the human mismatch repair pathway, MutSα, PCNA, and RFC cooperate to activate the latent MutLα endonuclease.

Human Mismatch Repair System

- This MutLα endonuclease then nicks the discontinuous strand of a DNA duplex on both the 5’ and 3’ sides of the mismatch.

Human Mismatch Repair System

- When the original nick is on the 3’ side of the mismatch, MutLα incisions produce a new 5’ terminus on the far side of the mismatch that can serve as an entry site for MutSα-activated ExoI.

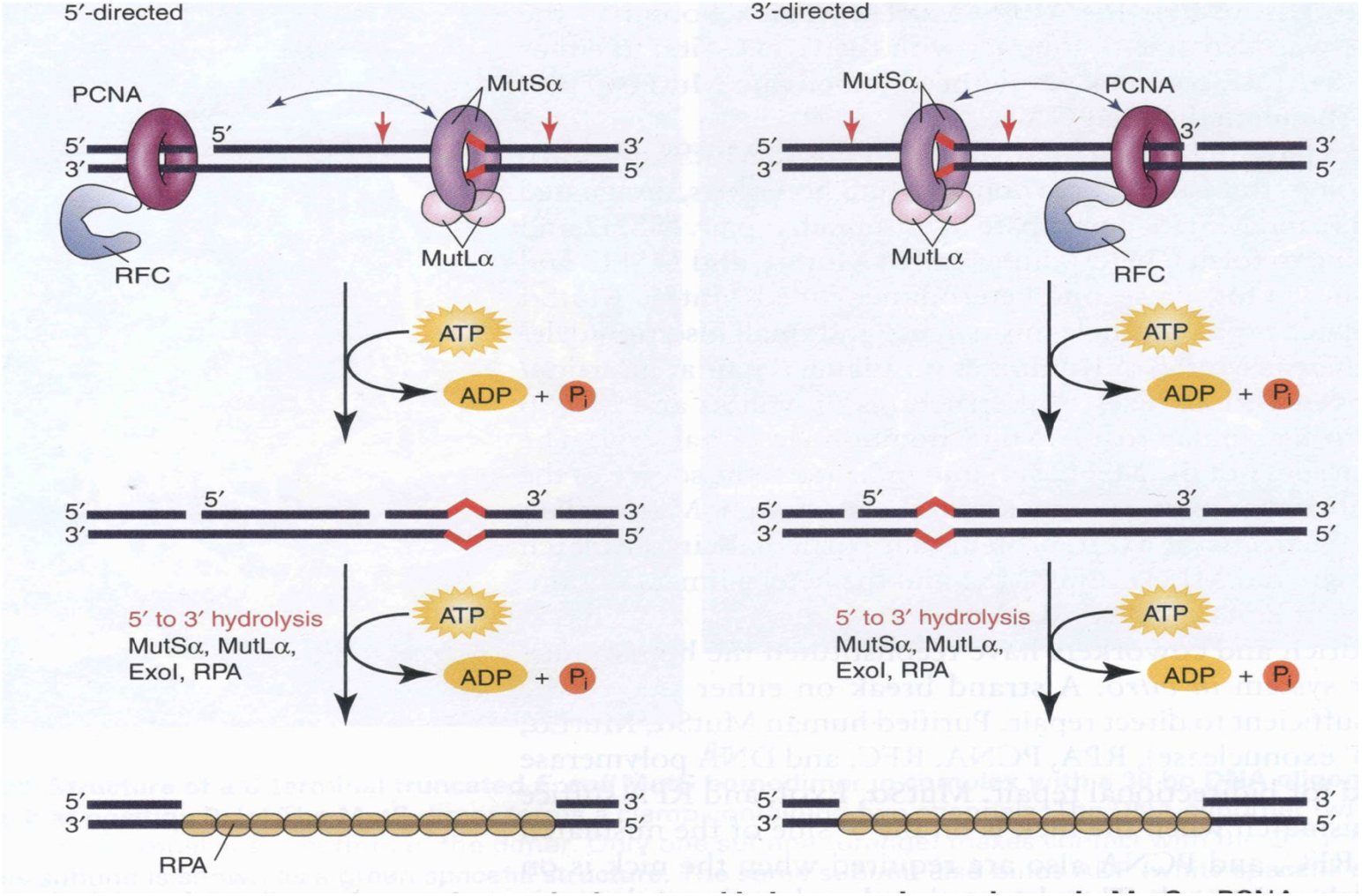
Human Mismatch Repair System

- This activated ExoI then removes the mismatch using its 5’ → 3’ exonuclease activity.
- RPA stimulates ExoI activity in the presence of MutSα as long as the mismatch is present.

Human Mismatch Repair System

- Once the mismatch has been removed, RPA inhibits the exonuclease, probably by displacing ExoI from the DNA.

Human Mismatch Repair System



Human Mismatch Repair System

- Individuals with a non-functional mismatch repair system due to defective MutSα or MutLα suffer from nonpolyposis colon cancer (HNPCC), an autosomal recessive syndrom.

end

The Central Dogma

- The central dogma outlines the flow of genetic information during growth and division of the cells.
- Genetic information flows from DNA to RNA to protein during cell growth.

The Central Dogma

- In addition, all living cells must replicate their DNA when they divide.
- During cell division each daughter cell receives a copy of the genome of the parent cell.

The Central Dogma

- Replication is the process by which two identical copies of DNA are made from an original molecule of DNA.
- So Replication occurs in the cells prior to cell division.

The Central Dogma

- An important point is that information does not flow from protein to RNA or DNA.
- However, flow of information from RNA “backwards” to DNA is possible in certain special circumstances due to the operation of reverse transcriptase.

The Central Dogma

- By the end of 1953, the working hypothesis was adopted that chromosomal DNA functions as the template for the synthesis of RNA molecules.

The Central Dogma

- These RNA molecules, the subsequently move to the cytoplasm, where they determine the arrangement of amino acids within proteins.

The Central Dogma

- In 1956, Francis Crick referred to this pathway for the flow of genetic information as the Central Dogma.

The Central Dogma

The Central Dogma

- An important point in the above equation is that the two arrows are unidirectional which means that RNA sequences are never determined by protein templates nor was DNA then imagined ever to be made on RNA templates.

The Central Dogma

- The idea that proteins never serve as templates for RNA has stood the test of time.
- However, RNA chains sometimes do act as templates for the synthesis of DNA chains of complementary sequence.

The Central Dogma

- Such reversals of the normal flow of information are very rare events compared with the enormous number of RNA molecules made on DNA templates.

The Central Dogma

- Thus, the central dogma as originally proclaimed more than 50 years ago still remains essentially valid.

end

The Adaptor Hypothesis of Crick

- At first it seemed simplest to believe that the RNA templates for protein synthesis were folded up to create cavities on their outer surfaces specific for the 20 different amino acids.

The Adaptor Hypothesis of Crick

- The cavities would be so shaped that only one given amino acid would fit, and in this way RNA would provide the information to order amino acids during protein synthesis.

The Adaptor Hypothesis of Crick

- By 1955, however, Crick became disenchanted with this conventional wisdom, arguing that it would never work.

The Adaptor Hypothesis of Crick

- In the first place, the specific chemical groups on the four bases of RNA (A, U, G, and C) should mostly interact with water-soluble groups.

The Adaptor Hypothesis of Crick

- Yet, the specific side groups of many amino acids (e.g., leucine, valine, and phenylalanine) strongly prefer interactions with water-insoluble (hydrophobic) groups.

The Adaptor Hypothesis of Crick

- In the second place, even if somehow RNA could be folded so as to display some hydrophobic surfaces, it seemed unlikely that

The Adaptor Hypothesis of Crick

- an RNA template would be used to discriminate accurately between chemically very similar amino acids like glycine and alanine or valine and isoleucine, both pairs differing only by the presence of single methyl groups.

The Adaptor Hypothesis of Crick

- Crick thus proposed that prior to incorporation into proteins, amino acids are first attached to specific adaptor molecules, which in turn possess unique surfaces that can bind specifically to bases on the RNA templates.

end

Discovery of Transfer RNA

- The discovery of how proteins are synthesized required the development of cell-free extracts capable of making proteins from amino acid precursors as directed by added RNA molecules.

Discovery of Transfer RNA

- These were first effectively developed in 1953 by Paul C. Zamecnik and his collaborators. Key to their success were the recently available radioactively tagged amino acids.

Discovery of Transfer RNA

- They used these labelled amino acids to mark the trace amounts of newly synthesized proteins.
- Early on, the cellular site of protein synthesis was pinpointed to be the ribosomes.

Discovery of Transfer RNA

- Several years later, Zamecnik, with his collaborator Mahlon B. Hoagland, made an important discovery that prior to their incorporation into proteins, amino acids are first attached to what we now call transfer RNA (tRNA) molecules.

Discovery of Transfer RNA

- Transfer RNA accounts for some 10% of all cellular RNA.
- To nearly everyone except Crick, this discovery was totally unexpected.

Discovery of Transfer RNA

- He had, of course, previously speculated that his proposed “adaptors” might be short RNA chains,

Discovery of Transfer RNA

- because their bases would be able to base-pair and “read” the appropriate groups on the RNA molecules that served as the templates for protein synthesis.

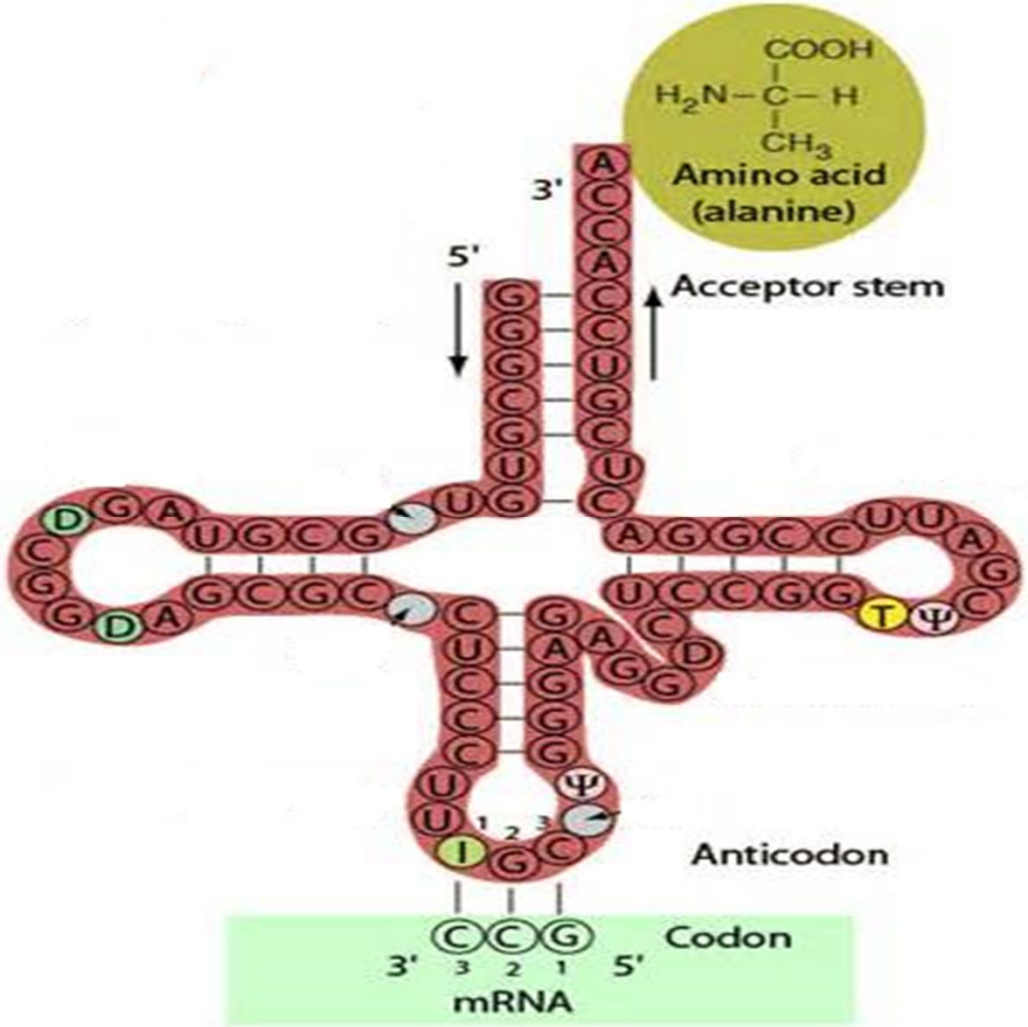
Discovery of Transfer RNA

- The transfer RNA molecules of Zamecnik and Hoagland are in fact the adaptor molecules postulated by Crick.

Discovery of Transfer RNA

- Each transfer RNA contains a sequence of adjacent bases (the anticodon) that bind specifically during protein synthesis to successive groups of bases (codons) along the RNA template.

Discovery of Transfer RNA



Discovery of Messenger RNA

- Cells infected with phage T4 provided the ideal system to find the true template for protein synthesis.
- Following infection by this virus, cells stop synthesizing *E. coli* RNA; the only RNA synthesized is transcribed off the T4 DNA.

Discovery of Messenger RNA

- Most strikingly, not only does T4 RNA have a base composition very similar to T4 DNA, but it does not bind to the ribosomal proteins that normally associate with rRNA to form ribosomes.

Discovery of Messenger RNA

- Instead, after first attaching to previously existing ribosomes, T4 RNA moves across their surface to bring its bases into positions where they can bind to the appropriate tRNA–amino acid precursors for protein synthesis.

Discovery of Messenger RNA

- In so acting, T4 RNA orders the amino acids and is thus the long-sought-for RNA template for protein synthesis.

Discovery of Messenger RNA

- Because it carries the information from DNA to the ribosomal sites of protein synthesis, it is called messenger RNA (mRNA).

Discovery of Messenger RNA

- The observation of T4 RNA binding to *E. coli* ribosomes, first made in 1960, was soon followed with evidence for a separate messenger class of RNA within uninfected *E. Coli* cells.

Discovery of Messenger RNA

- Instead, the rRNA components of ribosomes, together with some 50 different ribosomal proteins that bind to them, serve as the factories for protein synthesis.

Discovery of Messenger RNA

- They function to bring the tRNA–amino acid precursors into positions where they can read off the information provided by the mRNA templates.

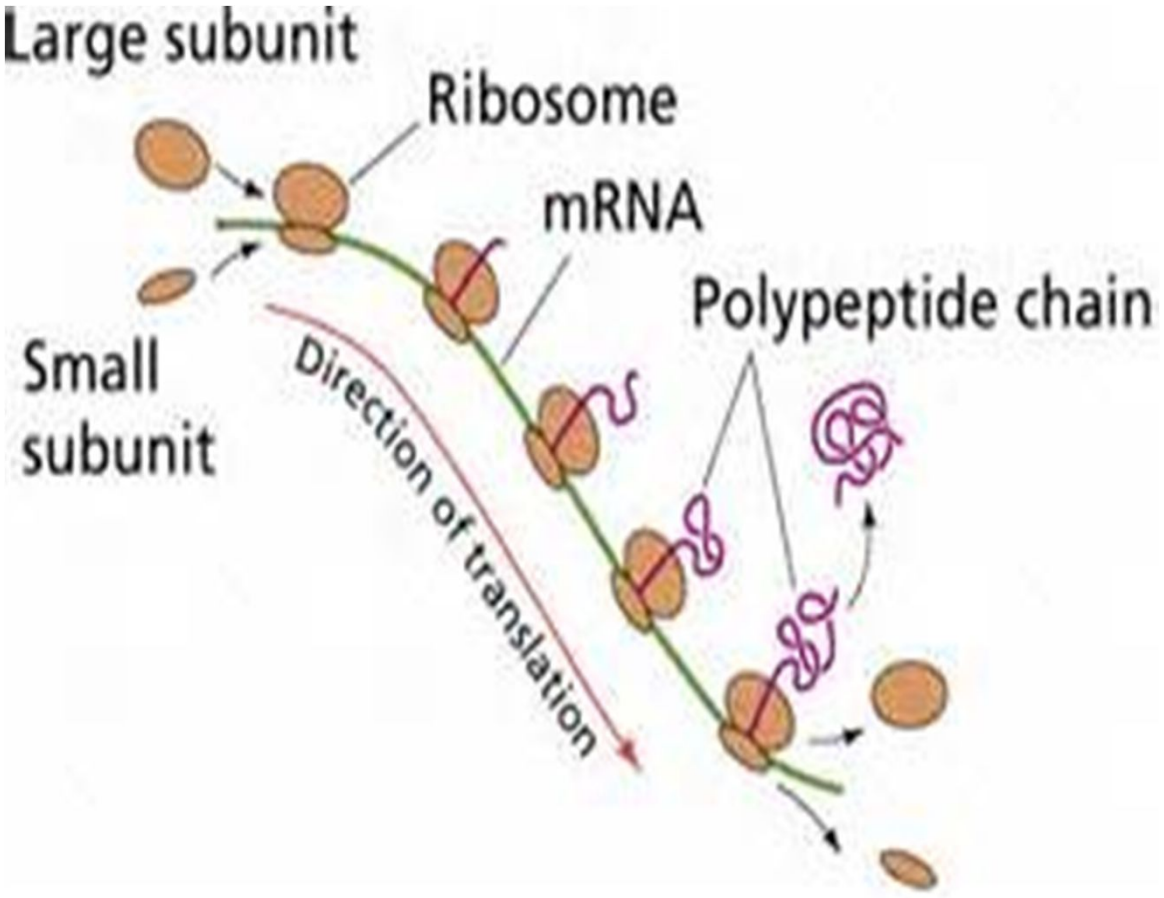
Discovery of Messenger RNA

- Only a few percent of total cellular RNA is mRNA.
- This RNA shows the expected large variations in length and nucleotide composition required to encode the many different proteins found in a given cell.

Discovery of Messenger RNA

- Hence, it is easy to understand why mRNA was first over looked. Because only a small segment of mRNAis attached at a given moment to a ribosome, a single mRNA molecule can simultaneously be read by several ribosomes.

Discovery of Messenger RNA



Synthesis of RNA upon DNA Templates

- As mRNA was being discovered, the first of the enzymes that synthesize (or transcribe) RNA using DNA templates was being independently isolated in the labs of biochemists Jerard Hurwitz and Samuel B. Weiss.

Synthesis of RNA upon DNA Templates

- These enzymes called RNA Polymerases function only in the presence of DNA, which serves as the template upon which single-stranded RNA chains are made, and use the nucleotides ATP, GTP, CTP, and UTP as precursors.

Synthesis of RNA upon DNA Templates

- These enzymes make RNA using appropriate segments of chromosomal DNA as their templates.

Synthesis of RNA upon DNA Templates

- Direct evidence that DNA lines up the correct ribonucleotide precursors came from seeing how the RNA base composition varied with the addition of DNA molecules of different AT/GC ratios.

Synthesis of RNA upon DNA Templates

- In every enzymatic synthesis, the RNA AU/GC ratio was roughly similar to the DNA's AT/GC ratio.
- During transcription, only one of the two strands of DNA is used as a template to make RNA.

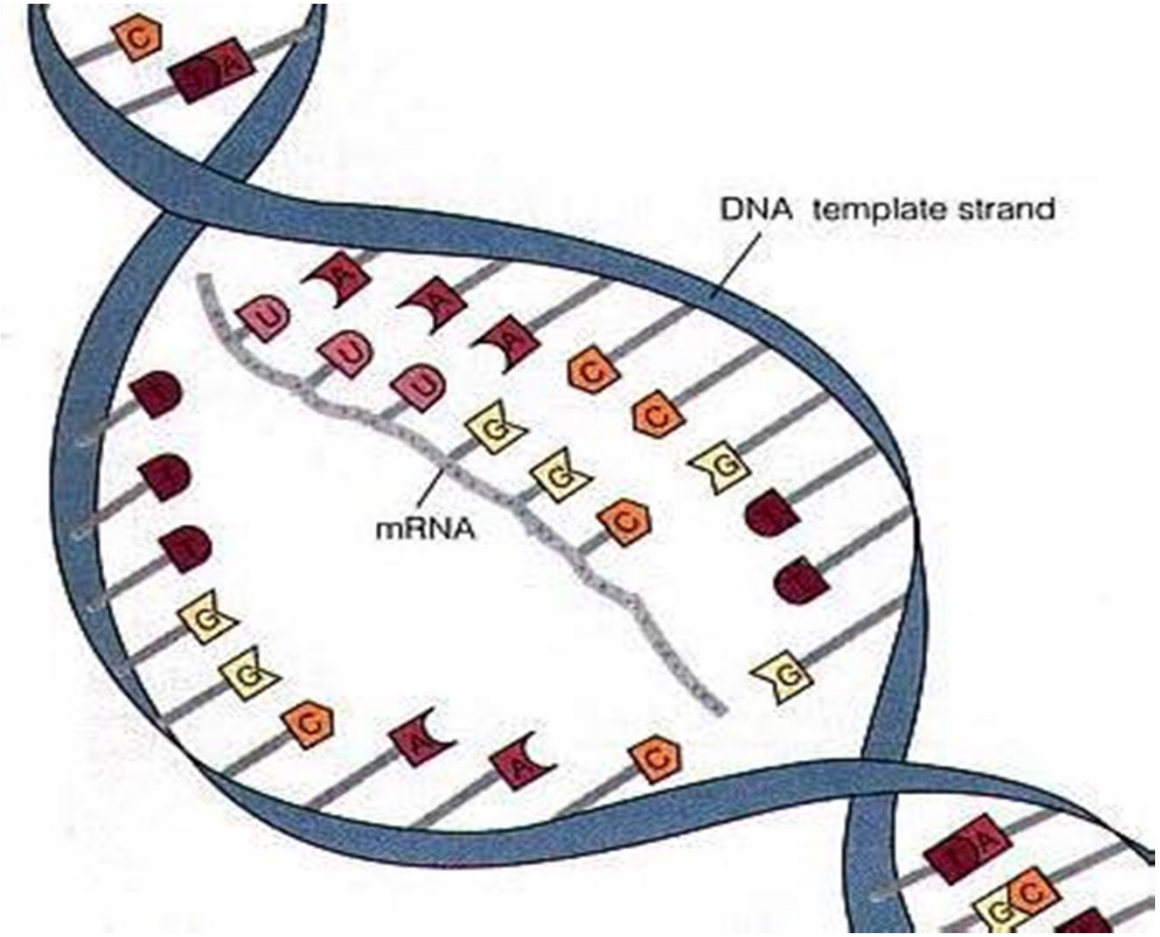
Synthesis of RNA upon DNA Templates

- This makes sense, because the messages carried by the two DNA strands, being complementary but not identical, are expected to code for completely different polypeptides.

Synthesis of RNA upon DNA Templates

- The synthesis of RNA always proceeds in a fixed direction, beginning at the 5’ end and concluding with the 3’-end nucleotide.

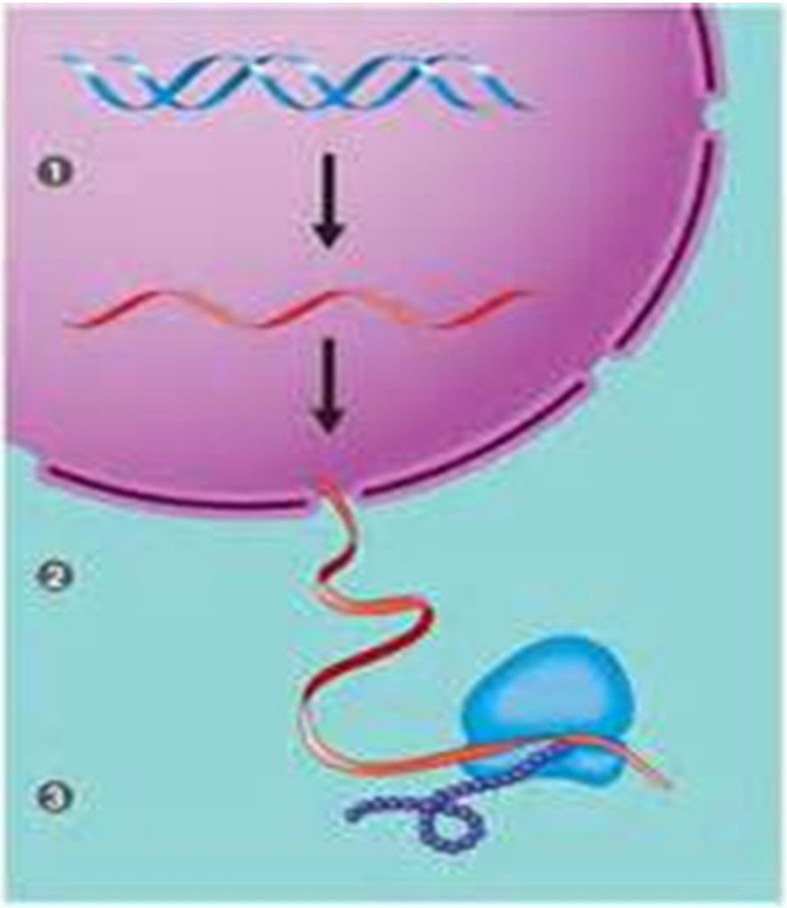
Synthesis of RNA upon DNA Templates



Synthesis of RNA upon DNA Templates

- By this time, there was firm evidence for the postulated movement of RNA from the DNA-containing nucleus to the ribosome-containing cytoplasm of eukaryotic cells.

Synthesis of RNA upon DNA Templates



Synthesis of RNA upon DNA Templates

- By briefly exposing cells to radioactively labelled precursors, then adding a large excess of unlabeled ribonucleotides (a “pulse chase” experiment), mRNA synthesized during a short time window was labelled.

Synthesis of RNA upon DNA Templates

- These studies showed that mRNA is synthesized in the nucleus.
- Within an hour, most of this RNA had left the nucleus and was observed in the cytoplasm.

end

Establishing the Genetic Code

- Given the existence of 20 amino acids but only four bases, groups of several nucleotides must somehow specify a given amino acid.
- Groups of two nucleotides, however, would specify only 16 (4x 4) amino acids.

Establishing the Genetic Code

- So from 1954, most attention was given to how triplets (groups of three) might work, even though they obviously would provide more permutations (4 x 4 x 4) than needed if each amino acid was specified by only a single triplet.

Establishing the Genetic Code

- It was assumed that successive groups of nucleotides along a DNA chain code for successive amino acids along a given polypeptide chain.

Establishing the Genetic Code

- An elegant mutational analysis on bacterial proteins, carried out in the early 1960s by Charles Yanofsky and Sydney Brenner, showed that co-linearity does in fact exist.

Establishing the Genetic Code

- Equally important were the genetic analyses by Brenner and Crick, which in 1961 first established that groups of three nucleotides are used to specify individual amino acids.

Establishing the Genetic Code

- But which specific groups of three bases (codons) determine which specific amino acids could only be learned by biochemical analysis.

Establishing the Genetic Code

- The major breakthrough came when Marshall Nirenberg and Heinrich Matthaei, then working together, observed in 1961 that the addition of the synthetic polynucleotide poly U (UUUUU . . .)

Establishing the Genetic Code

- to a cell-free system capable of making proteins leads to the synthesis of polypeptide chains containing only the amino acid phenylalanine.
- The nucleotide groups UUU thus must specify phenylalanine.

Establishing the Genetic Code

- Use of increasingly more complex polynucleotides as synthetic messenger RNAs rapidly led to the identification of more and more codons.

Establishing the Genetic Code

- Particularly important in completing the code was the use of polynucleotides like AGUAGU, put together by organic chemist Har Gobind Khorana.
- In this way, more specific sets of codons were tested.

Establishing the Genetic Code

- Completion of the code in 1966 revealed that 61 out of the 64 possible permuted groups corresponded to amino acids, with most amino acids being encoded by more than one nucleotide triplet.

Establishing the Genetic Code

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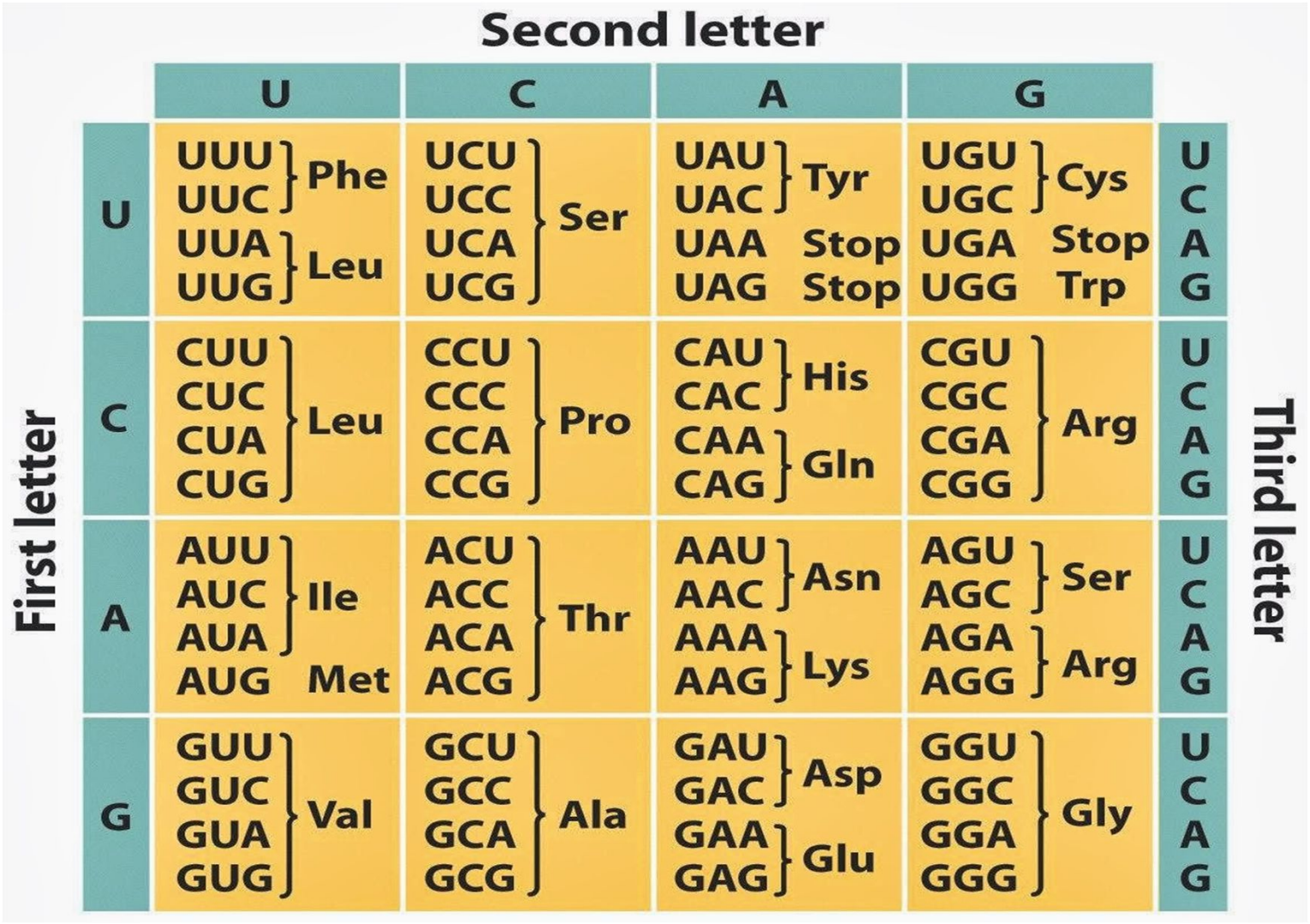
Establishing the Genetic Code



TRANSCRIPTION

- Up to this point, we have learnt that how the genome has been maintained i,e., how the genetic material is organized, protected, and replicated.

TRANSCRIPTION

- We now turn to the question of how that genetic material is expressed—that is, how the series of bases in the DNA directs the production of the RNAs and proteins that perform cellular functions and define cellular identity.

TRANSCRIPTION

- The basic processes responsible for gene expression include transcription, RNA processing, and translation.
- Transcription is chemically and enzymatically, very similar to DNA replication.

TRANSCRIPTION

- Both involve enzymes that synthesize a new strand of nucleic acid complementary to a DNA template strand.

TRANSCRIPTION

- However, the main difference is that in case of transcription, the new strand is made from ribonucleotides rather than deoxyribonucleotides.
- There are some other important differences include:-

TRANSCRIPTION

- RNA Polymerase (the enzyme that catalyzes RNA synthesis) does not need a primer; rather, it can initiate transcription de novo.
- The RNA product does not remain base-paired to the template DNA strand.

TRANSCRIPTION

- Transcription is less accurate than replication (one mistake/10,000 nucleotides, compared with one / 10 million). This difference reflects the lack of extensive proofreading mechanisms for transcription.

TRANSCRIPTION

- Transcription selectively copies only certain parts of the genome and makes any number of copies of that section. In contrast, replication must copy the entire genome and do so once (and only once) every cell division.

end

RNA POLYMERASES

- RNA polymerase performs essentially the same reaction in all cells, from bacteria to humans that is the synthesis of RNA.
- From bacteria to mammals, the cellular RNA polymerases are made up of multiple subunits.

RNA POLYMERASES

- Bacteria have only a single RNA polymerase, whereas eukaryotic cells have three: RNA polymerases I, II, and III (RNA Pol I, II, and III).

RNA POLYMERASES

- Recently, two more DNA-dependent RNA polymerases have been identified in recent years, and have been called as Pol IV and Pol V.
- These are found only in plants, where they transcribe small interfering RNAs.

RNA POLYMERASES

- The bacterial RNA polymerase core enzyme alone is capable of synthesizing RNA and comprises two copies of the α subunit and one each of the β, β' and ω subunits. This enzyme is closely related to the eukaryotic polymerases.

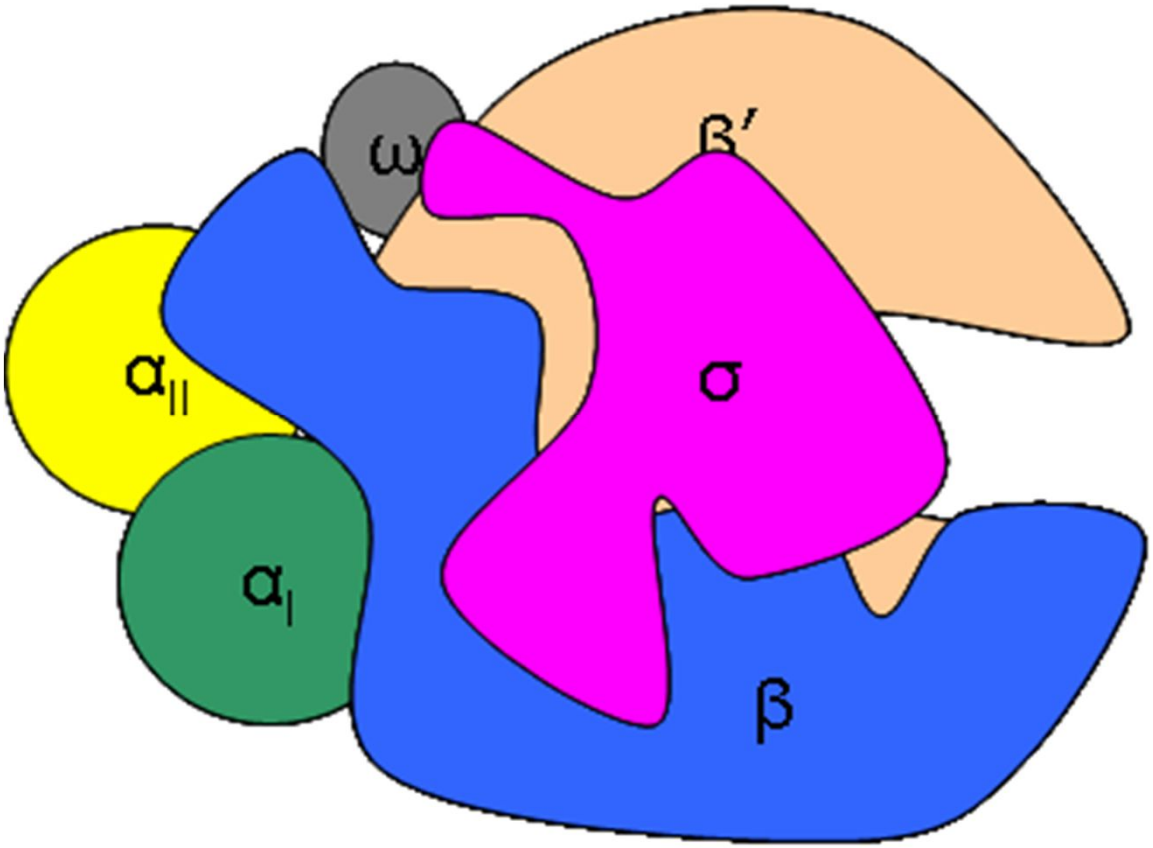
RNA POLYMERASES

- The structure of a bacterial RNA polymerase core enzyme is similar to that of the yeast Pol II enzyme.
- Overall, the shape of each enzyme resembles a crab claw.

RNA POLYMERASES

- This is reminiscent of the “hand” structure of DNA polymerases.
- The two pincers of the crab claw are made up predominantly of the two largest subunits of each enzyme (β and β’).

RNA POLYMERASES



RNA POLYMERASES

- The active site, which is made up of regions from both these subunits, is found at the base of the pincers within a region called the "active center cleft".

END

Transcription Proceeds in Series of Steps

- To transcribe a gene, RNA polymerase proceeds through a series of well defined steps grouped into three phases:
  - **Initiation**
  - **Elongation**
  - **Termination**

Transcription Proceeds in Series of Steps

- **Initiation**
- A promoter is the DNA sequence that initially binds to the RNA polymerase (together with any initiation factors required).

Transcription Proceeds in Series of Steps

- Once formed, the promoter–polymerase complex undergoes structural changes required for initiation to proceed.
- Then the DNA around the point where transcription will start unwinds.

Transcription Proceeds in Series of Steps

- The base pairs are disrupted, producing a "transcription bubble" of single-stranded DNA.
- Again, like DNA replication, transcription always occurs in a 5'-to-3' direction: the new ribonucleotide is added to the 3' end of the growing chain.

Transcription Proceeds in Series of Steps

- However, unlike replication, only one of the DNA strands acts as a template on which the RNA strand is built.
- The initiation can itself be broken down into a series of defined steps.

Transcription Proceeds in Series of Steps

- The first step is the initial binding of polymerase to a promoter to form what is called a closed complex.
- In this form, the DNA remains double-stranded, and the enzyme is bound to one face of the helix.

Transcription Proceeds in Series of Steps

- In the second step of initiation, the closed complex undergoes a transition to the open complex in which the DNA strands separate over a distance of 13 bp around the start site to form the transcription bubble.

Transcription Proceeds in Series of Steps

- In the next stage of initiation, polymerase enters the phase of initial transcription followed by promoter escape.
- The opening up of the DNA frees the template strand.

Transcription Proceeds in Series of Steps

- The first two ribonucleotides are brought into the active site, aligned on the template strand, and joined together.
- In the same way, subsequent ribonucleotides are incorporated into the growing RNA chain.

Transcription Proceeds in Series of Steps

- Incorporation of the first 10 or so ribonucleotides is a rather inefficient process, and at that stage, the enzyme often releases short transcripts (each of less than 10 or so nucleotides) and then begins synthesis again.

Transcription Proceeds in Series of Steps

- In this phase, the polymerase–promoter complex is called the initial transcribing complex.
- Once an enzyme makes a transcript longer than 10 nucleotides, it is said to have escaped the promoter.

Transcription Proceeds in Series of Steps

- At this point, it has formed a stable ternary complex, containing enzyme, DNA, and RNA.
- This is the transition to the elongation phase.

END

Transcription Proceeds in Series of Steps

- **Elongation**
- Once the RNA polymerase has synthesized a short stretch of RNA (10 bases), it shifts into the elongation phase.

Transcription Proceeds in Series of Steps

- During elongation, the enzyme performs an impressive range of tasks in addition to the catalysis of RNA synthesis.
- 1) It unwinds the DNA in front and re-anneals it behind.

Transcription Proceeds in Series of Steps

- 2) It dissociates the growing RNA chain from the template as it moves along.
- 3) And it performs the proofreading functions.

Transcription Proceeds in Series of Steps

- Recall that during replication, in contrast, several different enzymes are required to catalyze a similar range of functions.

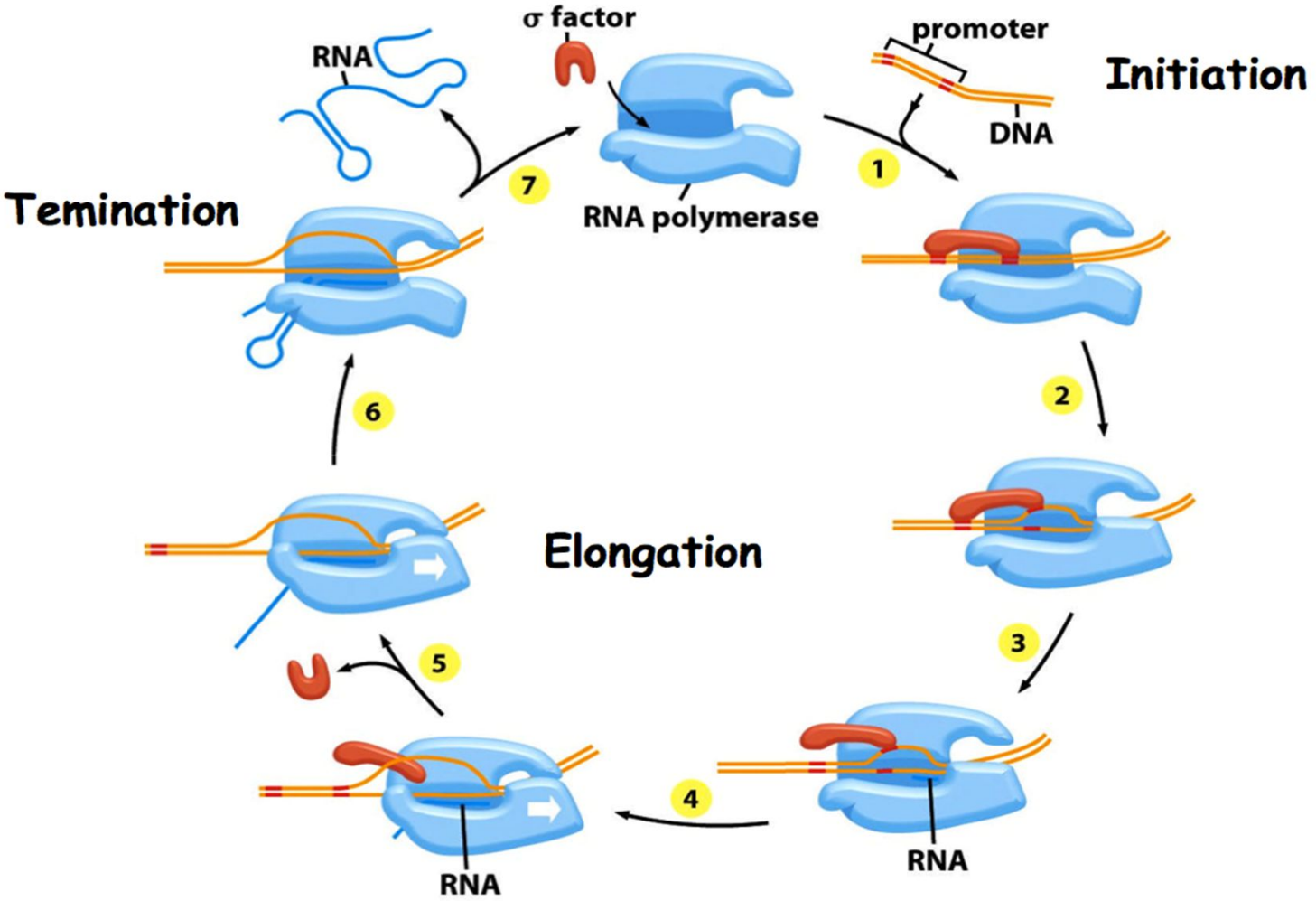
Transcription Proceeds in Series of Steps

- **Termination**
- Once the polymerase has transcribed the length of the gene (or genes), it must stop and release the RNA product (as well as dissociating from the DNA itself). This step is called termination.

Transcription Proceeds in Series of Steps

- In some cells, specific, well characterized sequences trigger termination. In others, it is less clear what instructs the enzyme to cease transcribing and dissociate from the template.

Transcription Proceeds in Series of Steps



Transcription Cycle in Bacteria

- The bacterial core RNA polymerase can, in principle, initiate transcription at any point on a DNA molecule.
- It, however, initiates transcription only at promoters.

Transcription Cycle in Bacteria

- It is the addition of an initiation factor called σ that converts core enzyme (α_2 ββ'ω) into the form that initiates only at promoters. This form of the enzyme is called the RNA polymerase holoenzyme.

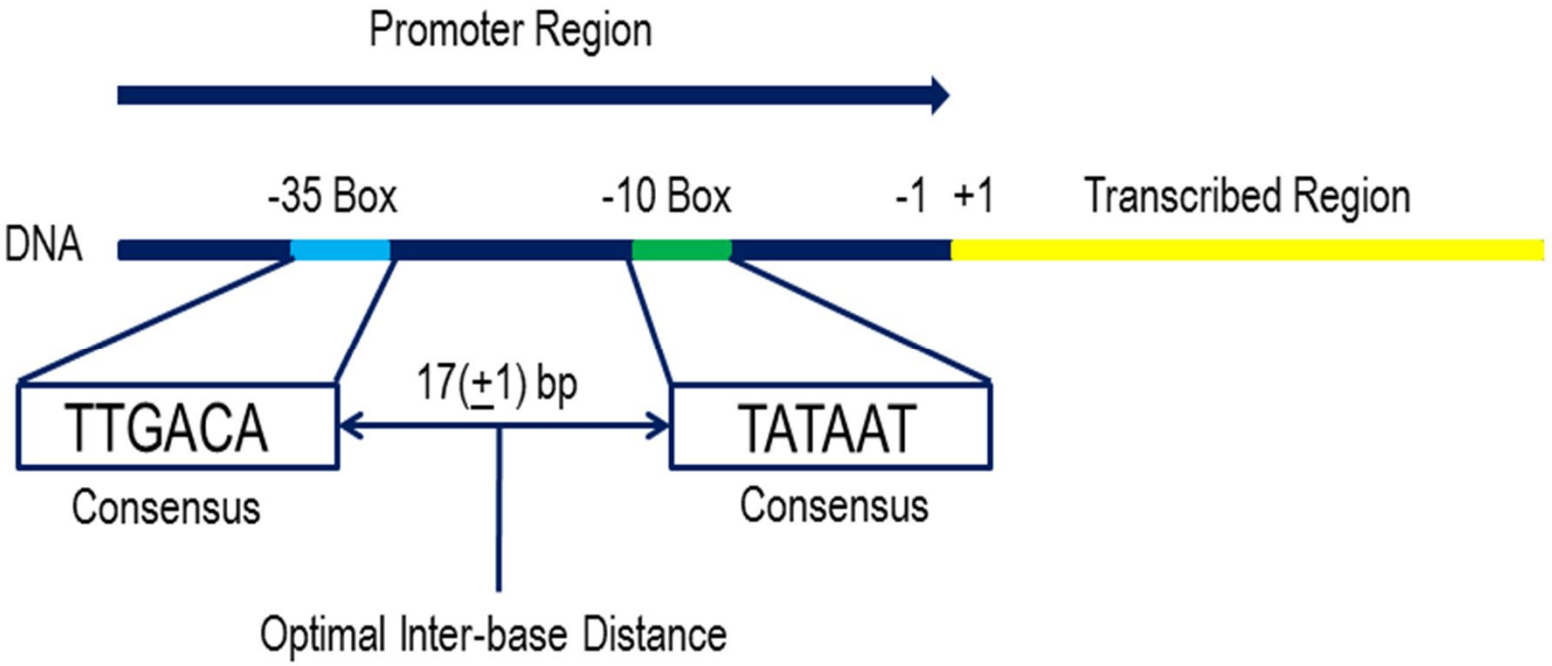
Transcription Cycle in Bacteria

- In the case of *Escherichia coli*, the predominant σ factor is called σ^{70}.
- Promoters recognized by polymerase have two conserved sequences each of 6 nucleotides, separated by a nonspecific stretch of 17–19 nucleotides.

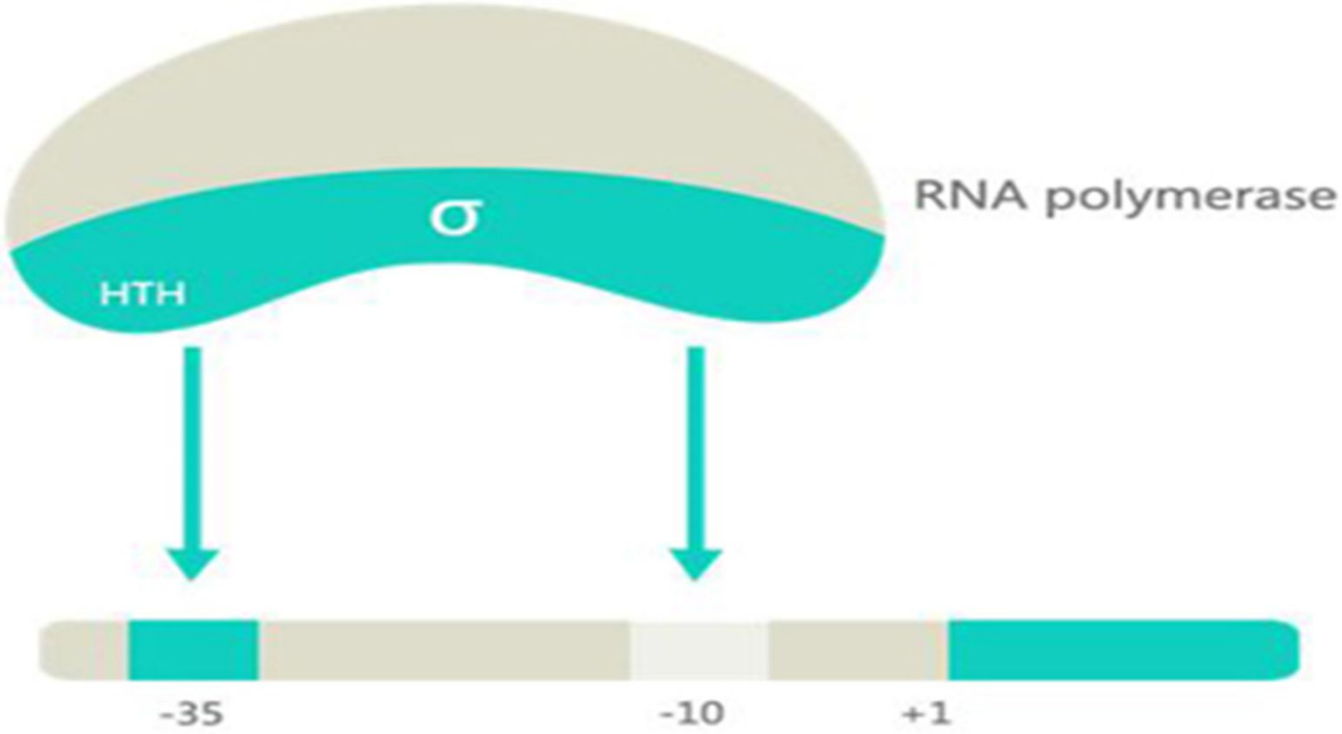
Transcription Cycle in Bacteria

- The two defined sequences are centered, respectively, at ~10 bp and at ~35 bp upstream of the site where RNA synthesis starts.
- The sequences are thus called the minus 35 and minus 10 regions or element.

Transcription Cycle in Bacteria



Transcription Cycle in Bacteria



Transcription Cycle in Bacteria

- The σ^{70} factor can be divided into four regions called σ region 1 through σ region 4.
- The regions that recognize the –10 and –35 elements of the promoter are regions 2 and 4, respectively.

Transcription Cycle in Bacteria

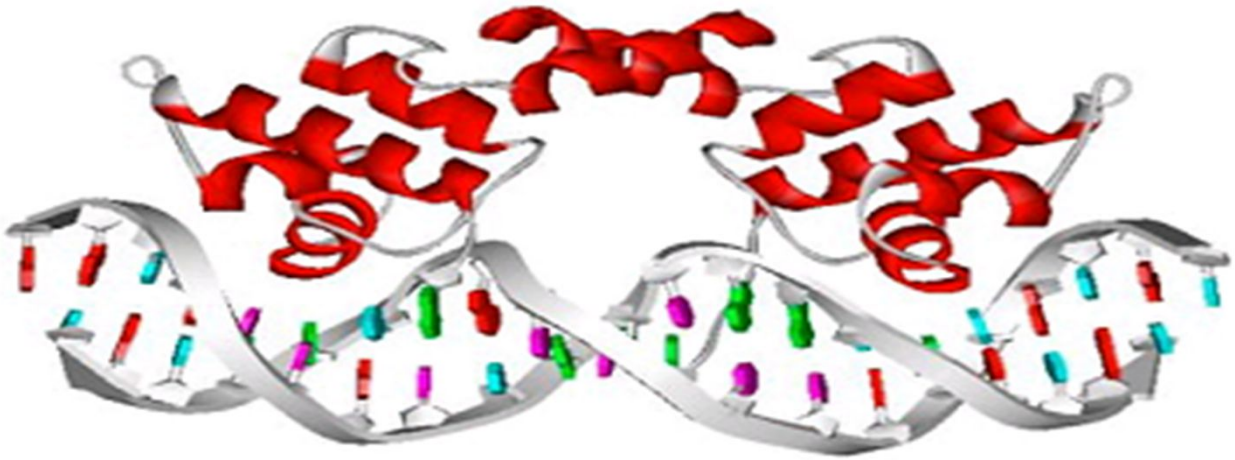
- Two helices within region 4 form a common DNA-binding motif called a helix-turn-helix.
- One of these helices inserts into the major groove and interacts with bases in the –35 region.

Transcription Cycle in Bacteria

- The other helix lies across the top of the groove, making contacts with the DNA backbone.

Transcription Cycle in Bacteria

Helix-turn-helix



Transcription Cycle in Bacteria

- The –10 region has a more elaborate role in transcription initiation, because it is within that element that DNA melting is initiated in the transition from the closed to the open complex.

Transcription Cycle in Bacteria

- Thus, the region of σ that interacts with the –10 region is doing more than simply binding DNA.

Transcription Cycle in Bacteria

- The a helix involved in recognition of the –10 region contains several essential aromatic amino acids that can interact with bases on the non-template strand in a manner that stabilizes the melted DNA.

Transcription Cycle in Bacteria

- Two bases in the non-template strand are flipped out and inserted into pockets within the σ protein where they make favorable contacts that stabilize the unwound state of the promoter region.

Transcription Cycle in Bacteria

- The σ subunit is positioned within the holoenzyme structure in such a way as to make feasible the recognition of various promoter elements.

Transcription Cycle in Bacteria

- Thus, the DNA-binding regions point away from the body of the enzyme, rather than being embedded.

END

Transition to the Open Complex

- Transition to the Open Complex involves structural changes in RNA Polymerase and in the Promoter DNA.
- The initial binding of RNA polymerase to the promoter DNA in the closed complex leaves the DNA in double-stranded form.

Transition to the Open Complex

- The next stage in initiation requires the enzyme to become more intimately engaged with the promoter, in the open complex.

Transition to the Open Complex

- The transition from the closed to the open complex involves structural changes in the enzyme and the opening of the DNA double helix to reveal the template and non-template strands.

Transition to the Open Complex

- This “melting” occurs between positions –11 and +2, with respect to the transcription start site.
- In case of the bacterial enzyme bearing σ70, this transition is often called isomerization.

Transition to the Open Complex

- This transition does not require energy derived from ATP hydrolysis and is instead the result of a spontaneous conformational change in the DNA–enzyme complex to a more energetically favourable form.

Transition to the Open Complex

- Two bases in the non-template strand of the –10 element flip out from their base-stacking interactions and instead insert into pockets within the σ protein where they make more favourable interactions.

Transition to the Open Complex

- By stabilizing the single-stranded form of the –10 element, these interactions drive melting of the promoter region.

Transition to the Open Complex

- Isomerization is essentially irreversible and, once complete, typically guarantees that transcription will subsequently initiate.
- Formation of the closed complex, in contrast, is readily reversible.

Transition to the Open Complex

- The examination of the structure of the holoenzyme in more detail reveals the active site of the enzyme, which is made up of regions from both the β and β' subunits, is found at the base of the pincers within the active center

END

Transition to the Open Complex

- There are five channels into the enzyme, as shown in the illustration of the open complex.
- The NTP-uptake channel allows ribonucleotides to enter the active center.

Transition to the Open Complex

- The RNA-exit channel allows the growing RNA chain to leave the enzyme as it is synthesized during elongation.
- The remaining three channels allow DNA entry and exit from the enzyme.

Transition to the Open Complex

- The downstream DNA (i.e., DNA ahead of the enzyme, yet to be transcribed) enters the active center cleft in double-stranded form through the downstream DNA channel (between the pincers).

Transition to the Open Complex

- Within the active center cleft, the DNA strands separate from position +3.

Transition to the Open Complex

- The non-template strand exits the active center cleft through the non-template-strand (NT) channel and travels across the surface of the enzyme.

Transition to the Open Complex

- The template strand, in contrast, follows a path through the active center cleft and exits through the template-strand (T) channel.
- The double helix re-forms at –11 in the upstream DNA behind the enzyme.

Transition to the Open Complex

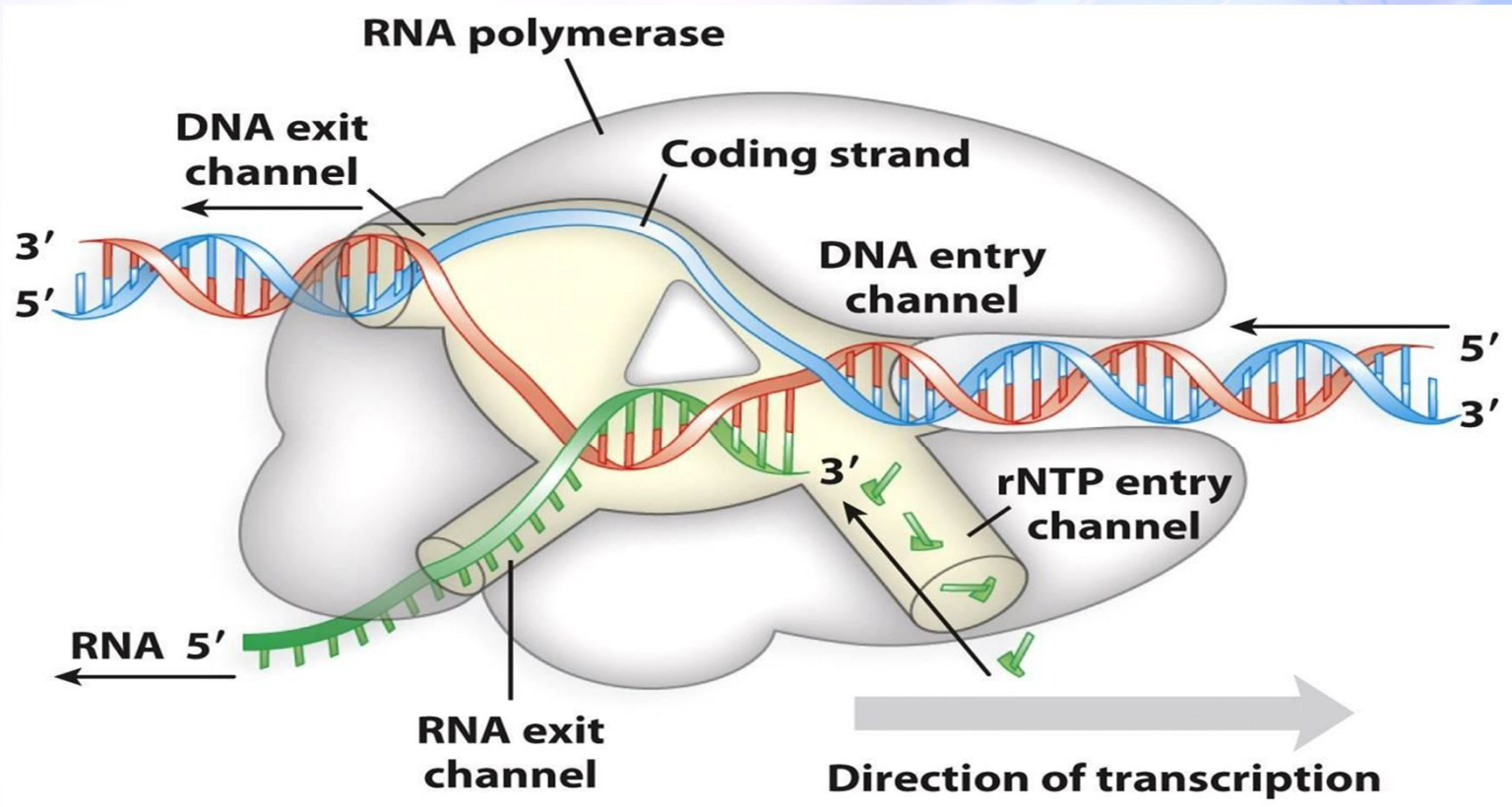


Figure 15-14
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Transition to the Open Complex

- Two striking structural changes are seen in the enzyme upon isomerization from the closed to the open complex.

Transition to the Open Complex

- First, the pincers at the front of the enzyme clamp down tightly on the downstream DNA.
- Second, there is a major shift in the position of the amino-terminal region of σ.

Transition to the Open Complex

- When not bound to DNA, σ region 1.1 lies within the active center cleft of the holoenzyme, blocking the path that, in the open complex, is followed by the template DNA strand.

Transition to the Open Complex

- In the open complex, region 1.1 shifts some 50A˚ and is now found on the outside of the enzyme, allowing the DNA access to the cleft.

Transition to the Open Complex

- Region 1.1 of σ is highly negatively charged (just like DNA). Thus, in the holoenzyme, region 1.1 acts as a molecular mimic of DNA.

Transition to the Open Complex

- The space in the active center cleft, which may be occupied either by region 1.1 or by DNA, is highly positively charged.

END

Initial Transcription

- RNA polymerase can initiate a new RNA chain on a DNA template and thus does not need a primer.

Initial Transcription

- This requires that the DNA template be brought into the polymerase active site and held stably in a helical conformation.

Initial Transcription

- And that the initiating ribonucleotide be brought into the active site and held stably on the template while the next NTP is presented with correct geometry for the polymerization to take place.

Initial Transcription

- This is particularly difficult because RNA polymerase starts most transcripts with an A, and that ribonucleotide binds the template nucleotide (T) with only two hydrogen bonds.

Initial Transcription

- Thus, the enzyme has to make specific interactions with the DNA template strand, the initiating ribonucleotide, and the second ribonucleotide to hold them all rigidly in the correct orientation to allow chemical attack by the incoming NTP.

Initial Transcription

- This is the reason that most transcripts start with the same nucleotide.

Initial Transcription

- During initial transcription, RNA polymerase produces and releases short RNA transcripts of <10 nucleotides (abortive synthesis) before escaping the promoter, entering the elongation phase, and synthesizing the proper transcript.

Initial Transcription

- It has long been unclear how the enzyme's active site translocates along the DNA template during initial abortive cycles of transcription.
- Three general models were proposed to explain this.

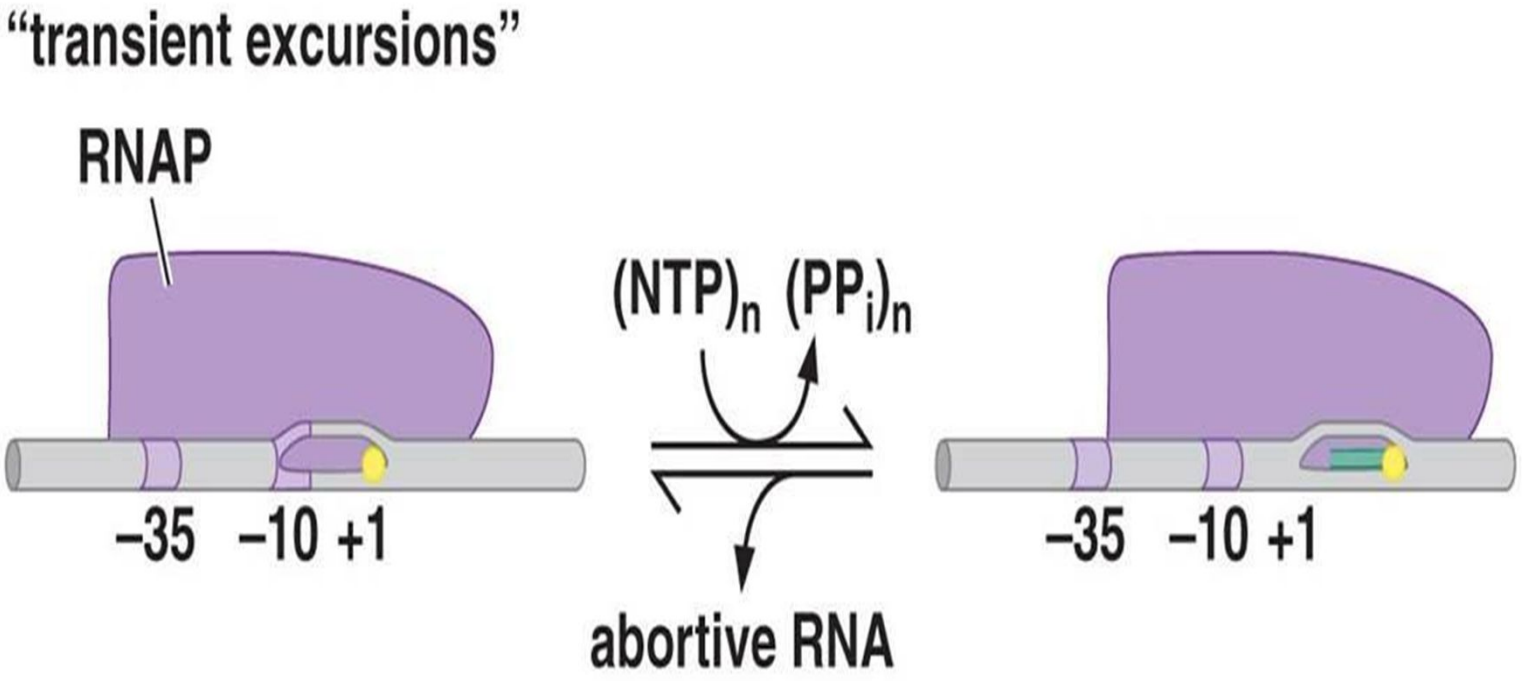
Initial Transcription

- 1. The "transient excursion" model proposes transient cycles of forward and reverse translocation of RNA polymerase.

Initial Transcription

- Thus, polymerase is thought to leave the promoter and translocate a short way along the DNA template, synthesizing a short transcript before aborting transcription, releasing the transcript, and returning to its original location on promoter.

Initial Transcription



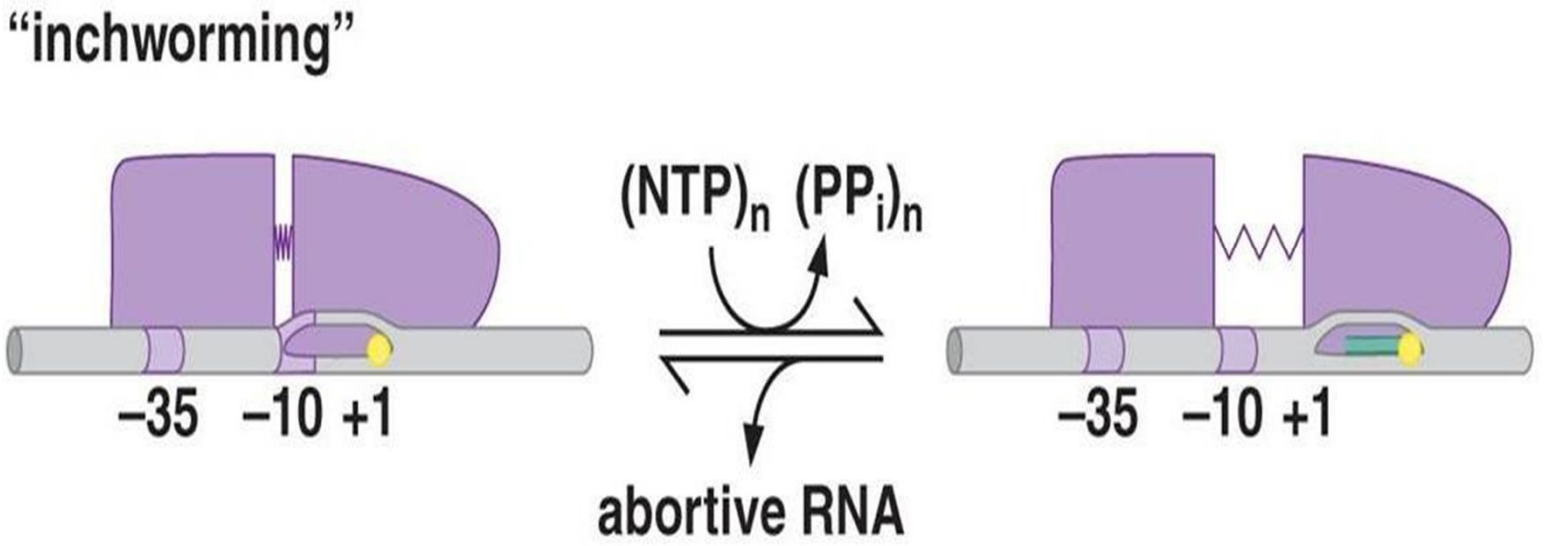
Initial Transcription

- 2. “Inchworming” invokes a flexible element within the polymerase that allows a module at the front of the enzyme, containing the active site.

Initial Transcription

- The module moves downstream, synthesizing a short transcript before aborting and retracting to the body of the enzyme still at the promoter.

Initial Transcription



Initial Transcription

- 3. "Scrunching" proposes that DNA downstream from the stationary, promoter-bound, polymerase is unwound and pulled into the enzyme.

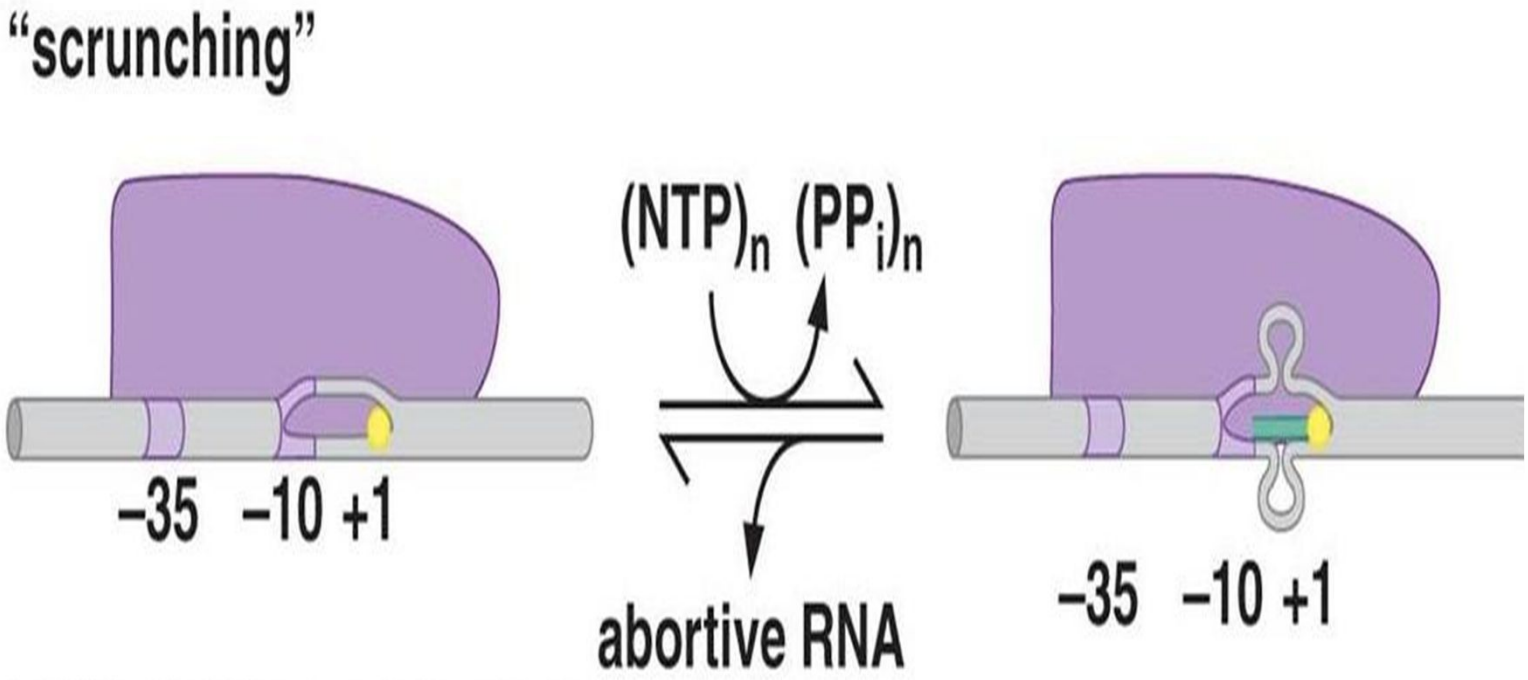
Initial Transcription

- The DNA thus accumulated within the enzyme is accommodated as single-stranded bulges.

Initial Transcription

- The DNA thus accumulated within the enzyme is accommodated as single-stranded bulges.

Initial Transcription



Initial Transcription

- It is now believed that the third model—scrunching — reflects what actually happens.

Initial Transcription

- The experiments have shown that during initial transcription, the polymerase remains stationary on the promoter, unwinds downstream DNA, and pulls that DNA into itself.

Initial Transcription

- Only the scrunching model is consistent with these results.

END

The Elongating Polymerase

- The Elongating Polymerase Is a processive machine that synthesizes and proofreads RNA.
- DNA passes through the elongating enzyme in a manner very similar to its passage through the open complex.

The Elongating Polymerase

- Thus, double-stranded DNA enters the front of the enzyme between the pincers.
- At the opening of the catalytic cleft, the strands separate to follow different paths through the enzyme.

The Elongating Polymerase

- The strands exit via their respective channels and re-form a double helix behind the elongating polymerase.

The Elongating Polymerase

- Ribonucleotides enter the active site through their defined channel and are added to the growing RNA chain under the guidance of the template DNA strand.

The Elongating Polymerase

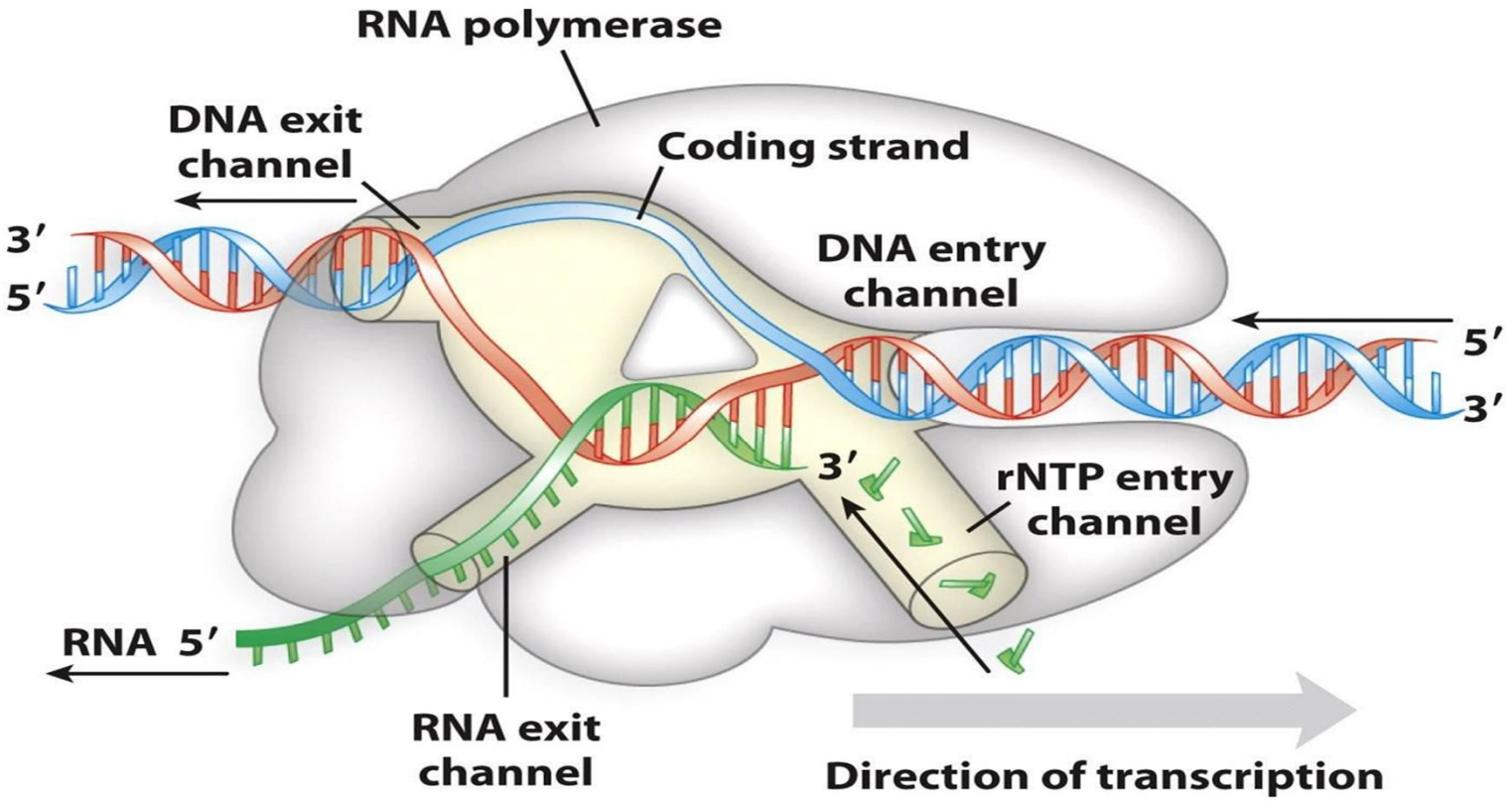


Figure 15-14
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The Elongating Polymerase

- Only 8 or 9 nucleotides of the growing RNA chain remain base-paired to the DNA template at any given time.
- The remainder of the RNA chain is peeled off and directed out of the enzyme through the RNA exit channel.

The Elongating Polymerase

- During elongation, the enzyme adds one nucleotide at a time to the growing RNA transcript.
- During elongation, polymerase uses a step mechanism: using single-molecule techniques.

The Elongating Polymerase

- It was shown that the enzyme steps forward as a molecular motor, advancing in a single step a distance equivalent to a base pair for every nucleotide it adds to the growing RNA chain.

The Elongating Polymerase

- In addition, the size of the bubble, i.e., the length of DNA that is not double-helical, remains constant throughout elongation.
- 1 bp is separated ahead of the processing enzyme, 1 bp is formed behind it.

The Elongating Polymerase

- As well as synthesizing the transcript, RNA polymerase performs two proofreading functions on that growing transcript.
- The first of these is called pyrophosphorolytic editing.

The Elongating Polymerase

- In this, the enzyme uses its active site, in a simple back-reaction, to catalyze the removal of an incorrectly inserted ribonucleotide, by reincorporation of PPi.

The Elongating Polymerase

- The enzyme can then incorporate another ribonucleotide in its place in the growing RNA chain.

The Elongating Polymerase

- In the second proofreading mechanism, called hydrolytic editing, the polymerase backtracks by one or more nucleotides and cleaves the RNA product, removing the error-containing sequence.

The Elongating Polymerase

- Another group of proteins—the Nus proteins—joins polymerase in the elongation phase and promotes the processes of elongation and termination.

END

Termination of Transcription

- The termination of transcription is a normal and important function at the ends of gene/s.
- However, an unusual termination may be triggered by damaged DNA or by other unanticipated hindrances.

Termination of Transcription

- The sequences called terminators trigger the elongating polymerase to dissociate from the DNA and release the RNA chain it has made.

Termination of Transcription

- In bacteria, terminators come in two types:-
  - 1. Rho-Dependant
  - 2. Rho-Independant

Termination of Transcription

- The first, requires a protein called Rho to induce termination.
- The second causes termination without the involvement of other factors.

Termination of Transcription

- Rho-dependent terminators have rather ill-defined RNA elements called rut sites.
- The work of these sites requires the action of the Rho factor.

Termination of Transcription

- Rho, which is a ring-shaped protein with six identical subunits, binds to single-stranded RNA as it exits the polymerase.

Termination of Transcription

- The protein also has an ATPase activity, and once attached to the transcript, Rho uses the energy derived from ATP hydrolysis to induce termination.

Termination of Transcription

- The precise mechanism of termination remains to be determined, and models include the following:-

Termination of Transcription

- i. Rho pushes polymerase forward relative to the DNA and RNA, resulting in termination in a manner analogous to termination by the protein TRCF (transcription-repair coupling factor).

Termination of Transcription

- Rho binds double-stranded DNA upstream of the polymerase and translocates along the DNA until it collides the RNA polymerase.

Termination of Transcription

- The collision pushes polymerase forward, causing dissociation of the ternary complex of RNA polymerase, template DNA, and RNA transcript.

END

Termination of Transcription

- ii. Rho pulls RNA out of the polymerase, resulting in termination; or
- iii. Rho induces a conformational change in polymerase, causing the enzyme to terminate the transcription.

Termination of Transcription

- Most recent experiments suggest that the last of the above may be correct and that the conformational change causes the elongating complex to stall, with dissociation following slowly.

Termination of Transcription

- Recent studies also have suggested that Rho binds to RNA polymerase throughout the transcription cycle.
- Thus, Rho doesn't reach polymerase by translocating along a nascent, rut-containing transcript like TRCF.

Termination of Transcription

- Rather, Rho binds polymerase early in transcription and then at some point also binds the RNA transcript being exuded from that elongating enzyme.

Termination of Transcription

- The role of translocation by Rho is thus perhaps to tighten the resulting RNA loop, and when sufficiently tight, polymerase elongation ceases.
- How is Rho directed to work on particular RNA transcripts?

Termination of Transcription

- First, there is some specificity in the sites it binds (the rut sites).
- These sites consist of stretches of ~40 nucleotides that do not fold into a secondary structure. They are also rich in C residues.

Termination of Transcription

- The second level of specificity is that Rho fails to bind any transcript that is being translated.

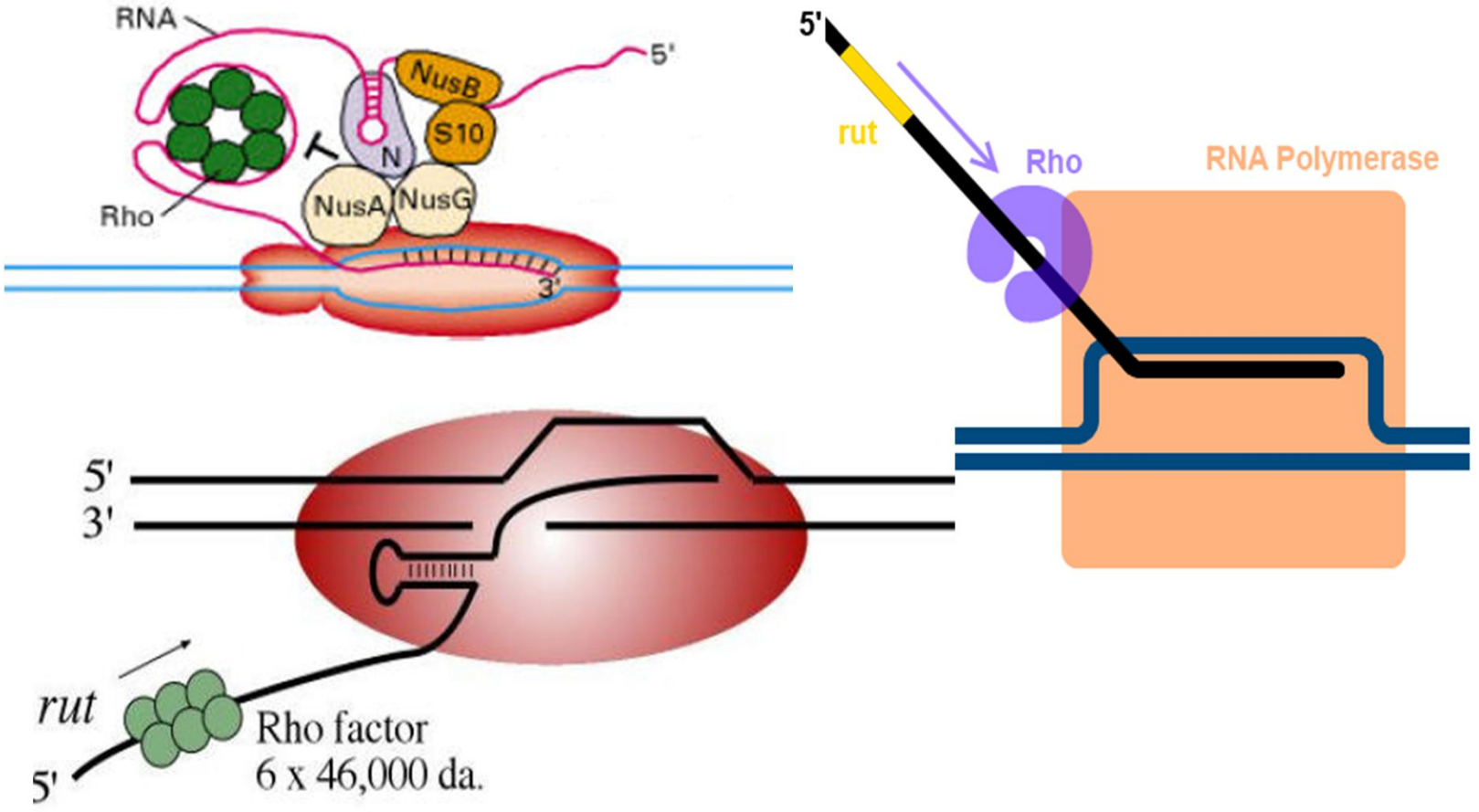
Termination of Transcription

- In bacteria, transcription and translation are tightly coupled—translation initiates on growing RNA transcripts as soon as they start exiting polymerase, while they are still being synthesized.

Termination of Transcription

- Thus, Rho typically terminates only those transcripts which are still being transcribed beyond the end of a gene or operon.

Termination of Transcription



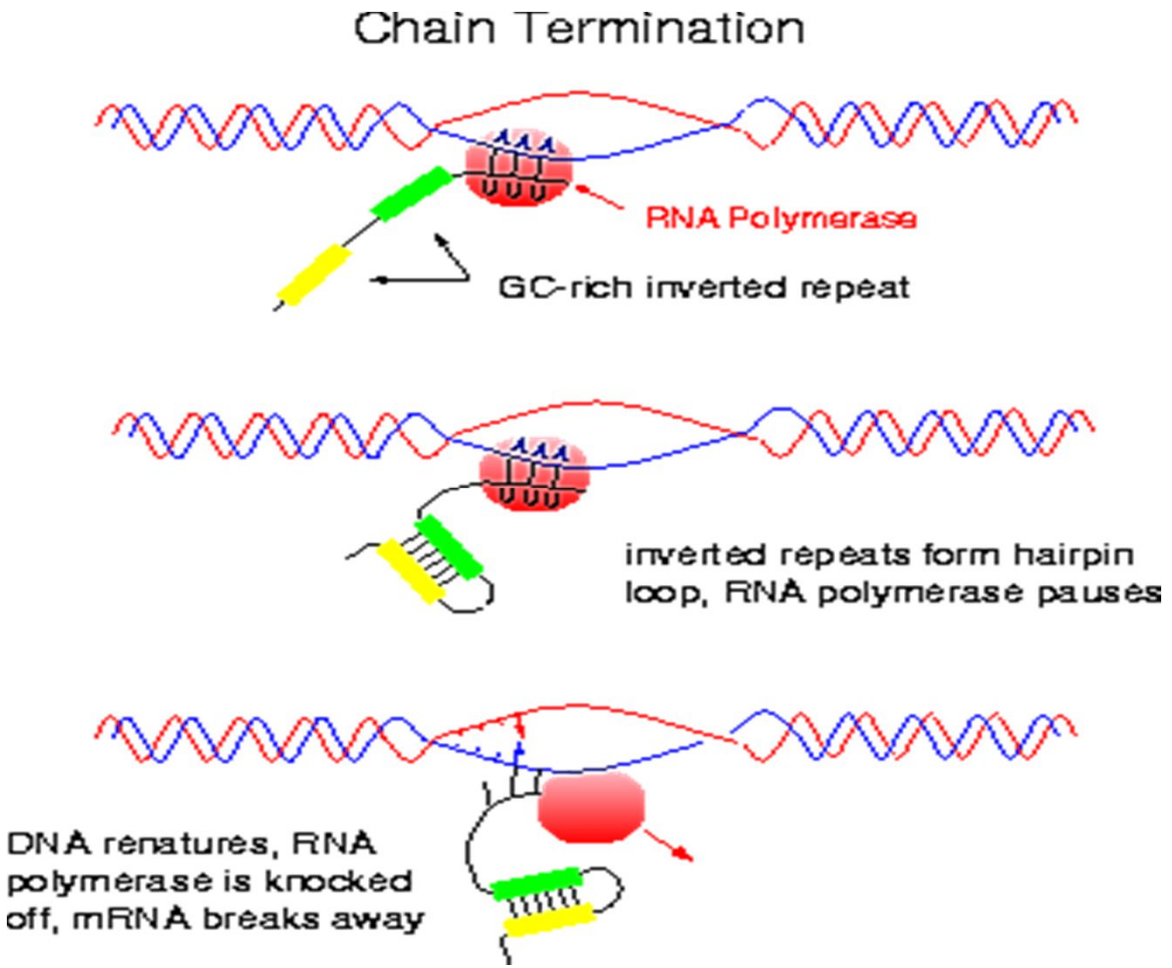
Rho-independent Terminators

- Rho-independent terminators are also called intrinsic terminators because they need no other factors to work.

Rho-independent Terminators

- They consist of two sequence elements:-
- A short inverted repeat of 20 nucleotides followed by a stretch of about eight A:T base pairs.

Rho-independent Terminators



Rho-independent Terminators

- These elements do not affect the polymerase until they have been transcribed—that is, they function in the RNA rather than in the DNA.

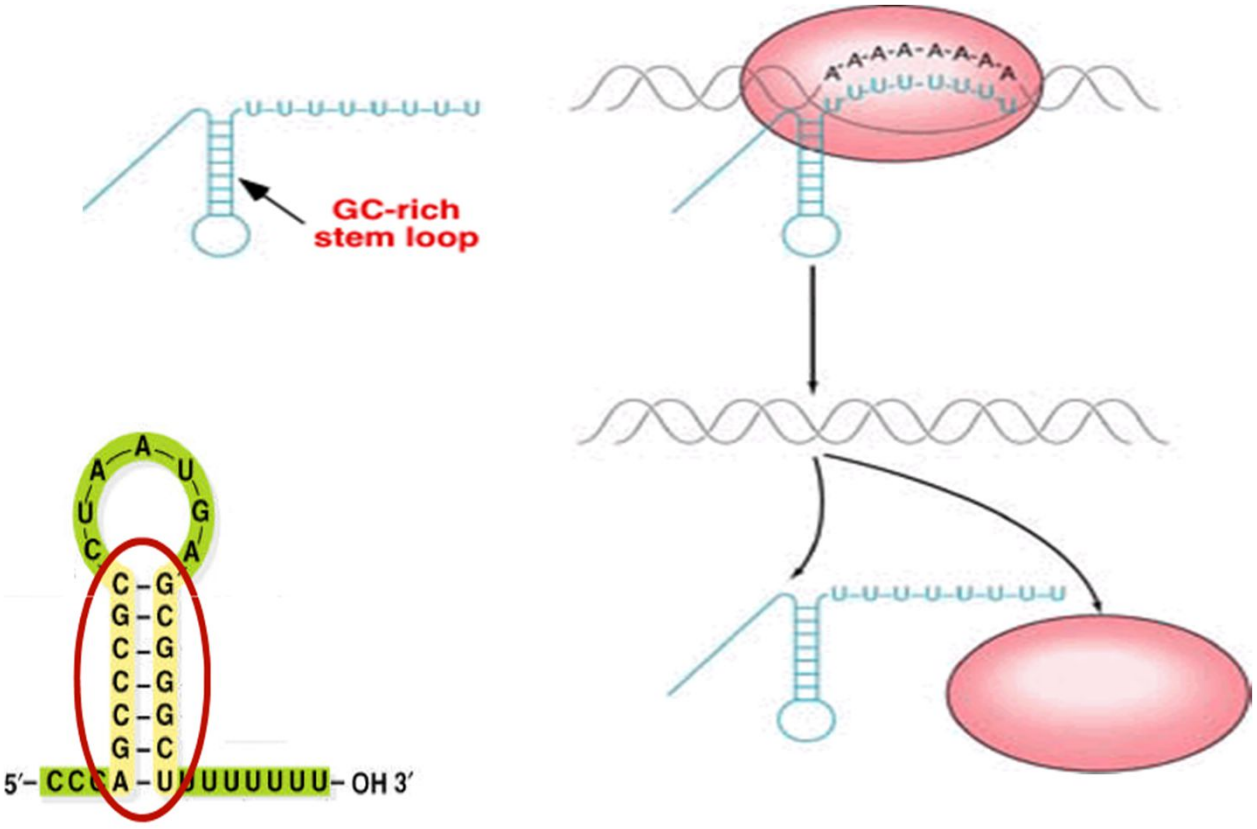
Rho-independent Terminators

- When polymerase transcribes an inverted repeat sequence, the resulting RNA can form a stem-loop structure often called a “hairpin” by base-pairing with itself.

Rho-independent Terminators

- Formation of the hairpin causes termination by disrupting the elongation complex.

Rho-independent Terminators



Rho-independent Terminators

- The hairpin induces termination by either pushing polymerase forward relative to the DNA and RNA, wresting the transcript from polymerase, or inducing a conformational change in polymerase.

Rho-independent Terminators

- The hairpin works as an efficient terminator only when it is followed by a stretch of A:U base pairs.

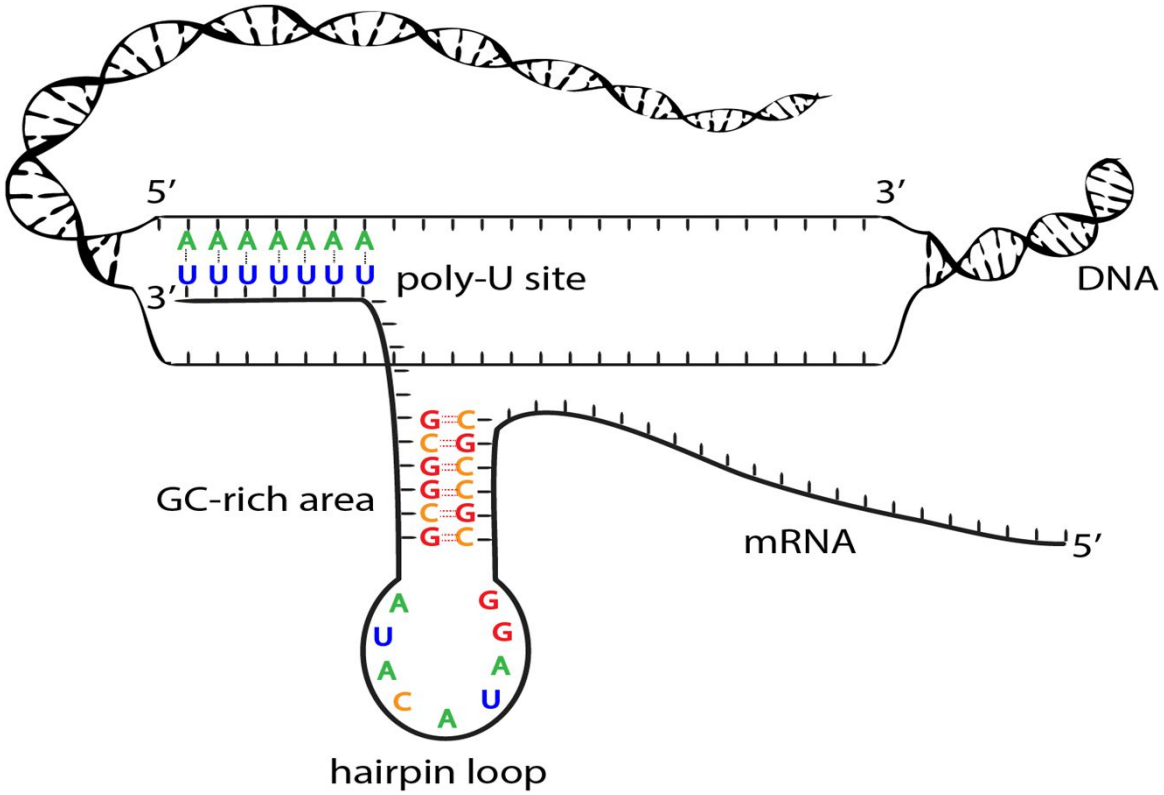
Rho-independent Terminators

- This is because, under those circumstances, at the time the hairpin forms, the growing RNA chain will be held on the template at the active site by only A:U base pairs.

Rho-independent Terminators

- Because A:U base pairs are the weakest of all base pairs, they are more easily disrupted by the effects of the stem-loop on the transcribing polymerase, and thus the RNA will more readily dissociate.

Rho-independent Terminators



TRANSCRIPTION IN EUKARYOTES

- Transcription in eukaryotes is undertaken by polymerases closely related to the RNA polymerases found in prokaryotes.

TRANSCRIPTION IN EUKARYOTES

- The process of transcription is identical in both the prokaryotes and eukaryotes. There are, however, differences in the machinery used in each case.

TRANSCRIPTION IN EUKARYOTES

- Bacteria have only one RNA polymerase but all eukaryotes have at least three different ones i.e., Pol I, II, and III; and plants also have a Pol IV and a Pol V.

TRANSCRIPTION IN EUKARYOTES

- In addition, whereas bacteria require only one additional initiation factor (σ), several initiation factors are required for efficient and promoter-specific initiation in eukaryotes.

TRANSCRIPTION IN EUKARYOTES

- These are called the general transcription factors (GTFs).
- In vitro, the general transcription factors are all that are required, together with Pol II, to initiate transcription on a DNA template.

TRANSCRIPTION IN EUKARYOTES

- In vivo, however, the general transcription factors are not alone sufficient to bind promoter sequences and elicit significant expression.

TRANSCRIPTION IN EUKARYOTES

- Rather, additional factors are required, including DNA-binding regulatory proteins, the so-called Mediator complex, and often chromatin-modifying enzymes.

TRANSCRIPTION IN EUKARYOTES

- The eukaryotic core promotor refers to the minimal set of sequence elements required for accurate transcription initiation by the Pol II machinery.

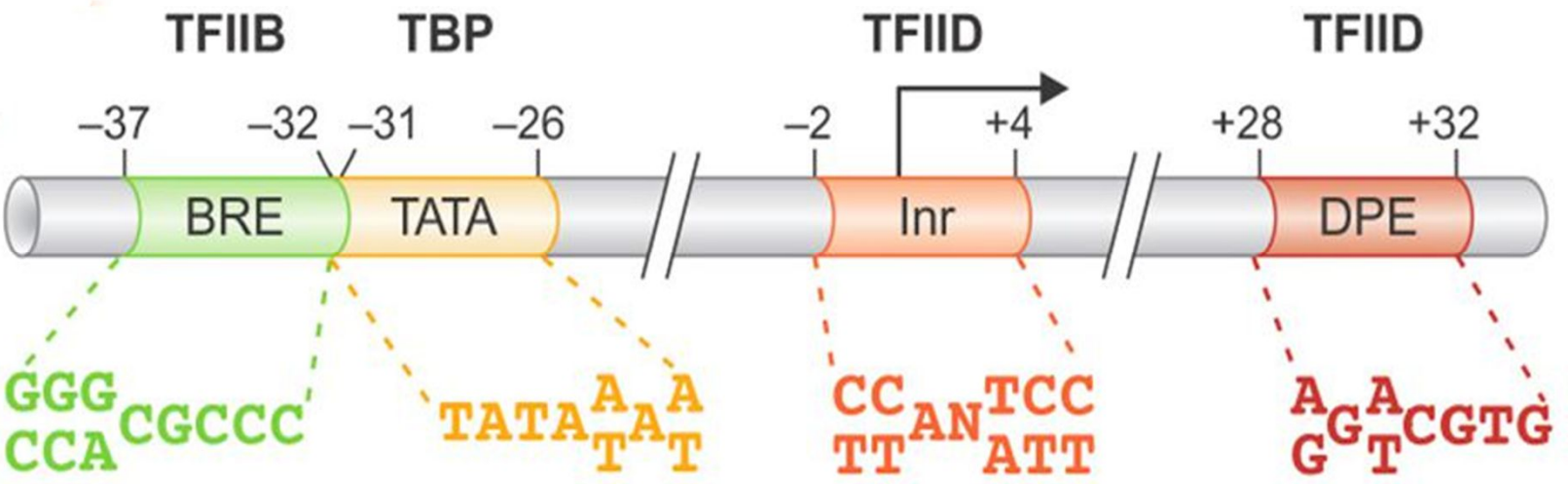
TRANSCRIPTION IN EUKARYOTES

- A core promoter is typically ~40–60 nucleotides long, extending either upstream or downstream from the transcription start site.

TRANSCRIPTION IN EUKARYOTES

- The elements found in Pol II core promoter include the TFIIB recognition element (BRE), the TATA element (or box), the initiator (Inr), and the downstream promoter elements (known as DPE, DCE, and MTE).

TRANSCRIPTION IN EUKARYOTES



Pol II core promoter

TRANSCRIPTION IN EUKARYOTES

- Typically, a promoter includes some subset of these elements. Thus, for example, promoters typically have either a TATA element or a DPE element, not both.

TRANSCRIPTION IN EUKARYOTES

- Often, a TATA-containing promoter also contains a DCE.
- The Inr is the most common element, found in combination with both TATA and DPEs.

TRANSCRIPTION IN EUKARYOTES

- Typically upstream of the core promoter, there are other sequence elements required for accurate and efficient transcription in vivo.
- Together, these elements constitute the regulatory sequences.

TRANSCRIPTION IN EUKARYOTES

- These elements include promoter proximal elements, upstream activator sequences (UASs), enhancers, and a series of other elements called silencers, boundary elements, and insulators.

TRANSCRIPTION IN EUKARYOTES

- All of these DNA elements bind regulatory proteins (activators and repressors), which help or hinder transcription from the core promoter.

TRANSCRIPTION IN EUKARYOTES

- Some of these regulatory sequences can be located many tens or even hundreds of kilobases from the core promoters on which they act.

END

Formation of pre-initiation Complex

- The general transcription factors collectively perform the functions performed by σ in bacterial transcription.

Formation of pre-initiation Complex

- Thus, the general transcription factors help polymerase bind to the promoter and melt the DNA.
- This is comparable to the transition from the closed to the open complex in the bacterial case.

Formation of pre-initiation Complex

- They also help polymerase escape from the promoter and embark on the elongation phase.

Formation of pre-initiation Complex

- The complete set of general transcription factors and polymerase, bound together at the promoter and poised for initiation, is called the pre-initiation complex.

Formation of pre-initiation Complex

- Many Pol II promoters contain a so-called TATA element (some 30 bp upstream of the transcription start site).
- This is where preinitiation complex formation begins.

Formation of pre-initiation Complex

- The TATA element is recognized by the general transcription factor called TFIID ("TFII" denotes a transcription factor for Pol II).

Formation of pre-initiation Complex

- Like many of the general transcription factors, TFIID is, in fact, a multi-subunit complex.
- The component of TFIID that binds to the TATA DNA sequence is called TBP (TATA-binding protein).

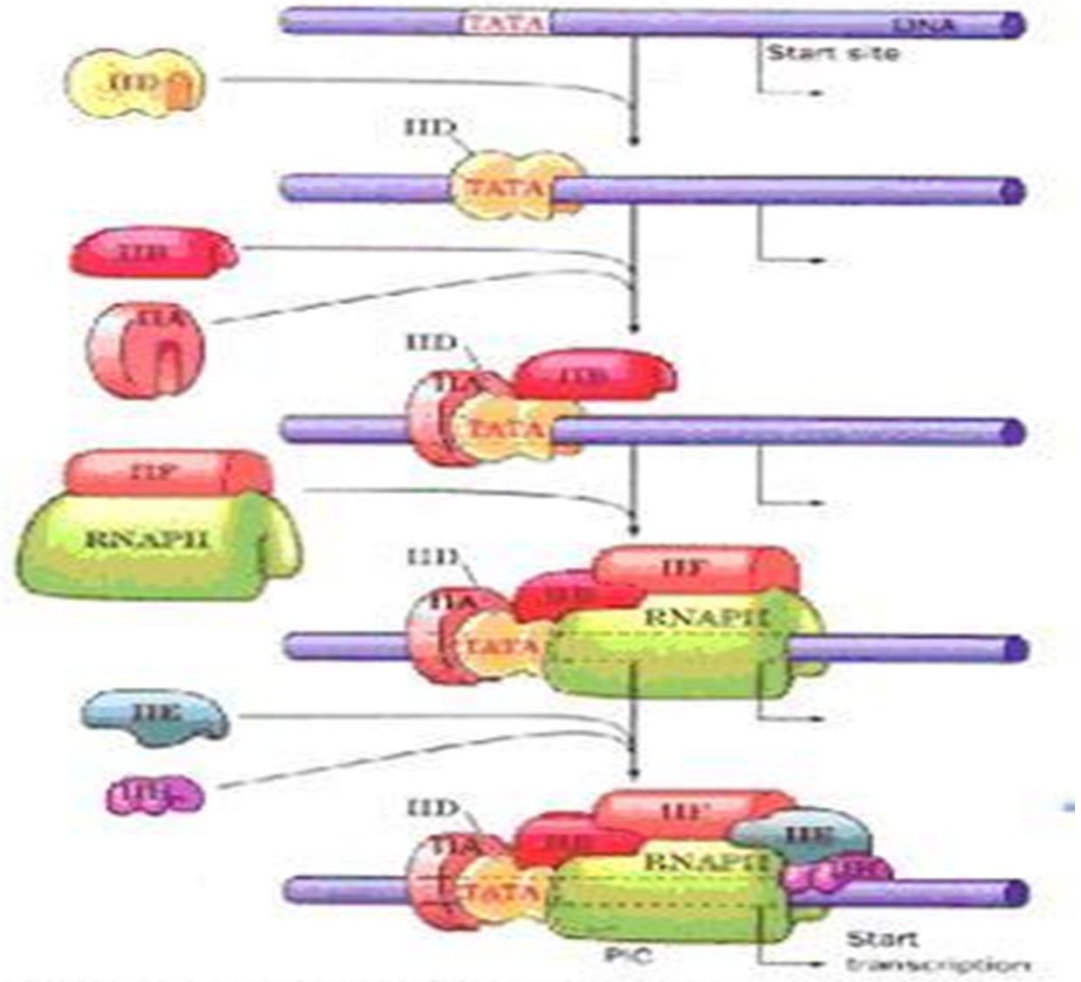
Formation of pre-initiation Complex

- The other subunits in this complex are called TAFs, for TBP-associated factors.
- Some TAFs recognize other core promoter elements such as the Inr, DPE, and DCE, although the strongest binding is between TBP and TATA.

Formation of pre-initiation Complex

- Thus, TFIID is a critical factor in promoter recognition and establishment of pre-initiation complex.

Formation of pre-initiation Complex



Formation of pre-initiation Complex

- Upon binding DNA, TBP extensively distorts the TATA sequence.
- The resulting TBP–DNA complex provides a platform to recruit other general transcription factors and polymerase itself to the promoter.

Formation of pre-initiation Complex

- In vitro, these proteins assemble at the promoter in the following order: TFIIA, TFIIB, TFIIF together with polymerase, and then TFIIE and TFIIH.

Formation of pre-initiation Complex

END

- Formation of the preinitiation complex containing these components is followed by promoter melting.
- In contrast to the situation in bacteria, promoter melting in eukaryotes requires hydrolysis of ATP and is mediated by TFIIH.

Promoter Escape

- The formation of pre-initiation complex follows a period of abortive initiation before the polymerase escapes the promoter and enters the elongation phase.

Promoter Escape

- During abortive initiation, the polymerase synthesizes a series of short transcripts.
- In eukaryotes, promoter escape involves two steps not seen in bacteria:-

Promoter Escape

- One is ATP hydrolysis (in addition to the earlier ATP hydrolysis needed for DNA melting), and the other is phosphorylation of the polymerase.

Promoter Escape

- The large subunit of Pol II has a carboxy-terminal domain (CTD), which is referred to as the “tail”.
- The CTD contains a series of repeats of the heptapeptide sequence: Tyr-Ser-Pro-Thr-Ser-Pro-Ser.

Promoter Escape

- There are 27 of these repeats in the yeast Pol II CTD, 32 in the worm *Caenorhabditis elegans*, 45 in the fly Drosophila, and 52 in humans.
- The number of repeats correlate with the complexity of the genome.

Promoter Escape

- Each repeat contains sites for phosphorylation by specific kinases, including one that is a subunit of TFIIH.

Promoter Escape

- The form of Pol II recruited to the promoter initially contains a largely unphosphorylated tail, but the species found in the elongation complex bears multiple phosphoryl groups on its tail.

Promoter Escape

- Addition of these phosphates helps polymerase shed most of the general transcription factors used for initiation, and which the enzyme leaves behind as it escapes the promoter.

Promoter Escape

- Regulating the phosphorylation state of the CTD of Pol II controls subsequent steps—elongation and even processing of the RNA— as well.

Promoter Escape

- Indeed, in addition to TFIIH, several other kinases have been identified that act on the CTD, as well as a number of phosphatases that remove the phosphates added by those kinases.

END

General Transcription Factors

- We do not know in detail the functions of all of the other general transcription factors.
- Some of these factors are in fact complexes made up of two or more subunits.

General Transcription Factors

- **TAFs**
- TBP is associated with about 10 TAFs. Two of the TAFs bind DNA elements at the promoter, for example, the initiator element (Inr) and the downstream promoter elements.

General Transcription Factors

- Several of the TAFs have structural homology with histone proteins, and it has been proposed that they might bind DNA in a similar manner, although evidence for such a binding has not been obtained.

General Transcription Factors

- For example, TAF42 and TAF62 from *Drosophila* have been shown to form a structure similar to that of the H3.H4 tetramer of histones.

General Transcription Factors

- Another TAF appears to regulate the binding of TBP to DNA. It does this using an inhibitory flap that binds to the DNA-binding surface of TBP.
- This flap must be displaced for TBP to bind TATA.

General Transcription Factors

- **TFIIB**
- This protein, a single polypeptide chain, enters the preinitiation complex after TBP.
- The crystal structure of the ternary complex of TFIIB–TBP–DNA shows specific TFIIB–TBP and TFIIB–DNA contacts.

General Transcription Factors

- These include base-specific interactions with the major groove upstream (to the BRE) and the minor groove downstream of the TATA element.

General Transcription Factors

- The asymmetric binding of TFIIB to the TBP–TATA complex accounts for the asymmetry in the rest of the assembly of the preinitiation complex and the unidirectional transcription that results.

General Transcription Factors

- TFIIB also contacts Pol II in the preinitiation complex.
- Thus, this protein appears to bridge the TATA-bound TBP and polymerase.

General Transcription Factors

- Structural studies suggest that segments of TFIIB insert into the RNA-exit channel and active center cleft of Pol II in a manner analogous to the σ region ¾ linker in the bacterial case.

General Transcription Factors

- These regions of TFIIB(called the linker and reader) aid in open complex formation, perhaps by stabilizing the melted DNA until the RNA:DNA hybrid takes over that role.

General Transcription Factors

- **TFIIF**
- This two-subunit factor associates with Pol II and is recruited to the promoter together with that enzyme (and other factors).

General Transcription Factors

- Binding of Pol II–TFIIF stabilizes the DNA–TBP–TFIIB complex and is required before TFIIE and TFIIH are recruited to the preinitiation complex.

General Transcription Factors

- In yeast, this factor includes a third subunit, but the function of the third subunit is not known.

END

General Transcription Factors

- **TFIIE and TFIIH**
- TFIIE, which consists of two subunits, binds next and has roles in the recruitment and regulation of TFIIH.

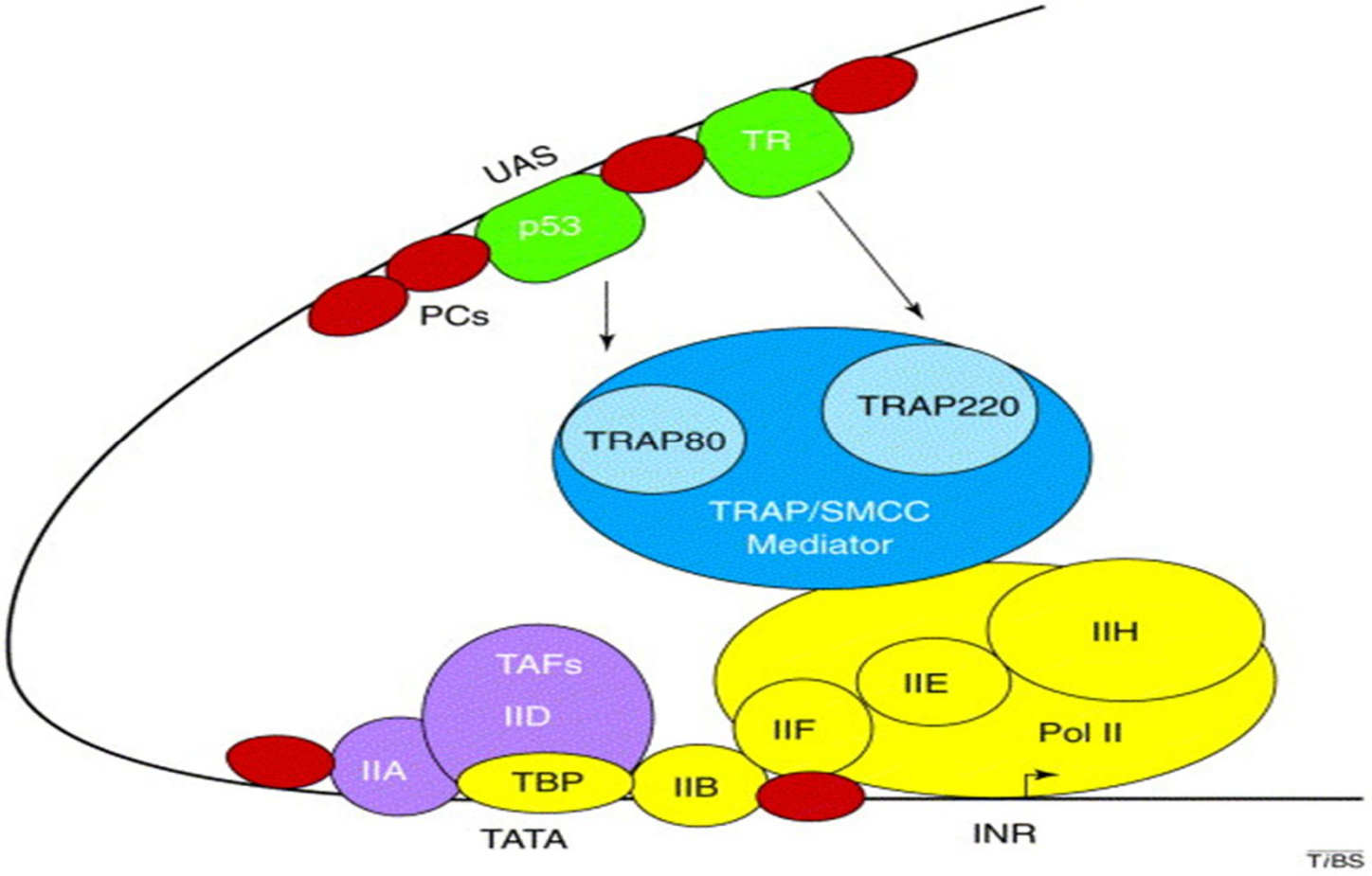
General Transcription Factors

- TFIIH controls the ATP-dependent transition of the preinitiation complex to the open complex.

General Transcription Factors

- It is the largest and most complex of the general transcription factors having 10 subunits and a molecular mass comparable to that of the polymerase itself.

General Transcription Factors



General Transcription Factors

- Within TFIIH are two subunits that function as ATPases and another that is a protein kinase, with roles in promoter melting and escape.

General Transcription Factors

- Together with other factors, the ATPase subunits are also involved in nucleotide excision repair.
- How does TFIIH mediate promoter melting?

General Transcription Factors

- It is now believed that a subunit of TFIIH acts as an ATP-driven translocator of double-stranded DNA.

General Transcription Factors

- This subunit binds to DNA downstream from polymerase and feeds double-stranded DNA, with a right-handed threading, into the cleft of the polymerase.

General Transcription Factors

- This action drives the melting of the DNA because the upstream promoter DNA is held in a fixed position by TFIID and the rest of the GTFs.

END

The Mediator Complex

- High regulated levels of transcription in vivo, require transcriptional regulatory proteins, the Mediator complex, and nucleosome modifying enzymes in addition to the above described general transcription factors.

The Mediator Complex

- One reason for these additional requirements is that the DNA template in vivo is packaged into chromatin.
- This condition complicates binding to the promoter of polymerase and its associated factors.

The Mediator Complex

- Transcriptional regulatory proteins called activators help recruit polymerase to the promoter, stabilizing its binding there.

The Mediator Complex

- This recruitment is mediated through interactions between DNA-bound activators, chromatin-modifying and -remodeling factors, and parts of the transcription machinery.

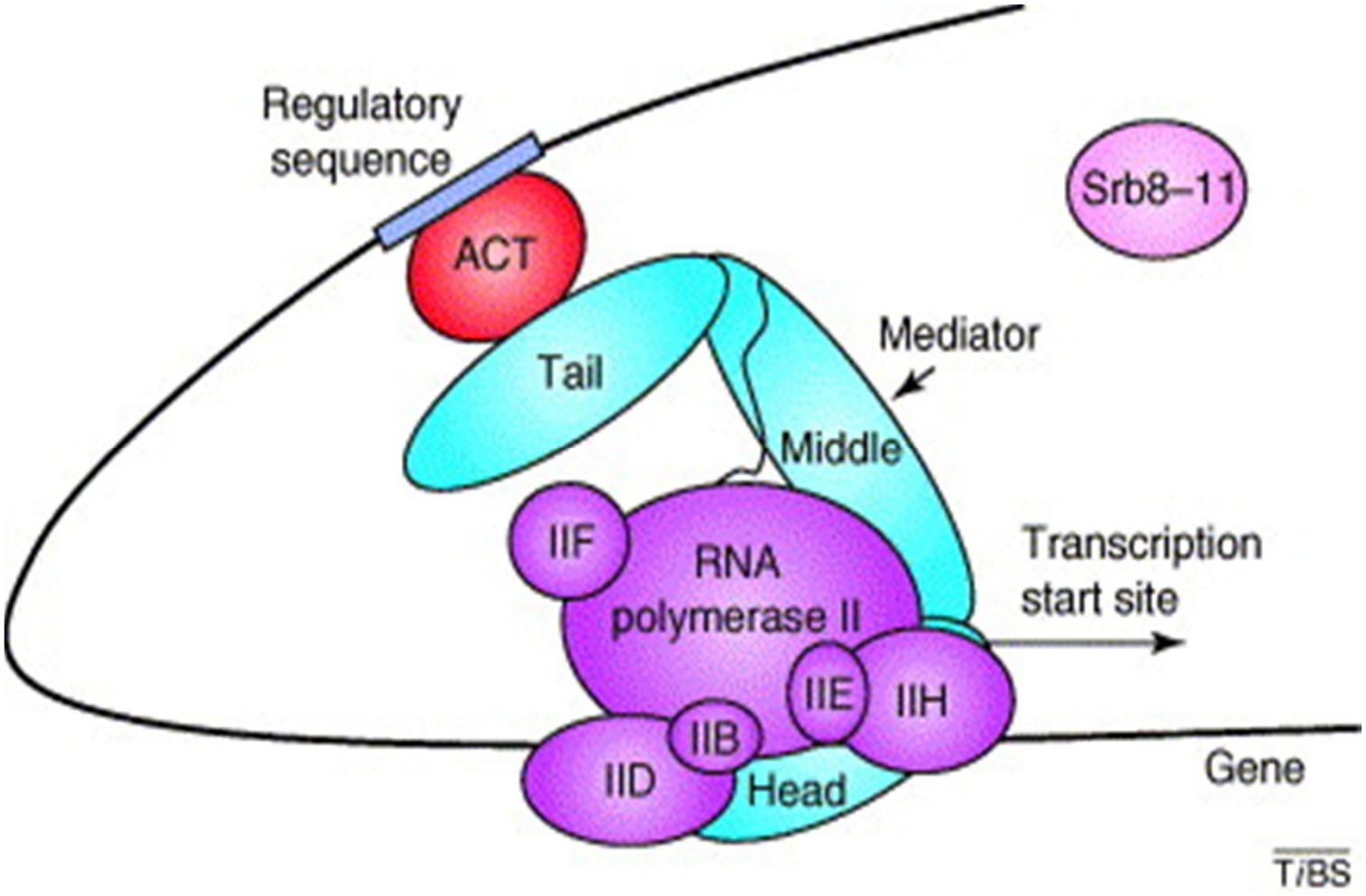
The Mediator Complex

- One such interaction is with the Mediator complex.
- Mediator is associated with the basic transcription machinery, most likely touching the CTD “tail” of the large polymerase subunit through one surface,

The Mediator Complex

- while presenting other surfaces for interaction with DNA-bound activators.
- This explains the need for Mediator to achieve significant transcription in vivo.

The Mediator Complex



The Mediator Complex

- Despite this central role in transcriptional activation, deletion of individual subunits of Mediator often leads to loss of expression of only a small subset of genes, different for each subunit.

The Mediator Complex

- This result likely reflects the fact that different activators are believed to interact with different Mediator subunits to bring polymerase to different genes.
- Mediator also aids initiation by regulating the CTD kinase in TFIIH.

The Mediator Complex

- The need for nucleosome modifiers and remodelers also differs at different promoters or even at the same promoter under different circumstances.

The Mediator Complex

- When and where required, these complexes are also typically recruited by the DNA-bound activators, or sometimes by regulatory RNAs.

The Mediator Complex

- The yeast and human Mediators each include more than 20 subunits, of which seven show significant sequence homology between the two organisms.

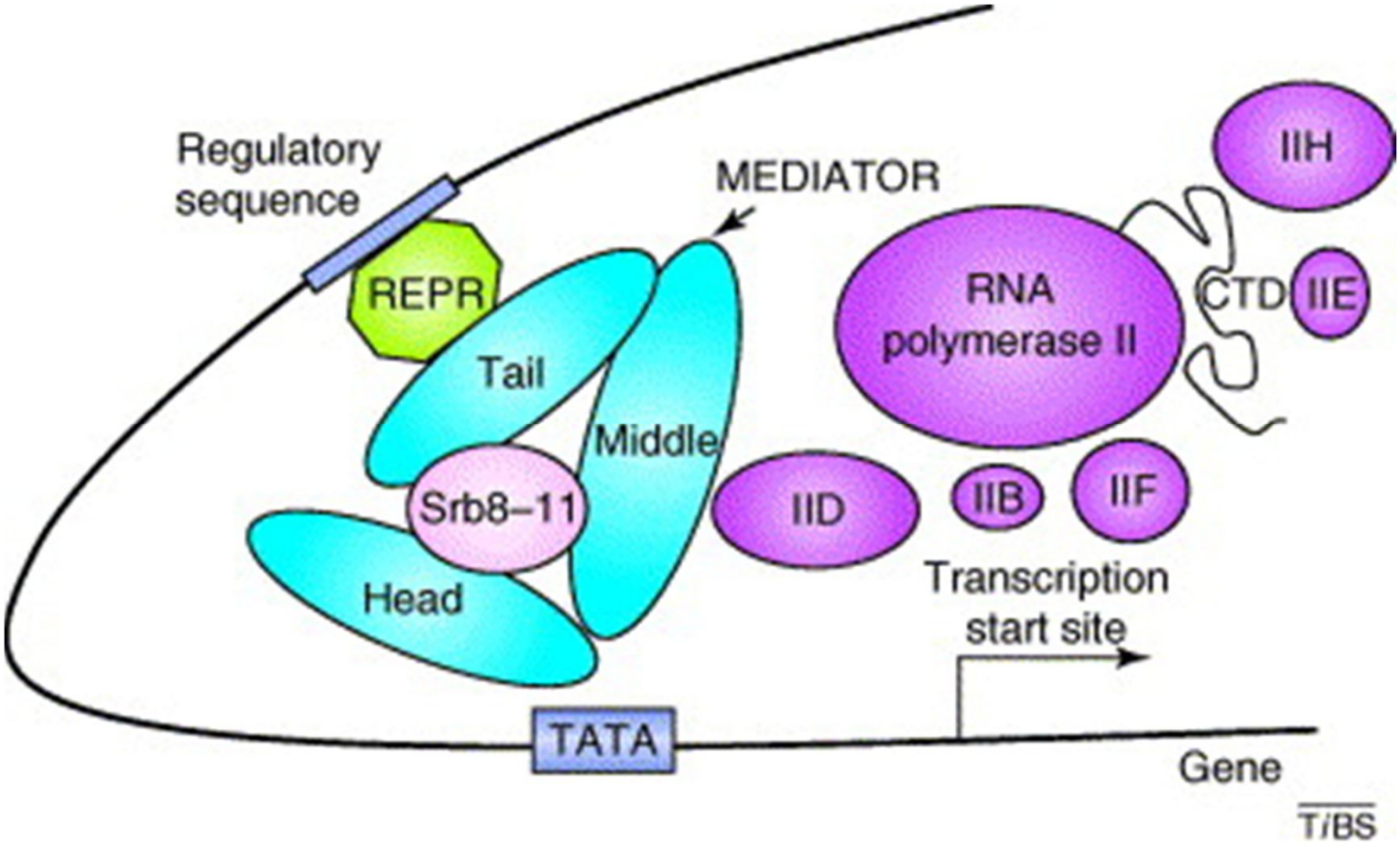
The Mediator Complex

- The Mediator from both yeast and humans is organized in modules, each containing a subset of the subunits.

The Mediator Complex

- These modules—called head, middle (or arm), and tail—can be dissociated from one another under certain conditions in vitro.

The Mediator Complex



The Mediator Complex

- Crystal structure of the head module of yeast Mediator reveals that it contains seven subunits (Med17/Srb4, Med11, Med22/Srb6, Med6, Med8, Med18/Srb5, and Med20/Srb2).

The Mediator Complex

- It forms a three-domain structure that binds the transcription complex in such a way as to juxtapose TFIIH and the CTD tail of RNA polymerase, promoting phosphorylation of the latter by the former.

END

RNA Elongation and Proofreading

- Once polymerase has escaped the promoter and initiated transcription, it shifts into the elongation phase.

RNA Elongation and Proofreading

- This transition involves the Pol II enzyme shedding most of its initiation factors such as, the general transcription factors and Mediator.

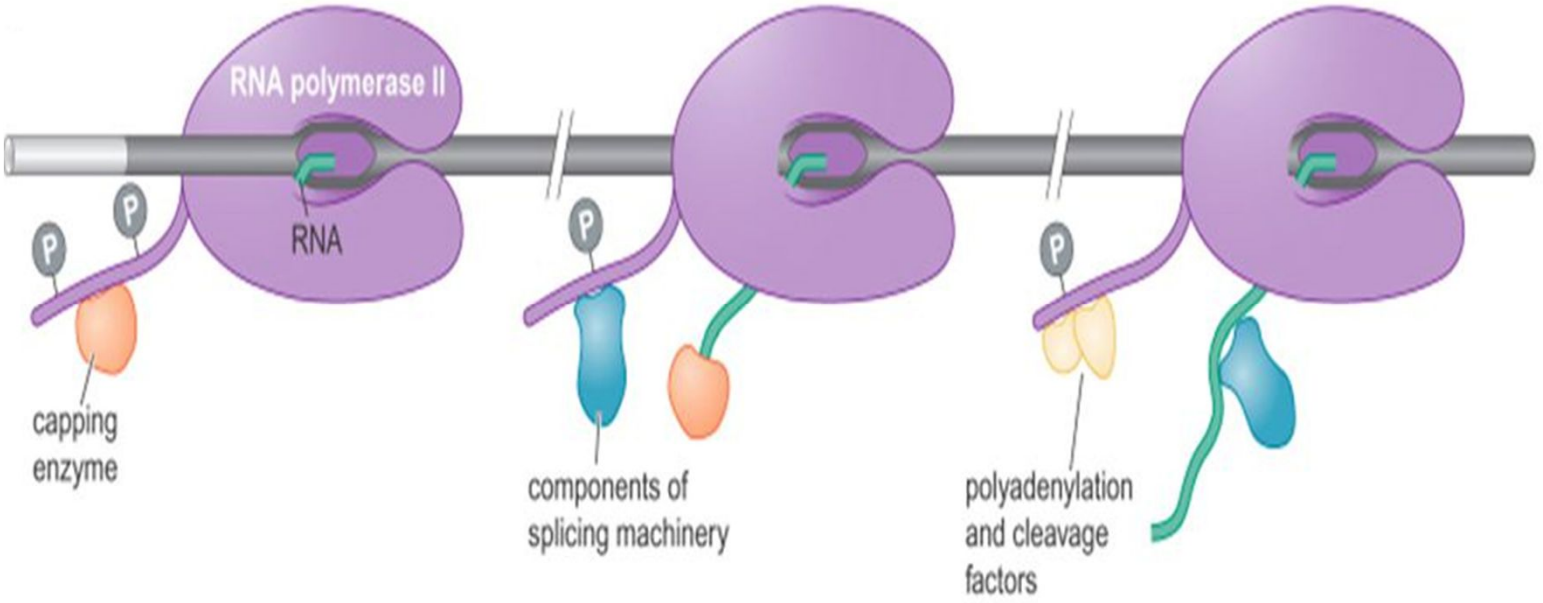
RNA Elongation and Proofreading

- In their place, another set of factors is recruited. Some of these (such as TFIIS and SPT5) are elongation factors (i.e., factors that stimulate elongation).
- Others are required for RNA processing.

RNA Elongation and Proofreading

- The enzymes involved in RNA processing are, like several of the initiation factors recruited to the carboxy-terminal (CTD) tail of the large subunit of Pol II

RNA Elongation and Proofreading



RNA Elongation and Proofreading

- In this case, however, the factors favor the phosphorylated form of the CTD. Thus, phosphorylation of the CTD leads to an exchange of initiation factors for those factors required for elongation and RNA processing.

RNA Elongation and Proofreading

- As is evident from the crystal structure of yeast Pol II, the polymerase CTD lies directly adjacent to the channel through which the newly synthesized RNA exits the enzyme.

RNA Elongation and Proofreading

- The CTD tail is also very long.
- It could potentially extend 800 A˚ from the body of the enzyme—that is, about seven times the length of the rest of the enzyme.

RNA Elongation and Proofreading

- Together, these features allow the tail to bind several components of the elongation and processing machinery and deliver them to the emerging RNA.

RNA Elongation and Proofreading

- Various proteins are thought to stimulate elongation by Pol II.
- One of these, the kinase P-TEFb, is recruited to polymerase by transcriptional activators.

RNA Elongation and Proofreading

- Once bound to Pol II, this protein phosphorylates the serine residue at position 2 of the CTD repeats. That phosphorylation event correlates with elongation.

RNA Elongation and Proofreading

- In addition, P-TEFb phosphorylates and thereby activates another protein, called SPT5, itself an elongation factor.
- Finally, TAT-SF1, yet another elongation factor, is recruited by P-TEFb.

RNA Elongation and Proofreading

- Thus, P-TEFb stimulates elongation in three separate ways.
- SPT5 is comparable to the bacterial elongation factor NusG.

RNA Elongation and Proofreading

- Indeed, this is the only universally conserved transcription factor across all three kingdoms of life—from bacteria, through Archaea, to eukaryotes.

END

RNA Elongation and Proofreading

- NusG/SPT5 factors bind to their respective RNA polymerases at the tip of the clamp, overlapping the region contacted by σ region 4 (in bacteria) and TFIIB (in eukaryotes).

RNA Elongation and Proofreading

- This overlapping and binding raises the interesting possibility that displacing initiation factors may be part of the function of these elongation regulators.

RNA Elongation and Proofreading

- This also suggests that regulating the rate of elongation is an ancient mechanism of regulating gene expression.

RNA Elongation and Proofreading

- There are some promoters in higher eukaryotes where the preinitiation complex is recruited effectively, but polymerase remains paused just after initiating transcription.

RNA Elongation and Proofreading

- Such promoters seem to be associated with genes poised to be expressed either rapidly or in a highly coordinated fashion.

RNA Elongation and Proofreading

- And their expression is regulated through recruitment by specific activators of the PTEFb kinase, which then releases them from their pause.

RNA Elongation and Proofreading

- Yet another class of elongation factor is the so-called ELL family.
- These also bind to elongating polymerase and suppress transient pausing by the enzyme.

RNA Elongation and Proofreading

- The first human ELL protein was originally identified as the product of a gene that undergoes translocations in acute myeloid leukemia.

RNA Elongation and Proofreading

- Another factor that does not affect initiation, but stimulates elongation, is TFIIS.

RNA Elongation and Proofreading

- This factor, like ELL, stimulates the overall rate of elongation by limiting the length of time that polymerase pauses when it encounters sequences that would otherwise tend to slow the enzyme's progress.

RNA Elongation and Proofreading

- It is a feature of polymerase that it does not transcribe through all sequences at a constant rate.
- Rather, it pauses periodically, sometimes for rather long periods, before resuming transcription.

RNA Elongation and Proofreading

- In the presence of TFIIS, the length of time that polymerase pauses at any given site is reduced.
- TFIIS also contributes to proofreading by polymerase.

RNA Elongation and Proofreading

- TFIIS stimulates an inherent Rnase activity in polymerase (not part of the active site), allowing an alternative approach to removing misincorporated bases through local limited RNA degradation.

RNA Elongation and Proofreading

- This feature is comparable to the hydrolytic editing in the bacterial case stimulated by the Gre factors.

END

Transcription Termination

- The final RNA processing event, polyadenylation of the 3’ end of the mRNA, is intimately linked with the termination of transcription, although exactly how is still not quite clear.

Transcription Termination

- Recently, however, an enzyme that degrades the second RNA as it emerges from the polymerase has been identified, and this enzyme may itself trigger termination. This is called the torpedo model of termination.

Transcription Termination

- The free end of the second RNA is uncapped and thus can be distinguished from genuine transcripts.

Transcription Termination

- This new RNA is recognized by an RNase called, in yeast, Rat1 (in humans, Xrn2) that is loaded onto the end of the RNA by another protein (Rtt103) that binds the CTD of RNA polymerase.

Transcription Termination

- The Rat1 enzyme is very processive and quickly degrades the RNA in a 5’-to-3’ direction, until it catches up to the still-transcribing polymerase from which the RNA is being spewed.

Transcription Termination

- Termination may not require very specific interaction between Rat1 and polymerase and might, in fact, be triggered in a manner rather similar to that of Rho-dependent termination in bacteria.

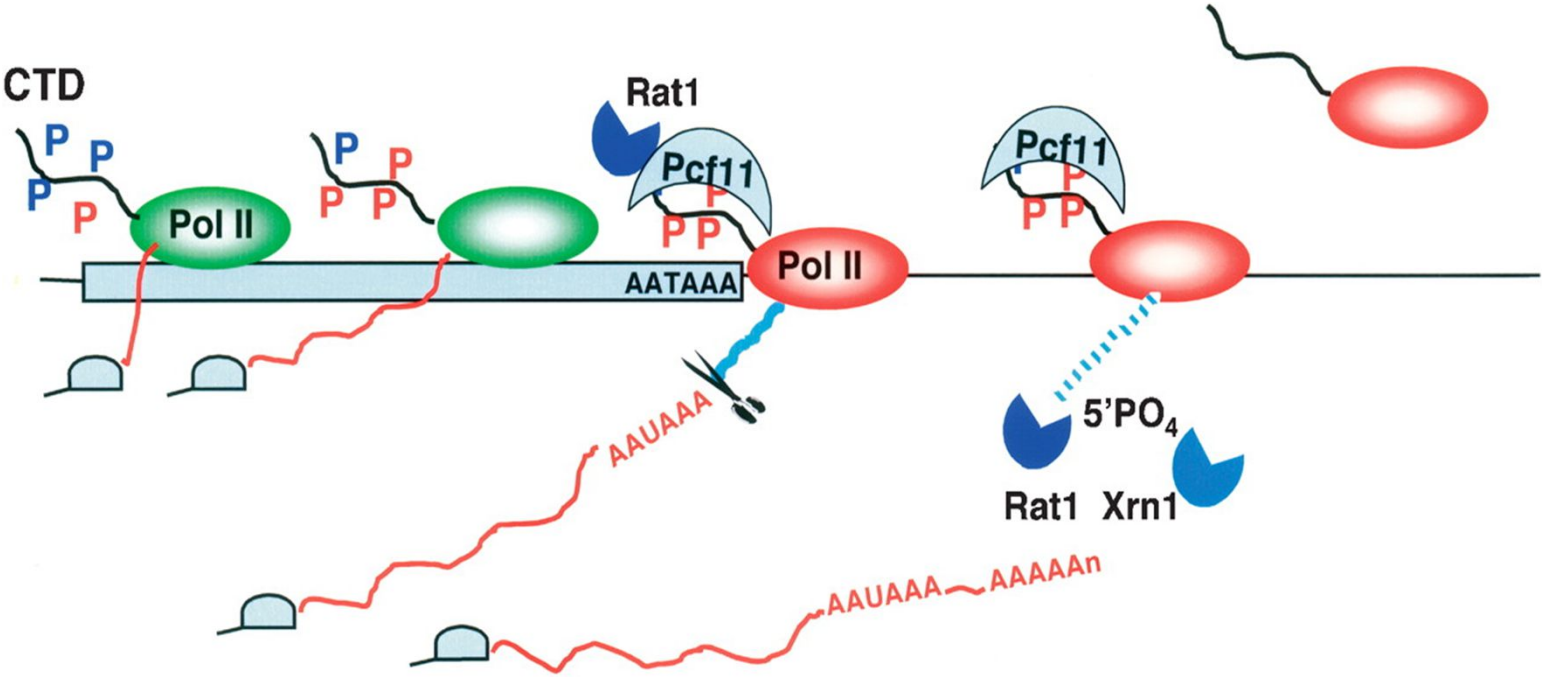
Transcription Termination

- The highly processing RNase polymerase either pushes polymerase forward and/or pulls the remains of the nascent RNA transcript from the enzyme.

Transcription Termination

- It is also possible that other factors are needed in addition to Rat1 to dislodge polymerase as, in vitro, Rat1 is alone insufficient to carry out this function, even after it has degraded the transcript.

Transcription Termination



Transcription Termination

- Although the torpedo model for termination is now the favored one, there is an alternative called the allosteric model.

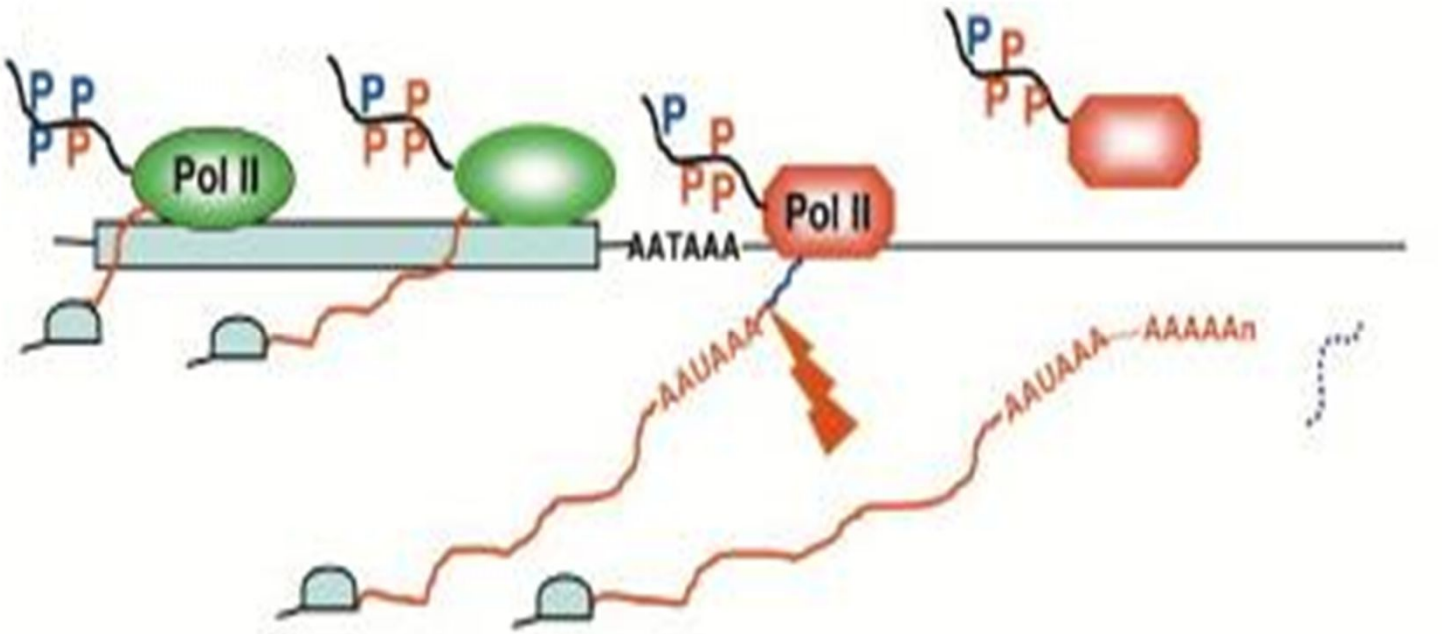
Transcription Termination

- According to this model, termination depends on a conformational change in the elongating polymerase that reduces the processivity of the enzyme leading to spontaneous termination soon afterward.

Transcription Termination

- This conformational change would be linked to polyadenylation and could be triggered by the transfer of the 3'-processing enzymes from the CTD tail of polymerase to the RNA.

Transcription Termination



Polymerases I and III

- All eukaryotes have two other RNA polymerases— Pol I and Pol III—in addition to Pol II.

Polymerases I and III

- These enzymes are related to Pol II and even share several subunits but they initiate transcription from distinct promoters and transcribe distinct genes.

Polymerases I and III

- Those genes encode specialized RNAs rather than proteins. Each of these enzymes also works with its own unique set of general transcription factors.

Polymerases I and III

- TBP, however, is universal—it is involved in initiating transcription by Pol I and Pol III, as well as Pol II.

Polymerases I and III

- Although TBP is the only GTF that is used by Pol I and Pol III aswell as by Pol II, it has emerged recently that some of the other GTFs, in fact, have structurally and functionally equivalent components in the other systems.

Polymerases I and III

- Thus, for example, TFIIF seems to have a counterpart in two subunits within Pol I (A49/34.5), and also in Pol II (C37/53).
- Likewise, TFIIE-like subunits are found in Pol I and Pol III enzymes.

Polymerases I and III

- In addition, both these other systems include additional factors comparable to TFIIB: the TAF1B factor in the Pol I system, and the Brf1 subunit of TFIIIB in the case of Pol III.

Polymerases I and III

- Pol I is required for the expression of only one gene, that encoding the rRNA precursor.

Polymerases I and III

- There are many copies of that gene in each cell, and, indeed, it is expressed at far higher levels than any other gene, perhaps explaining why it has its own dedicated polymerase.

Polymerases I and III

- The promoter for the rRNA gene comprises two parts: the core element and the UCE (upstream control element).

END

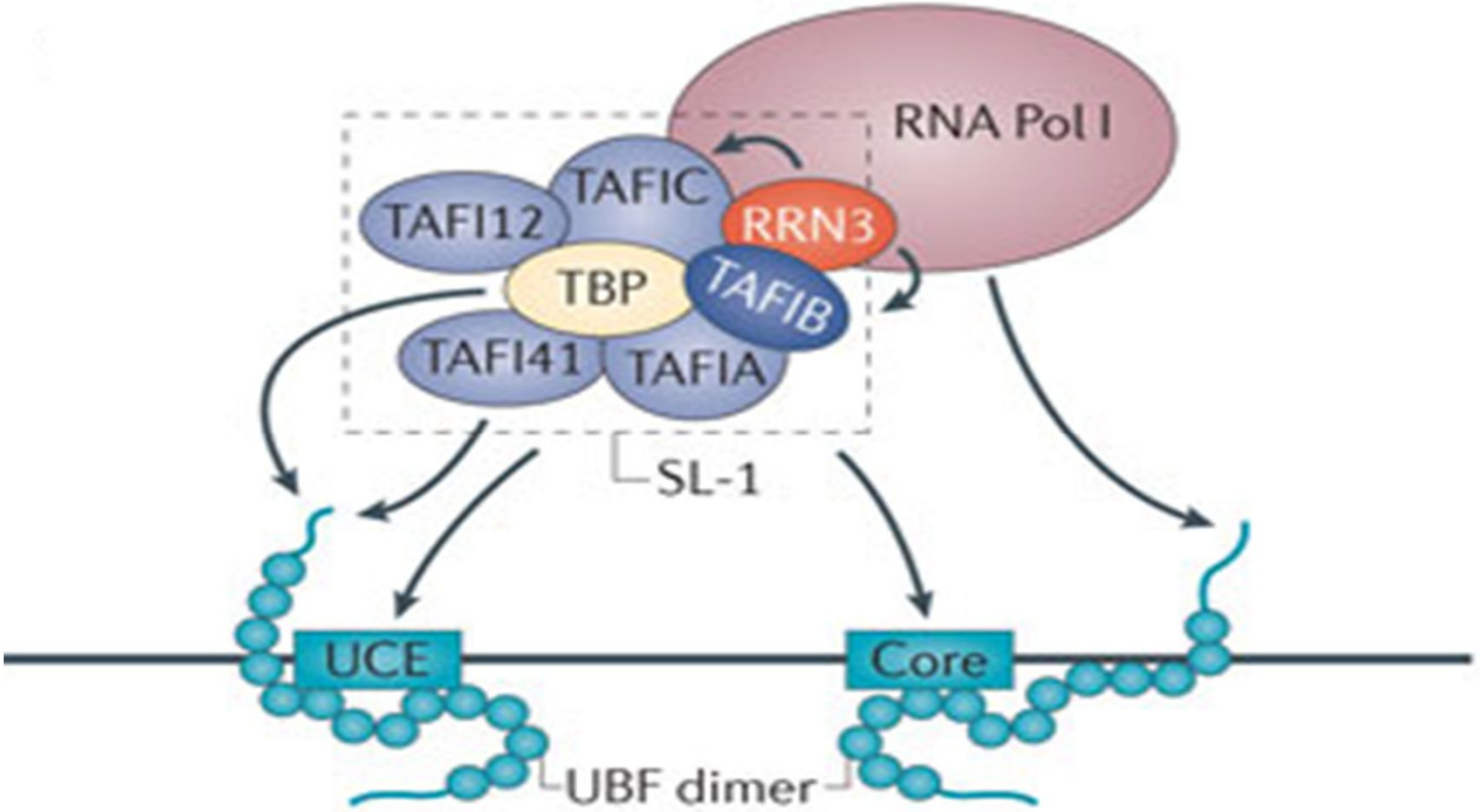
Polymerases I and III

- The former is located around the start site of transcription, and the latter between 100 and 150 bp upstream (in humans).
- In addition to Pol I, initiation requires two other factors, called SL1 and UBF.

Polymerases I and III

- SL1 comprises TBP and three TAFs specific for Pol I transcription. This complex binds to the core element.

Polymerases I and III



Polymerases I and III

- SL1 binds DNA only in the presence of UBF. This factor binds to UCE, bringing in SL1 and stimulating transcription from the core promoter by recruiting Pol I.

Polymerases I and III

- Pol III promoters come in various forms, and the vast majority have the unusual feature of being located downstream from the transcription start site (i.e., within the coding region of the gene).

Polymerases I and III

- Some Pol III promoters (e.g., those for the tRNA genes) consist of two regions, called Box A and Box B, separated by a short element.

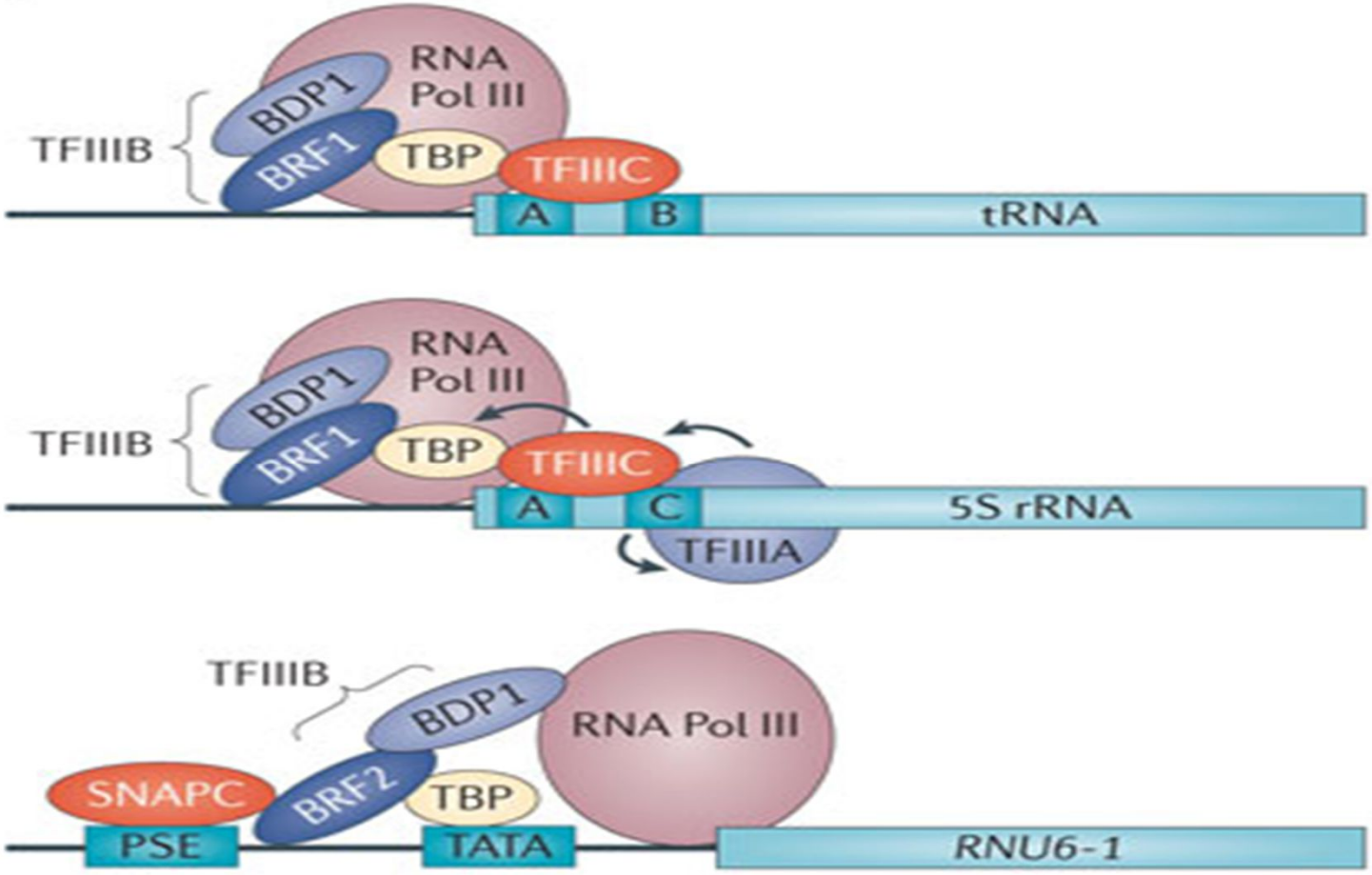
Polymerases I and III

- Others contain Box A and Box C (e.g., the 5S rRNA gene); and still others contain a TATA element like those of Pol II.

Polymerases I and III

- Transcription by Pol III also requires transcription factors in addition to polymerase.
- In this case, the factors are called TFIIIB and TFIIIC for the tRNA genes and those plus TFIIIA for the 5S rRNA gene.

Polymerases I and III



Polymerases I and III

- The TFIIIC complex binds to the promoter region. This complex recruits TFIIIB to the DNA just upstream of the start site, where it, in turn, recruits Pol III to the start site of transcription.

Polymerases I and III

END

- The enzyme then initiates, presumably displacing TFIIIC from the DNA template as it goes.
- Pol III also uses TBP like other polymerases. In this case, that ubiquitous factor is found within the TFIIIB complex.